Vanguard

Vanguard Brokerage Account (Retirement)

Application to Remove Brokerage Account Restrictions

Effective September 2024

Your fund holdings were recently transitioned from our legacy mutual fund-only platform to an identically registered Vanguard Brokerage Individual Retirement Account (IRA). You are currently restricted from making additional purchases or exchanges in your accounts. To remove these restrictions from your account, complete this application to provide Vanguard Brokerage Services with information we're required by FINRA, the SEC, and/or other state and federal regulations and laws to collect, as well as acknowledge and agree to the terms and conditions of relevant agreements such as the enclosed Vanguard Brokerage Account Agreement.

To avoid additional delays in gaining full access to your account, **be sure to include all required documentation**. If the account is to be administered by a court appointed guardian or conservator please make sure Vanguard has a copy of the court appointment on file.

See the most recent **Vanguard Brokerage Services® commission and fee schedules** for any fees that may apply.

Print in capital letters and use black ink.

1. Type of IRA

Select the appropriate retirement account and provide the account number(s). You can check more than one IRA type, if the IRAs are owned by the same owner.

☐ **Traditional IRA**	Account number(s)
☐ **Roth IRA**	Account number(s)
☐ **Rollover IRA**	Account number(s)
☐ **SEP-IRA**	Account number(s)
☐ **Inherited Traditional IRA**	Account number(s)
☐ **Inherited Roth IRA**	Account number(s)

2. Investment objectives

Industry regulations require us to collect the information below.

Review and select your investment objectives for this account from the list that follows.

- **Capital preservation.** Seeking to keep the original investment amount (the principal) from decreasing in value.
- **Income.** Seeking current cash flow in exchange for a reduction in potential capital appreciation.
- **Growth.** Seeking investments with capital appreciation potential in exchange for a reduction in income and increased volatility.
- **Speculative investing.** Seeking capital appreciation. This objective involves investments that can have significant fluctuations in value. It involves a high level of risk, which can lead to a significant loss of principal.

Note: If the investment objectives differ among multiple accounts, copy this section for each account number.

Primary investment objective *(required) Check only one.*

☐ Capital preservation ☐ Income ☐ Growth ☐ Speculative investing

Secondary investment objective *(optional) Check only one.*

☐ Capital preservation ☐ Income ☐ Growth ☐ Speculative investing

3. Account owner information

Complete A if the owner of the account is an individual person.

Complete B if the owner of the IRA will be a trust, estate, or organization.

A. Information for individuals

If the account owner isn't of legal adult age for the state in which he or she resides (18 for most states, 19 in Alabama and Nebraska, and 21 in Mississippi), then please check the box indicating the account owner is a minor.

If the account that you are attempting to regain full access to is registered as a minor account but the account owner is now of legal adult age, **then you will not be able to restore full access**. Please have the former minor contact us to open a new account.

Check below if the account owner is

☐ A minor ☐ An incapacitated person

Important: If the owner is a minor or incapacitated, you must also complete Section 4: Fiduciary information.

For Guardian accounts additional documentation is required. Please see the instructions at the front of this application.

Provide your full legal name. >

Name *first, middle initial, last*	
Birth date *mm/dd/yyyy*	Email address *required for e-delivery*
Social Security (SSN) number or individual taxpayer ID number	Gender *optional* ☐ Male ☐ Female

You must complete this entire section. >

You must provide at least one phone number. >

Daytime phone *area code, number, extension* ☐ Mobile	Evening phone *area code, number, extension* ☐ Mobile

Return ALL pages of this application, even if some sections are left blank.

Citizenship and tax residency

Check and complete only one.

☐ U.S. citizen		
☐ Resident alien	Country of citizenship	
☐ Nonresident alien*	Country of citizenship	Country of tax residence

Primary residence

Street *A P.O. box or rural route number is NOT acceptable; address can be military APO or FPO.*	
City, state, zip	Country *if not U.S.*

Mailing address

This is required if the mailing address is different from the street address. >

Street or P.O. box	
City, state, zip	Country *if not U.S.*

Important: If the mailing and/or street address you provide differ from the addresses listed on your account(s), the addresses on your brokerage account(s) will be updated accordingly.

The following information is required by industry regulations.

Employment of owner *(not required for minors) Check only one.*

☐ Employed ☐ Self-employed ☐ Retired ☐ Not employed ☐ Vanguard employed

If you checked *Employed* or *Self-employed*, you must complete this entire section. If you don't, we won't be able to process your application. >

Occupation	
Name of employer or nature of your business if self-employed	
Street address of employer or your business	
City, state, zip	Country *if not U.S.*

*If you're a nonresident alien, you must complete an IRS Form W-8 electronically to certify your tax status, and to claim treaty benefits if applicable. We'll mail you instructions for completing the electronic Form W-8 once your request has been processed. Additional restrictions apply to accounts owned by nonresident aliens.

Association *Check all that apply.*

☐ Check this box if you are a Vanguard employee.

☐ Check this box if your spouse is a Vanguard employee.	Vanguard employee's name

☐ Check this box if:
- an employee of FINRA,
- or a person associated with a member of a stock exchange, FINRA member firm, or a municipal securities dealer ("associated person"),
- or such associated person's spouse,
- or such associated person's minor children, will have a financial interest in or trading/discretionary authority over this account.

If you check this box, you must provide a letter of account approval from a compliance officer of such member firm along with this application to open this account. Failure to include the required approval letter may delay the processing of your request to open this account. Upon written request of the member firm, we'll automatically send the firm duplicate copies of confirmations, statements, and other information.

By way of example, account approval is required whenever an employee of a broker-dealer, or such employee's spouse or minor children, seeks to open an account with us, or when such employee is a custodian or trustee or granted investment authority under a power of attorney for an account. An account approval letter isn't required for FINRA, NYSE, or Vanguard employees to open an account.

Name of FINRA member firm you are associated with	☐ My firm has provided prior blanket approval for this application

Control person *Check if applicable.*

☐ Check this box if you are, or a household member is, a control person or an affiliate of a public company, as defined in SEC Rule 144 (this would include, but is not limited to, 10% shareholders, policymaking executives, and members of the board of directors). If this box is checked, you must provide the names and trading symbols of the companies for which such person serves as a control person or an affiliate.

Name of company	Trading symbol
Name of company	Trading symbol

Industry regulations require us to collect the information below

Annual income and liquid net worth questions do not apply to trustees, executors, or organization representatives.

Required for all individual accounts*

Annual income **Required*** *Check one.* Include ALL sources of income.	☐ $0–$19,999 ☐ $20,000–$49,999 ☐ $50,000–$74,999 ☐ $75,000–$99,999	☐ $100,000–$499,999 ☐ $500,000–$999,999 ☐ $1,000,000 or above
Liquid net worth (cash, securities, etc.) **Required*** *Check one.* Don't include your home.	☐ $0–$19,999 ☐ $20,000–$49,999 ☐ $50,000–$74,999 ☐ $75,000–$99,999	☐ $100,000–$499,999 ☐ $500,000–$999,999 ☐ $1,000,000 or above

*This information isn't required if your account is a trust or an organization, such as an estate or a limited liability company (LLC).

Return ALL pages of this application, even if some sections are left blank.

B. Information for Trusts/Estates/Organizations

Important: If the owner is a trust, estate or organization, you must also

- provide all information requested below,
- include the appropriate documentation listed below with your application, and
- complete Section 4: Custodian/Guardian/Trustee/Executor/Authorized Signer information.

The mailing address listed below will be the primary address for any mailings

Type of owner *Check one of the following four options.*

☐ **Trust under will/testamentary.** Provide to Vanguard copies of the pages of the will that confirm the creation of the trust and those that contain the trustee name(s) and the testator's (decedent's) signature. The pages of the will should indicate it was filed with the probate court and include a court stamp (the stamp doesn't need to be original or include a date). If the will doesn't contain a court stamp, attach a copy of Letters Testamentary or court document appointing the trustee of the trust.

Note: For trustees that are organizations, we require our Organization Resolution form. If an organization is acting as the only trustee, complete our Organization Resolution form and our Certification of Beneficial Owners form and submit them with this application. **All currently serving trustees must sign in Section 8.**

☐ **Any other type of trust.** Provide to Vanguard copies of the pages of the trust agreement that contain the trust name and date, and a listing of the trustees' names and the signature page(s), signed by the persons who were required to sign under state law.

Note: For trustees that are organizations, we require our Organization Resolution form. If an organization is acting as the only trustee, complete our Organization Resolution form and our Certification of Beneficial Owners form and submit them with this application. **All currently serving trustees must sign in Section 8.**

☐ **Estate.** Provide to Vanguard a copy (dated within 180 days) of the court appointment of fiduciary, such as Letters Testamentary or Letters of Administration.

☐ **Charity/Organization.** Provide to Vanguard copies of the documents evidencing the existence of the organization, such as articles of incorporation or similar document. In addition, you must attach a completed Organization Resolution form and Certification of Beneficial Owners form and submit them with this application.

You must complete this entire section. Only grantor trusts can use the grantor's SSN. All other trusts must provide an EIN.

Name of trust, estate, or charity/organization
Taxpayer ID number

Type of trust *if trust is owner* *Check one.*

☐ Trust under agreement		Date of trust agreement *mm/dd/yyyy*
☐ Grantor trust	Name of grantor	Date of trust agreement *mm/dd/yyyy*
☐ Trust under will/testamentary		

You must complete this entire section. >

Country where trust, organization, or estate was established

☐ U.S. ☐ Other*

Country of establishment *if not U.S.*

Tax residency

☐ U.S. ☐ Other*

Country of tax residence *if not U.S.*

Primary residence

Street address *A P.O. box or rural route number is NOT acceptable; address can be military APO or FPO.*

City, state, zip

Country *if not U.S.*

Mailing address

This is required if the mailing address is different from the street address. >

Street address

City, state, zip

Country *if not U.S.*

4. Custodian/Guardian/Trustee/Executor/Authorized signer information

You must provide all requested information for individual fiduciaries. If the fiduciary is an organization skip to 4B.

A. Information for individuals

If you need more space to list additional fiduciaries, photocopy pages 6, 7, and 8

Provide your full legal name. >

Name of individual *first, middle initial, last*

You must complete this entire section. >

Birth date *mm/dd/yyyy*

Email address *required for e-delivery*

Social Security (SSN) number or individual taxpayer ID number

Gender *optional*
☐ Male ☐ Female

You must provide at least one phone number. >

Daytime phone *area code, number, extension*
☐ Mobile

Evening phone *area code, number, extension*
☐ Mobile

*__For trusts.__ If the country of establishment or tax residence is other than the U.S., you'll be required to complete either an official IRS Form W-9 on paper to certify the trust's tax status as U.S. or a Form W-8 electronically to certify the trust's tax status as foreign, and to claim treaty benefits if applicable. We'll mail you instructions once your request has been processed. Additional restrictions apply to accounts owned by nonresident aliens.

For estates, charities, and organizations. If the country of establishment is other than the U.S., you must complete an IRS Form W-8 electronically to certify your foreign status, and to claim treaty benefits if applicable. We'll mail you instructions for completing the electronic Form W-8 once your request has been processed. Additional restrictions apply to accounts owned by nonresident aliens.

Return ALL pages of this application, even if some sections are left blank.

Citizenship and tax residency

Check and complete only one.

☐ U.S. citizen		
☐ Resident alien	Country of citizenship	
☐ Nonresident alien*	Country of citizenship	Country of tax residence

Primary residence

Street *A P.O. box or rural route number is NOT acceptable; address can be military APO or FPO.*	
City, state, zip	Country *if not U.S.*

The following information is required by industry regulations.

Employment of fiduciary *Check only one. (not needed for Organization authorized signers)*

☐ Employed ☐ Self-employed ☐ Retired ☐ Not employed ☐ Vanguard employed

If you checked *Employed* or *Self-employed*, you must complete this entire section. If you don't, we won't be able to process your application. >

Occupation	
Name of employer or nature of your business if self-employed	
Street address of employer or your business	
City, state, zip	Country *if not U.S.*

*If you're a nonresident alien, you must complete an IRS Form W-8 electronically to certify your tax status, and to claim treaty benefits if applicable. We'll mail you instructions for completing the electronic Form W-8 once your request has been processed. Additional restrictions apply to accounts owned by nonresident aliens.

Association *Check all that apply.*

☐ Check this box if you are a Vanguard employee.

☐ Check this box if your spouse is a Vanguard employee.	Vanguard employee's name

☐ Check this box if:
- an employee of FINRA,
- or a person associated with a member of a stock exchange, FINRA member firm, or a municipal securities dealer ("associated person"),
- or such associated person's spouse,
- or such associated person's minor children, will have a financial interest in or trading/discretionary authority over this account.

If you check this box, you must provide a letter of account approval from a compliance officer of such member firm along with this application to open this account. Failure to include the required approval letter may delay the processing of your request to open this account. Upon written request of the member firm, we'll automatically send the firm duplicate copies of confirmations, statements, and other information.

By way of example, account approval is required whenever an employee of a broker-dealer, or such employee's spouse or minor children, seeks to open an account with us, or when such employee is a custodian or trustee or granted investment authority under a power of attorney for an account. An account approval letter isn't required for FINRA, NYSE, or Vanguard employees to open an account.

Name of FINRA member firm you are associated with	☐ My firm has provided prior blanket approval for this application

Control person *Check if applicable.*

☐ Check this box if you are, or a household member is, a control person or an affiliate of a public company, as defined in SEC Rule 144 (this would include, but is not limited to, 10% shareholders, policymaking executives, and members of the board of directors). If this box is checked, you must provide the names and trading symbols of the companies for which such person serves as a control person or an affiliate.

Name of company	Trading symbol
Name of company	Trading symbol

Industry regulations require us to collect the information below

Annual income and liquid net worth questions do not apply to trustees, executors, or organization representatives.

Required for all individual accounts*

Annual income **Required*** *Check one.* Include ALL sources of income.	☐ $0–$19,999 ☐ $20,000–$49,999 ☐ $50,000–$74,999 ☐ $75,000–$99,999	☐ $100,000–$499,999 ☐ $500,000–$999,999 ☐ $1,000,000 or above
Liquid net worth (cash, securities, etc.) **Required*** *Check one.* Don't include your home.	☐ $0–$19,999 ☐ $20,000–$49,999 ☐ $50,000–$74,999 ☐ $75,000–$99,999	☐ $100,000–$499,999 ☐ $500,000–$999,999 ☐ $1,000,000 or above

*This information isn't required if your account is a trust or an organization, such as an estate or a limited liability company (LLC).

Return ALL pages of this application, even if some sections are left blank.

B. Information for organizations as fiduciary

If the beneficiary is an organization (or a trust for which organizations are the sole trustees), you must also complete both our Organization Resolution form and our Certification of Beneficial Owners form and submit them with this application.

You must complete this entire section. >

Name of organization

Taxpayer ID number

You must complete this entire section. >

Country where the organization was established

☐ U.S. ☐ Other*

Country of establishment *if not U.S.*

Tax residency

☐ U.S. ☐ Other*

Country of tax residence *if not U.S.*

Primary residence

Street address *A P.O. box or rural route number is NOT acceptable; address can be military APO or FPO.*

City, state, zip

Country *if not U.S.*

Mailing address

This is required if the mailing address is different from the street address. >

Street address

City, state, zip

Country *if not U.S.*

Important: If the mailing and/or street address you provide differ from the addresses listed on your account(s), the addresses on your brokerage account(s) will be updated accordingly.

5. Your settlement fund

Your new brokerage account includes a settlement fund, a holding used to pay for investments and receive proceeds from investment sales and other transactions. We have established your settlement fund in the Vanguard Federal Money Market Fund, our standard settlement fund. If you prefer our other settlement fund option, Vanguard Cash Deposit, a bank sweep product, you may switch to it at any time by logging in to your account at **vanguard.com**.

6. Dividends and other distributions

Dividends and distributions from your securities have been set to reinvest. You may change this election at any time by logging in to your account at **vanguard.com** or contacting us..

7. Beneficiaries

The current beneficiary designations on your mutual fund IRA have been carried over to your new Vanguard Brokerage Account IRA. If you would like to update your beneficiaries, please complete the IRA Beneficiary Designation Kit or log in to **vanguard.com**. This does not apply to Inherited IRAs owned by a trust, estate, or other entity.

***For organizations.** If the country of establishment is other than the U.S., you must complete an IRS Form W-8 electronically to certify your foreign status, and to claim treaty benefits if applicable. We'll mail you instructions for completing the electronic Form W-8 once your request has been processed. Additional restrictions apply to accounts owned by nonresident aliens.

8. Authorization of account owner

Important information about opening a new account. Vanguard Marketing Corporation is required by federal law to obtain from each person who opens an account, and in some cases, regarding each owner or trustee, certain personal information—including name, primary residence, and taxpayer identification number, among other information—that will be used to verify identity. If You do not provide Us with this information, We will not be able to open the account. If We are unable to verify Your identity, Vanguard Marketing Corporation reserves the right to close Your account or take other steps We deem reasonable.

Read this section carefully and sign on page 12 exactly as indicated in Section 3 or 4.

By signing this Application to Remove Brokerage Account Restrictions (Retirement) (Application), You acknowledge that:

1. You have received and read a copy of the attached Vanguard Brokerage Account Agreement (Agreement), which contains a **predispute arbitration agreement**. You acknowledge that Your signature signifies and constitutes Your agreement that this Account and Your relationship with Vanguard Brokerage Services will be governed by and consists of the terms set forth in the Agreement and incorporates the terms set forth in the Vanguard Brokerage Services commission and fee schedules and the other disclosures, terms, and agreements relating to Your Account or to particular features or services offered in connection with Your Account, each as amended from time to time. You understand there are fees associated with establishing, maintaining, and engaging in transactions in and transferring assets out of this Account.
2. You have received and read the following documents which were delivered according to your delivery preference (email or United States Postal Service) in previous communications about the transition of your mutual fund-only account(s) to Vanguard Brokerage Account(s): VMC Form CRS, Vanguard's Privacy Policy, Vanguard Brokerage Services commission and fee schedules, and Business Contingency Planning and Disaster Recovery Programs at Vanguard.
3. **You hereby adopt the enclosed Vanguard Traditional and Roth IRA Custodial Account Agreement that is incorporated herein by reference, and which You acknowledge having received and read. You further acknowledge that You have received and read the enclosed Vanguard Traditional and Roth IRA Disclosure Statement.**
4. Your existing beneficiaries have been carried over to your new VBA Account(s). By signing in this section, you reaffirm the beneficiary information you have previously provided. If you would like to update your beneficiaries, please complete and send in the IRA Beneficiary Designation Kit or contact us.

Signing on behalf of a minor

If I am signing on behalf of a minor beneficiary, I understand that I also must confirm the following by signing in this section.

- The minor beneficiary has not yet reached the age of majority under the law of the state of the beneficiary's domicile.
- I am authorized to serve as the custodian for the minor beneficiary and, to the best of my knowledge, there are no other persons who are authorized to serve as custodian for the minor beneficiary. If requested by Vanguard, I agree to submit documentation supporting my authority to serve as custodian.

Signing on behalf of a trust

If I am/We are signing as a trustee of a trust beneficiary, I/We understand that I/We also must confirm the following by signing in this section.

- The trust is valid and in full force and effect as of the date on the next page; the trustees have full authority under the trust document and applicable law to enter into investment transactions on behalf of the trust, including the purchase, sale, exchange, transfer, and redemption of mutual funds; and the trustees may issue general instructions as well as execute and deliver documents on behalf of the trust.
- The trustees listed and signing this form are all of the currently serving trustees.

Return ALL pages of this application, even if some sections are left blank.

☐ **The number of trustee signatures required to take written action on behalf of the trust.** You understand that Vanguard will accept instructions online or by phone from any one of the trustees. If a specific number isn't provided, the signature of any one trustee will be accepted for written transactions.

Authorization for e-delivery *optional*

Complete this section if the trust indicated in Section 3 is established under an EIN with more than one trustee and You want to designate one of the trustees as authorized to elect e-delivery of the trust's Vanguard account documents (statements, confirmations, etc.) with email notification of document availability to that trustee's email address.

Important:

- If any of the trustees is an organization, e-delivery is NOT available for the account. Skip this section.
- If you're a trustee on more than one EIN trust with multiple trustees that has an account at Vanguard, you must submit a separate Trustee Certification form for each trust to elect e-delivery for that trust account.
- This service is available **only** for trust accounts registered under an EIN with more than one trustee.

Name of designated trustee *first, middle initial, last*

Email address *required for e-delivery*

Signing on behalf of an organization

If I am/We are signing as a authorized signer of an organization beneficiary, I/We understand that I/We also must confirm the following by signing in this section.

- You confirm that the organization is in existence and that You have full authority to enter into investment transactions on behalf of the organization and to execute and deliver documents on its behalf.
- The following statement applies to organizations that may engage in option trading: I/We understand the investment objectives and trading plan of the registered account owner(s) and will only use trading strategies that are consistent with these objectives and plans. I/We acknowledge that I/we have received from the registered account owner(s) (or I/we already possess) copies of the Vanguard Brokerage Option Application and Vanguard Brokerage Option Account Agreement (Option Agreement), and I/we have read and understand them. I/We also acknowledge that if the registered account owner(s) is/are approved for option trading, the owner(s) will be sent the booklet Characteristics and Risks of Standardized Options. I/We agree to read the booklet prior to conducting any trades in the registered account owner's(s') account. I/We also accept all the terms and conditions of the Option Agreement. I/We acknowledge receiving a copy of the Option Agreement, which contains a **predispute arbitration clause** highlighted in paragraphs 14 and 15 on page 3. By signing this application, I/we agree to be bound by the terms of the Option Agreement.

Important:

- If the new owner is a minor or incapacitated, the custodian/guardian listed in Section 3 must sign.
- If the new owner is a trust or estate, the trustee(s) or executor listed in Section 4 must sign.

For purposes of this Application and the attached Vanguard Brokerage Account Agreement, the terms "You," "Your," and "Account Owner" refer to the person who signs this Application. The terms "We," "Us," and "Our" refer to Vanguard Brokerage Services, a division of Vanguard Marketing Corporation, and its affiliates.

You must sign on page 12.

If I am a U.S. citizen, a U.S. resident alien, or a representative of a U.S. entity, I certify under penalties of perjury that:

1. **The taxpayer ID number I have given on this application is correct (or I am waiting for a number to be issued to me).**
2. **I am not subject to backup withholding because (a) I am exempt from backup withholding, or (b) I have not been notified by the Internal Revenue Service (IRS) that I am subject to backup withholding as a result of a failure to report all interest or dividends, or (c) the IRS has notified me that I am no longer subject to backup withholding.**
   Important: **Cross out item 2 if You have been notified by the IRS that You are currently subject to backup withholding because You have failed to report all interest or dividends on Your tax return.**
3. **I am a U.S. citizen or other U.S. person (as defined by the IRS in its Form W-9 instructions).**
4. **The Foreign Account Tax Compliance Act (FATCA) code(s) entered on this application (if any) indicating that I am exempt from FATCA reporting is correct.**

If I am not a U.S. person, I will complete the appropriate Form W-8BEN electronically to certify my foreign status, including my FATCA status, and to claim treaty benefits if applicable.

The IRS does not require Your consent to any provision of this document other than the certification required to avoid backup withholding.

NOTE THAT SECTION f ON PAGE 18 OF THE ATTACHED AGREEMENT CONTAINS A PREDISPUTE ARBITRATION CLAUSE.

All those listed in Section 3 or 4 (except minors or incapacitated persons) must sign with *today's date* here. >

If additional signatures are required, copy this page.

Signature of owner or fiduciary	Today's date *mm dd yyyy*
X	M M – D D – Y Y Y Y
Print name.	Title *if applicable*

Signature of owner or fiduciary	Today's date *mm dd yyyy*
X	M M – D D – Y Y Y Y
Print name.	Title *if applicable*

Vanguard Fiduciary Trust Company	Title
Armond E. Mosley	Officer

Return ALL pages of this application, even if some sections are left blank.

Mailing information

Mail ALL pages of this application—even if some sections are left blank—and any other required forms in the enclosed postage-paid envelope. The included Vanguard Brokerage Account Agreement and the Vanguard Traditional and Roth IRA Custodial Account Agreement are for your records.

Mail to: >
Vanguard
P.O. Box 982901
El Paso, TX 79998-2901

For registered or certified mail, or overnight delivery, mail to: >
Vanguard
5951 Luckett Court, Suite A1
El Paso, TX 79932-1882

Reminders

If you previously had a full agent on your Vanguard mutual fund account and wish to keep that agent on your account:

Complete our Agent Authorization form and submit it with this application.

If your account is owned by an organization or has an organization as the sole trustee:

Complete both our Organization Resolution form and our Certification of Beneficial Owners form and submit them with this application.

If you need to add or update banking information

Please submit our Electronic Bank Transfer Form or log in to vanguard.com

Return ALL pages of this application, even if some sections are left blank.

Vanguard Brokerage Services is a division of Vanguard Marketing Corporation, member FINRA and SIPC.

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Vanguard

Important update for retirement account owners

The Setting Every Community Up for Retirement Enhancement (SECURE) Act signed into law on December 20, 2019, and SECURE 2.0, signed into law on December 29, 2022, include several provisions that affect retirement accounts.

Highlights

IRA owners over age 70½ can contribute to traditional IRAs. The age limit for contributing to a traditional IRA (including employee contributions to a SEP-IRA) has been eliminated. If you're still working, you may be able to contribute to your traditional or Roth IRA (provided you're otherwise eligible to make IRA contributions).

IRA owners may be able to delay taking distributions until they reach age 73. If you turn 73 after December 31, 2022, the deadline for taking your first required minimum distribution (RMD) from your traditional IRA (including SEP IRAs) is now April 1 of the year after the year you turn 73. For each subsequent year, you must take your RMD by December 31. Keep in mind, if you delay your initial RMD until April 1, you'll be responsible for two withdrawals that year (one by April 1 and one by December 31), which could result in a larger tax liability.

The excise tax on the failure to take an RMD is reduced from 50% to 25%. This excise tax is further reduced to 10% if a taxpayer receives a distribution in the amount of the missed RMD in a timely manner. This reduced excise tax rate applies to missed RMDs after January 1, 2023. Special IRS reporting requirements apply.

Non-spouse beneficiaries (with certain exceptions) must take RMDs within 10 years. In general, non-spouse beneficiaries (with certain exceptions) who inherit IRAs (including traditional, Roth, and SEP IRAs) from owners who die after December 31, 2019 must distribute the entire account balance by the end of the 10th year after the IRA owner's death. A spouse beneficiary and certain non-spouse beneficiaries (including beneficiaries who are chronically ill or disabled, minor children of the IRA owner, beneficiaries not more than 10 years younger than the IRA owner, and certain trusts) must generally take RMDs annually over their own life expectancy. For more information about the RMD rules, refer to IRS Publication 590-B, Distributions from Individual Retirement Arrangements (available at irs.gov).

IRA owners are now allowed a one-time, $54,000 qualified charitable distribution to a charitable gift annuity, charitable remainder unitrust, or a charitable remainder annuity trust (provided you're otherwise eligible to make a qualified charitable distribution). This provision also indexes the current $108,000 qualified charitable distribution limit for inflation.

IRA owners younger than age 59½ can take penalty-free distributions for the birth or adoption of a child. You can take a penalty-free distribution from your IRA (including traditional, Roth, and SEP IRAs) up to $5,000 per parent for the birth or adoption of a child. Such distributions must be made within one year from the date of the birth or adoption and may later be repaid back into an IRA within three years of the date of distribution.

IRA owners younger than age 59½ can take penalty-free distributions up to $22,000 if impacted by a qualified federally declared disaster. Additionally, the gross income attached to qualified disaster distributions can be spread over three years and can be rolled back into an IRA within three years of the date of distribution. This provision applies to distributions and repayment contributions related to disasters whose incident period began on or after January 26, 2021.

IRA owners younger than age 59½ can take penalty-free distributions if they are victims of domestic abuse. After December 31, 2023, eligible victims may take penalty-free distributions to the lesser of $10,000 or 50% of their account. Such distributions must be made within one year from the date on which the individual is a victim of abuse and may later be repaid back into an IRA within three years of the date of distribution.

IRA owners younger than age 59½ can take penalty-free distributions if they are terminally ill. To qualify, documentation from a physician certifying the IRA owner's terminal illness may be required. Such distributions may later be repaid back into an IRA within three years of the date of distribution.

The IRA catch-up contribution limit for individuals aged 50 and over is indexed for inflation. Similar to the IRA contribution limit, beginning on taxable years after December 31, 2023, the current $1,000 additional catch-up contribution limit will be indexed for inflation.

Beneficiaries of 529 college savings accounts are permitted to make direct rollovers into their own Roth IRA without tax or penalty. Effective for distributions after December 31, 2023, beneficiaries are eligible to roll over up to a lifetime limit of $35,000 into a Roth IRA. Rollovers are subject to IRA annual contribution limits, but limitations based on the taxpayer's adjusted gross income do not apply. To qualify, the 529 account must have been open for more than 15 years and the rollover cannot exceed the aggregate amount contributed to the 529 plan (and any earnings on those funds) before the 5-year period ending on the date of the rollover.

IRA owners under 59½ who contribute too much to an IRA will no longer be assessed a 10% early distribution penalty when timely correcting their excess contribution. However, if you remove the excess contribution after the applicable IRS correction deadline, the principal amount of the excess contribution is subject to a 6% excise tax each year it remains in your account. New provisions also limit the number of years this 6% penalty can apply to a maximum of six years.

Please note that the Vanguard Traditional and Roth IRA Custodial Account Agreement and Disclosure Statement does not yet reflect the changes in the law. We'll notify you of the changes and how to access the updated documents once amended to reflect the new rules.

Learn more

For more information about the SECURE Act and its effect on IRA owners and their beneficiaries, visit **investor.vanguard.com/inheriting-accounts/rmd-rules-for-inherited-iras**.

Thank you for belonging to the Vanguard community of investors.

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Vanguard

Vanguard Brokerage Services® commission and fee schedules

Effective July 10, 2026

Stay focused on your financial goals with confidence that you're not paying too much.

About our commissions and fees

Vanguard Brokerage Services ("VBS")[1] aligns the commissions and fees associated with your account to your Qualified Assets ("QA")[2].

Qualifying Assets ("QA")

A client's fees and commissions are primarily determined by their total Qualifying Assets ("QA")[2]. In general, as a client's QA increases, the fees and commissions applicable to that client decrease. Clients with QA exceeding $1 million, as well as Wealth Management clients[3], may be eligible for certain fee or commission waivers that are not available to all clients. VBS periodically reviews QA calculations to confirm clients receive eligible fee and commission waivers, and may update their QA or waiver eligibility at any time based on these reviews. Changes resulting from these reviews may not be reflected immediately and may be applied after a period of continued eligibility or ineligibility, as determined by VBS.

Note: The costs applicable to a specific account will be determined by the QA of the owner of the account. For joint accounts, only the QA of the primary owner applies for purposes of all fees and commissions applicable to those accounts.

Broker-Assisted Commissions ("BAC")

Completing transactions online, when available, is generally the most cost-effective option.[4] If a client chooses to complete a transaction over the phone with the assistance of a VBS representative (i.e., "Broker-assisted"), a $25 broker-assisted commission ("BAC") may apply, in addition to other applicable fees or commissions. Generally, the BAC will not apply to transactions performed by clients with $1 million or more in QA or to accounts enrolled in a Vanguard-affiliated advisory service.

Orders subject to a BAC that execute over multiple days are charged separate commissions. Orders that are changed by the client and executed in multiple trades on the same day are charged separate commissions.

Review specific fee and commission disclosures below for more details.

[1]Brokerage assets are held through Vanguard Brokerage Services, a division of Vanguard Marketing Corporation, member FINRA and SIPC.

[2]Generally, each client's QA is calculated based on the value of assets they own at Vanguard as well as any assets associated with the client's residential address(es), if applicable. When calculating a client's QA, the following will be included: retirement and non-retirement accounts custodied through VBS, accounts enrolled in Vanguard-affiliated advisory services, retirement accounts provided by the Vanguard Group, Inc. ("VGI"), accounts held by VGI, Vanguard 529 accounts. QA does not include assets invested in HarbourVest Private Equity, Vanguard Retirement Investment Plan accounts ("VRIPs"), or brokerage assets held outside of VBS in an employer sponsored retirement account.

[3]Eligibility for Wealth Management is determined based on a variety of factors including, but not limited to, asset levels and the scope of a client's relationship with VBS. These factors are determined and applied by VBS at its sole discretion. Visit **www.vanguard.com/wealth** for more information.

[4]If online trading is unavailable and you must complete a transaction by phone, the BAC will not apply.

Fee by transaction or security type

The following sets forth the fees and/or commissions applicable to trading specific types of securities and to complete certain transactions:

ETF, equity & options transactions

Qualifying Assets held at Vanguard	Less than $1 million	$1 million – $4,999,999	$5 million or more
ETFs & stocks	**Online:** $0 **Broker-assisted:** $25 BAC per trade	$0	$0
Options (online or broker-assisted)	$0 + $1-per-contract fee	$0 for first 25 trades* $0 + $1-per-contract fee thereafter	$0 for first 100 trades* $0 + $1-per-contract fee thereafter

Other fees or commissions may apply; refer to **Fees for other services** on page 4 for details.

Investing in ETFs: All ETFs are subject to management fees and expenses; refer to each ETF's prospectus for more information. All stock and ETF sales are subject to a securities transaction fee. These also apply to sales of leveraged and inverse ETFs and ETNs (exchange-traded notes).

Investing in options: All options exercises and assignments are commission-free.

*VBS offers a limited number of commission-free options and transaction-fee ("TF") mutual fund trades for eligible clients as set forth here. This benefit resets at the end of each calendar year and applies to any combination of options and TF mutual fund transactions performed across all of the eligible client's VBS accounts.

Vanguard mutual fund transactions

Qualifying Assets held at Vanguard	Less than $1 million	$1 million or more
Vanguard Mutual Funds	**Online:** $0 **Broker-assisted:** $25 BAC per trade	$0

Other fees or commissions may apply; refer to **Fees for other services** on page 4 for details.

Investing in Vanguard mutual funds: A few Vanguard mutual funds charge special purchase and/or redemption fees that are paid directly to the funds to help cover higher transaction costs and protect long-term investors by discouraging short-term, speculative trading. Those fees vary from 0.25% to 1.00% of the amount of the transaction, depending on the fund. Refer to the fund prospectus for details.

Non-Vanguard mutual fund transactions

Qualifying Assets held at Vanguard	Less than $1 million	$1 million – $4,999,999	$5 million or more
No-transaction fee (NTF) mutual funds	**Online:** $0 **Broker-assisted:** $25 BAC per trade	$0	$0
Transaction-fee (TF) mutual funds	**Online:** $20 per trade **Broker-assisted:** $20 + $25 BAC per trade	$0 for first 25 trades* $8 per trade thereafter	$0 for first 100 trades* $8 per trade thereafter

Note: There is a fee of $3 per transaction for clients enrolled in a TF mutual fund dollar-cost-averaging purchase program.

Other fees or commissions may apply; refer to **Fees for other services** on page 4 for details.

Investing in non-Vanguard mutual funds: Regardless of whether it's placed online or broker-assisted, all sales executed within 60 calendar days of the trade date of a client's most recent purchase of the same fund will be subject to a $50 early redemption fee. This schedule also applies to sales of leveraged and inverse mutual funds. A purchase fee is added to an order's cost. A sales fee is subtracted from an order's proceeds. If you exchange shares of a fund for another fund in the same fund family and share class, the transaction fee will be paid from your settlement fund.

*VBS offers a limited number of commission-free options and transaction-fee ("TF") mutual fund trades for eligible clients as set forth here. This benefit resets at the end of each calendar year and applies to any combination of options and TF mutual fund transactions performed across all of the eligible client's VBS accounts.

CDs (certificates of deposit) & bonds

	New issues	Secondary market *Each broker-assisted trade is subject to a $25 BAC for clients with QA less than $1 million.*
CDs[1], U.S. Government agency securities and corporate bonds Minimum investment: $1,000	$0	$1 per $1,000 face amount ($250 maximum)
U.S. Treasury securities Minimum investment: $1,000	$0	$0
Municipal bonds Minimum investment: $5,000; also subject to dealer minimums	Not available	$1 per $1,000 face amount ($250 maximum)
Mortgage-backed securities	Not available	$35 per trade
Unit investment trusts (UITs)	Not available	$35 per trade

[1]All online sales of CDs before they mature are commission-free; however, if the sale is performed by phone with the assistance of a VBS representative, the $25 BAC will be applied for clients with QA of less than $1 million.

VBS may act as either an agent (executing your order at cost plus a commission) or a principal (adding markups to purchase prices or subtracting markdowns from sale prices). When acting as a principal for a primary market issue, VBS generally receives a fee concession from the issuer.

CD and bond prices are available at **www.vanguard.com/buycdsbonds** and are price indications only. Online prices for all secondary-market CDs and bonds are before commissions.

Fees for other services

	Description and amount
Account closure and full transfer out fee	VBS may charge a $100 processing fee for each account closure and full transfer of account assets to another firm. Electronic Bank Transfers and ACH are not subject to this fee. The fee will not apply to any accounts of clients with a QA of at least $5 million or Wealth Management clients. It also will not be charged to any brokerage accounts enrolled in a Vanguard-affiliated advisory service.
Account service fee	VBS may charge a $25 annual account service fee. The fee is waived for all Cash Plus Accounts, as well as accounts held by clients who meet one or more of the following conditions: • The client is an organization or a trust registered under an employee identification number (EIN). • The client has elected e-delivery of statements and annual privacy policy notice; confirmations; reports, prospectuses and proxy materials; and notices, amendments and other important account updates. • The client has a brokerage account enrolled in a Vanguard-affiliated advisory service. • The client has QA of at least $1 million. • The client is in Wealth Management.
American Depository Receipts (ADRs)	Banks that custody ADRs are permitted to charge ADR holders certain fees, as detailed in the ADR prospectuses. "Pass through" ADR fees are collected from VBS by the Depository Trust Company (DTC) and will be automatically deducted from your brokerage account and shown on your account statement. Other fees—including dividend processing fees—may be withheld by the DTC from the amount paid by the issuer.
Broker-assisted (closing transactions)	A $25 BAC may be assessed on closing transactions placed by VBS on behalf of clients to cover a margin call or debit obligation. The commission will not be assessed for clients with QA of at least $1 million or in brokerage accounts enrolled in a Vanguard-affiliated advisory service.
Class Action Service	A fee of 20% may be deducted from any class action settlement funds recovered on your behalf subsequent to the deposit of proceeds into your brokerage account. See the Vanguard Brokerage Account Agreement for more information about the service.
Deposit of physical certificates	VBS charges a $100 processing fee (per CUSIP) for the deposit of physical certificates. The fee does not apply to Certificates of Deposit (CDs).
Foreign securities and ADR dividends	VBS charges a fee of 1% on the gross dividend amount, up to $100 per dividend payment, when foreign tax withholding relief is received on a dividend of a foreign or ADR asset held in US dollars (USD).
Foreign securities transactions	$50 processing fee for non-DTC-eligible securities (not applicable to ADRs) plus commission, if applicable. If a trade executes over multiple days, the commission will be charged for each day on which an execution occurs. Additional fees may apply for trades executed directly on local markets.
Restricted security legend removal	VBS may charge a $250 processing fee for research and removal of a restriction on a security held in your brokerage account.
Wire transfers	VBS charges a $10 fee per wire transfer. However, this fee is waived for any wire transfers from retirement accounts and for clients with QA of at least $1 million. While VBS doesn't charge a fee for Electronic Bank Transfers, your bank may charge you a fee.
Tax filing for Master Limited Partnerships (MLPs) in an IRA	Vanguard Fiduciary Trust Company (VFTC), the custodian for IRAs held at VBS, is responsible for IRS 990-T tax filings for MLPs for the tax year 2020 and after. VFTC charges a fee of $500 per account for these filings. VBS will facilitate collection of the fee by deducting the fee amount from your brokerage account when a filing is required. This fee will not be assessed for clients with QA of at least $1 million or in brokerage accounts enrolled in a Vanguard-affiliated advisory service.
Miscellaneous	Other securities transaction or maintenance fees may apply. Call us for additional information.

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Vanguard®

For more information about Vanguard ETFs, Vanguard mutual funds, or non-Vanguard ETFs or mutual funds offered through Vanguard Brokerage Services, visit investor.vanguard.com to obtain a prospectus or, if available, a summary prospectus. Investment objectives, risks, charges, expenses, and other important information are contained in the prospectus; read and consider it carefully before investing.

You must buy and sell Vanguard ETF Shares through Vanguard Brokerage Services (we offer them commission-free online) or through another broker (which may charge commissions). Vanguard ETF Shares are not redeemable directly with the issuing fund other than in very large aggregations worth millions of dollars. ETFs are subject to market volatility. When buying or selling an ETF, you will pay or receive the current market price, which may be more or less than net asset value.

All investing is subject to risk, including the possible loss of the money you invest. Investments in bonds are subject to interest rate, credit, and inflation risk.

Advice services are provided by Vanguard Advisers, Inc., a registered investment advisor, or by Vanguard National Trust Company, a federally chartered, limited purpose trust company.

Options are a leveraged investment and are not suitable for every investor. Options involve risk, including the possibility that you could lose more money than you invest. Prior to buying or selling options, you must receive a copy of Characteristics and Risks of Standardized Options issued by OCC. A copy of this booklet is available at theocc.com. It may also be obtained from your broker, from any exchange on which options are traded, or by contacting OCC at 125 S. Franklin Street, Suite 1200, Chicago, IL 60606 (888-678-4667 or 888-OPTIONS). The booklet contains information on options issued by OCC. It is intended for educational purposes. No statement in the booklet should be construed as a recommendation to buy or sell a security or to provide investment advice. Call The Options Industry Council (OIC) Helpline at 888-OPTIONS or visit optionseducation.org for more information. The OIC can provide you with balanced options education and tools to assist you with your options questions and trading.

Brokerage assets are held by Vanguard Brokerage Services, a division of Vanguard Marketing Corporation, member FINRA and SIPC.

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Distributor of the Vanguard Funds.

CMSCHP 092026

Vanguard

Vanguard Brokerage Account Agreement

Effective October 1, 2026

Vanguard Brokerage Services

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Vanguard Brokerage Account Agreement

1. About the Vanguard Brokerage Account

a. Introduction

Vanguard Brokerage Services® ("VBS") is a division of Vanguard Marketing Corporation ("VMC") and was created to offer retail brokerage services to clients. VMC, a broker-dealer, is a member firm of the Financial Industry Regulatory Authority ("FINRA"). VBS offers its clients brokerage Accounts, including the Vanguard Brokerage Account ("VBA"), in which clients may hold, as well as buy and sell, mutual funds, ETFs, stocks, bonds, options (collectively, "Securities"), and Other Property, and the Vanguard Cash Plus Account, designed to operate as a cash management account in which clients may hold, buy, and sell select Vanguard money market funds. This document, the Vanguard Brokerage Account Agreement ("the Agreement" or "this Agreement"), contains terms and conditions, disclosures, and other important information about the Vanguard Brokerage Account, the Vanguard Cash Plus Account, and a client's relationship with VBS. This Agreement also explains some of the features and services available for these Accounts.

When You submit an Application to open a Vanguard Brokerage Account or Vanguard Cash Plus Account (hereafter each referred to as "an Account," "the Account," "Your brokerage Account," or "Your Account"), You agree that Your Account, if opened, and any product or service relating to it, and Your relationship with VBS, will be governed by the terms, conditions, disclosures, and other information in this Agreement and in any service- or feature-specific form or brochure, including but not limited to, The Vanguard Traditional IRA, SEP-IRA, and Roth IRA Disclosure Statement and Custodial Account Agreement applicable to IRAs. In addition, You agree to be bound by policies that can be found on **vanguard.com** or in other documents We provide to You, such as the Vanguard Brokerage Services commission and fee schedules, VBS' Privacy Policy, Frequent-Trading Policy, Market- Timing Policy, After-Hours Trading Policy, etc. VBS may amend this Agreement, change or alter its policies, or separately add, remove, or change any of the features and services offered on an Account. This Agreement supersedes any previous Agreement You, any of Your joint Account Owners, or any of Your Representatives (including fiduciaries appointed for You or Your estate) may have entered into with VMC or VBS relating to Your Account or Your relationship with VBS. Therefore, You should read this and other documents We provide to You carefully and retain them for future reference. If You do not understand any of the terms, conditions, or disclosures in this Agreement, please contact Us or seek the advice of Your attorney.

Client Benefits and Services

VBS may offer certain clients enhanced services, benefits, and support experiences. Eligibility is determined by VBS, in its sole discretion. VBS may modify, suspend, or discontinue any offering, benefit, eligibility criterion, or service level at any time. Information regarding service offerings and eligibility criteria is available on **vanguard.com**. For information about applicable fees and commissions, see the Vanguard Brokerage Services commission and fee schedules on **vanguard.com**.

b. Digital Interaction Expectations

VBS proactively evolves Our offerings and technologies to enable Us to better deliver on Our mission to help clients achieve investment success; core to achieving that is using more modern digital platforms. VBS is committed to helping You navigate toward simpler, more effective digital channels and practices that have proven to be quicker, safer, and more cost-effective.

We strongly urge all VBS Account holders to web register and use digital channels like **vanguard.com** and Our mobile application to manage Your Account.

To assist with Your understanding of VBS' digital-first service model, VBS has put together the following list of digital interaction expectations of which, by applying for an Account, You acknowledge Your understanding:

1. VBS expects You primarily use digital channels for interactions and communication regarding Your Account, inclusive of e-delivery. VBS provides You with various options to do so, including but not limited to **vanguard.com**, Our mobile app, instant chat, secure message center, and other electronic means to facilitate efficient and convenient interactions.
2. VBS expects You to provide Your mobile phone number and email address, so that We can contact You via text messaging and other digital means (e.g., secure messages, e-delivery) to service Your Account.
3. VBS expects that You access and use **vanguard.com**, Our mobile app, or Our automated answering system to perform certain tasks, including but not limited to Account management (e.g., checking Account balances), transaction execution (e.g., placing trades), and research (e.g., requesting quotes). On **vanguard.com** VBS provides FAQs, tutorials, and more to assist You in this process.
4. You understand that excessive reliance on Our phone associates for tasks that can be accomplished online may negatively impact Your customer service experience including but not limited to delayed response times, additional fees, and possible Account termination.
5. As set forth in Section 8(f) of this Agreement, VBS reserves the right to resolve Non-Trade Inquiries exclusively on its website or other available channels as opposed to providing such information by phone. Furthermore, at its discretion, consistent with Section 8(c) of this Agreement, VBS reserves the right to close Your Account, or terminate any feature or service at any time, for any reason, and without prior notice, inclusive of not meeting Our Digital Interaction Expectations.

2. How to Contact VBS

Online: vanguard.com

Many account interactions and communications can be accomplished efficiently using digital channels.

By mail: P.O. Box 982901
El Paso, TX 79998-2901

By phone: 800-992-8327

If You have a complaint about the services VBS provides You, You should direct it to:

Complaints
Vanguard Brokerage Services
P.O. Box 982901
El Paso, TX 79998-2901

3. FINRA's BrokerCheck

FINRA sponsors a program, called BrokerCheck, designed to give investors convenient access to information about the professional background, business practices, and conduct of FINRA member firms and brokers formerly or currently registered with FINRA or a national securities exchange. The BrokerCheck hotline number is 800-289-9999. You may obtain a brochure that describes BrokerCheck by calling the hotline number or at finra.org.

4. Account Protection

a. SIPC Coverage

Securities held in Your Account are held in custody by VBS, a division of VMC. VMC is a member of the Securities Investor Protection Corporation (SIPC), which protects securities customers of its members up to $500,000 (including $250,000 for claims for cash). Securities in Your brokerage account are eligible for SIPC coverage. To obtain information about SIPC, including an explanatory SIPC brochure, please contact SIPC at sipc.org or 202-371-8300.

VBS maintains additional coverage through an insurer that supplements the SIPC coverage available to Securities. Account protection, either under SIPC or the additional insurance maintained by VBS, does not cover fluctuations in the market value of the investments in Your Account. Any positions You may have in Securities held away from VBS are not in VBS' custody or control and are not covered by SIPC or VBS' additional insurance.

b. Eligibility for FDIC Insurance

Certain types of products offer insurance coverage provided by the Federal Deposit Insurance Corporation ("FDIC"). FDIC insurance is backed by the full faith and credit of the United States and protects eligible deposits held at FDIC-insured banks in the event that the bank fails. Coverage is generally available up to $250,000 per depositor, per FDIC-insured bank, per ownership category, subject to applicable FDIC rules, limits and conditions. Subject to applicable limits, FDIC insurance generally insures money in a deposit account at an FDIC-insured bank, including deposit products such as certificates of deposit ("CDs") and money placed through VBS' Bank Sweep Program, but FDIC insurance does not cover stocks, bonds, mutual funds, or other investment Securities. FDIC insurance is offered only by FDIC-insured banks and not broker-dealers. VMC, a registered broker-dealer, is not a bank.

If You have questions about FDIC insurance coverage, You may wish to seek advice from Your own attorney, including concerning FDIC insurance coverage of deposits held in more than one insurable capacity. You may also obtain information by visiting fdic.gov. The disclosures in this Agreement regarding FDIC insurance coverage are a summary only, do not state all of the requirements and conditions of FDIC insurance, and are subject to, and qualified in their entirety by, regulations and guidance of the FDIC.

You are solely responsible for monitoring the aggregate amount that You have on deposit with each bank to determine Your FDIC insurance coverage, including through other accounts with VBS and other financial institutions. For more information about the Bank Sweep Program, see the Vanguard Bank Sweep Products Terms of Use on vanguard.com. See also **Brokered Certificates of Deposit (Brokered CDs)** section in this Agreement.

5. Definitions

We define certain terms throughout this Agreement. Others are defined below:

- **Account:** The Vanguard Brokerage Account, including a Retirement Account, and/or Vanguard Cash Plus Account governed by this Agreement. The Account is opened on Your completion, and Our approval, of the appropriate Application.
- **Account Owner:** Each person who has an ownership interest in the Account. If the Account holds community property, an Owner's spouse, whether listed in the registration or not, may be considered an Account Owner.
- **Application or Brokerage Account Application:** The form (paper or electronic) used to open an Account.
- **Bank Sweep Program:** One of the options of VBS' Sweep Program. An FDIC-insured product that may be made available to invest Free Credit Balances and to pay for transactions and receive the proceeds from trades and other transactions in Your Account.
- **Brokerage Activity:** Trades, dividends, interest, and corporate actions activity that occurs in Your Account. Dividend reinvestment in a money market fund may not be considered Brokerage Activity.
- **Control or Restricted Security:** Securities defined in Rule 144 under the Securities Act of 1933 ("the '33 Act") as "restricted securities" or Securities issued by an issuer of which You are an "affiliate" (as that term is defined by Rule 144 of the '33 Act) or Securities that are being sold in reliance on Rule 701 or 145(d) under the '33 Act or Securities of which You and the issuer, or its underwriter, have entered into an agreement restricting the transferability of such Securities.
- **Debit Balance:** The amount of money You owe to VBS.
- **Discretionary Trading Authority:** Authority given by the Owner(s) of an Account to a third party (for example, a registered representative or an investment advisor), which permits the third party to decide which Securities to buy or sell in an Account and to make other transactions.

- **EBT:** Electronic Bank Transfer is an electronic movement of money or information using the Automated Clearing House (ACH) Network.
- **FDIC:** The Federal Deposit Insurance Corporation.
- **FINRA:** The Financial Industry Regulatory Authority.
- **Free Credit Balance:** The uninvested cash in Your Account, less what is needed to pay for unsettled purchase transactions, charges to Your Account, or credit balances designated as collateral. Free Credit Balances are payable on demand.
- **Fund Your Account:** Incoming money intended to be credited to an Account. If there is no Debit Balance, or if the outstanding Debit Balance is less than the incoming money, the funding (or the funding less the debit amount) will Sweep under the Sweep Program.
- **Good Delivery:** The delivery to VBS of freely transferable Securities (that is, properly registered, endorsed, and fully negotiable).
- **Marginable Securities:** Any registered Security, Over-the-Counter (OTC) margin stock, OTC margin bond, or National Market System (NMS) Security.
- **Market Data:** The last sale transaction data, bid and ask quotations, news reports, analysts' reports or research, or other information relating to Securities and the Securities markets.
- **Market Volatility:** Trading conditions characterized by heavy trading, illiquidity, fast markets, unpredictable or steep price fluctuations, or other volatile market conditions that may result in trading disruptions, suspensions, halts or other restrictions impacting either individual Securities or the market as a whole.
- **Other Property:** An investment in Your Account that is not commonly known as a Security.
- **Representative(s):** A person who has been properly appointed as a full or limited agent for a principal, or a fiduciary of, for example, an incompetent person, a trust, a minor, or an estate, or an officer or authorized signatory for an organization.
- **Retirement Account:** A tax-advantaged Account, either an individually owned Account, such as an Individual Retirement Account, or IRA, or one held as part of an employer-sponsored plan, intended to be used primarily as a savings vehicle to provide income in retirement. A Retirement Account can be opened using the appropriate retirement Application and agreeing to the terms in this Agreement and any other agreement that is specific to the type of Account opened, such as the Vanguard Custodial Account Agreement and the Disclosure Statement applicable to an IRA. A Retirement Account may be part of a retirement plan.
- **Security or Securities:** Any note, stock, bond, option, mutual fund, exchange-traded fund (ETF), unit investment trust (UIT), or other instrument of investment commonly known as a security.
- **Settlement Fund:** Together, the money market fund and/or Bank Sweep Program options under VBS' Sweep Program to or from which Free Credits are swept.
- **Sweep:** A transaction that moves cash under the Sweep Program.
- **Sweep Program:** The service VBS may provide to its clients to move Free Credit Balances into or out of money market funds and/or Bank Sweep Program options.
- **USPS:** The United States Postal Service.
- **Vanguard Fund:** Any mutual fund in the Vanguard family of mutual funds.
- **We/Us/Our/VBS:** Vanguard Brokerage Services, a division of Vanguard Marketing Corporation, and its affiliates.
- **You/Your/Yourself:** An Account Owner and his, her, or its properly authorized Representative.

6. General Terms of the Vanguard Brokerage Account

To open an Account, You must complete and submit the appropriate Application, and VBS must approve Your Application. You do not have an Account until VBS processes and approves Your Application. Once the Account is opened, You are responsible for complying with the terms of this Agreement.

This section of the Agreement explains VBS' Sweep Program, how to enter orders, how to pay for orders, and other information about Your Account and the Securities and Other Property held in it.

a. Sweep Program

VBS has a Sweep Program and may sweep Free Credit Balances into or out of money market funds and/or Bank Sweep Program options.

For the VBA, unless stated otherwise, VBS' default Sweep Program moves free credits in Your Settlement Fund into a money market fund. Clients with eligible Accounts may also choose to enroll in Vanguard Cash Deposit, which provides access to Vanguard's Bank Sweep Program that automatically sweeps Free Credit Balances in Your Settlement Fund into one or more Program Banks. For more information about the Bank Sweep Product, see the Vanguard Bank Sweep Products Terms of Use on **vanguard.com**.

For the Vanguard Cash Plus Account, the only Sweep Program option available is Vanguard's Bank Sweep Program that automatically sweeps Free Credit Balances into one or more Program Banks. If You opt out of all Program Banks, cash will not sweep and will be held as a Free Credit Balance. The Vanguard Cash Plus Account does not offer sweep of Free Credit Balances to a money market fund, but You may purchase and hold select Vanguard money market funds in the Cash Plus Account. For more information about the Bank Sweep Program and Vanguard Cash Plus Account, see the Vanguard Bank Sweep Products Terms of Use on **vanguard.com**.

Balances swept to a money market fund may be covered by SIPC, whereas balances under the Bank Sweep are eligible for FDIC insurance. See **Account Protection** for more information. Only cash can move into and out of the Sweep Program; if VBS receives directions to move something other than cash into the Sweep Program, including a direction to transfer shares in kind to or from Your Settlement Fund, VBS will treat the directions as an instruction to transfer cash.

There is no guarantee that cash will Sweep into or out of the Sweep Program. VBS's Sweep Program may decline to accept a purchase for any reason, including, for example, if the size of the purchase may negatively impact a Sweep Program option's operation or performance. If the Sweep Program declines Your purchase, the Sweep will not be processed and no Sweep Program option will be credited. VBS will attempt to notify You if the Sweep Program declines a purchase and to obtain alternative investment instructions. If VBS does not receive alternative investment instructions, or if it cannot process Your alternative investment instructions, the amount of Your attempted purchase will result in a Free Credit Balance in Your Account (which is another sweep option). VBS will not be liable for lost profits, or other losses and damages that allegedly result from a failure or delay to Sweep.

You agree that VBS may, on prior written notice to You, limit, change, or add options to its Sweep Program or end a Sweep Program. If a sweep option under the Sweep Program is temporarily made unavailable (because, for example, a money market fund closes or a fund determines that a customer is not an eligible investor or a Program Bank refuses deposits), then VBS may sweep to another available option under the Sweep Program or it may opt to reflect cash as a Free Credit Balance in Your Account. If the current money market fund is eliminated or restricted as an option in the VBS Sweep Program, VBS may seek a new money market mutual fund, considering such factors as yield, fees, investment objectives, risks, and current market conditions, or VBS may make available or use another available option or combination of options under the Sweep Program, such as an FDIC-insured Bank Sweep Program option or Free Credit Balances in Your Account.

After notice, whenever feasible, VBS may change Your investment made through the Sweep Program from one Sweep Program option to another.

Cash not swept under the Sweep Program will be reflected as a free credit or a Free Credit Balance in Your Account. VBS expects to receive compensation based on the amount of Free Credit Balances in VBS Accounts.

b. Trading or Disbursement Restrictions

VBS may, at any time and without prior notice, reject, cancel, modify, suspend, or impose conditions on any order or trading activity, including any automated, algorithmic, high-frequency, or electronic trading activity, or funding. These restrictions could relate to or result from many factors, including, without limitation:

- Court orders, tax levies, or garnishments;
- Market Volatility;
- Trade execution quality, including interference with VBS' ability to obtain best execution or provide price improvement opportunities;
- Activity that may create a risk to VBS, including financial, regulatory, legal, credit, operational or reputational risk;
- Activity that may violate applicable law, regulation, exchange rules or VBS' policies and procedures;
- Requests from, or sanctions or similar restrictions related to the purchase, sale, or transfer of any Securities and/or Other Property imposed by a government agency or law enforcement authority;
- Failure to pay for trades in a timely manner or for trading or trade-related violations;
- Notice of a dispute involving any party claiming interest in, or authority over, the Account. Such parties may include, but are not limited to, Account Owners, beneficiaries, authorized agents, legal or court-appointed representatives, fiduciaries, or any other individuals or entities asserting rights, authority, or claims with respect to the Account or its assets; or
- If VBS believes that doing so will prevent fraud, financial exploitation or abuse, or to protect vulnerable investors.

You agree to abide by any restriction and not to initiate trades or transactions that would violate any such restriction. See also **Satisfying Legal Obligations by Releasing Account Information or Liquidating Securities and Other Property; Account Restrictions and Closing an Account**.

c. Requests for Transactions and Order Entry Services

VBS will accept online or verbal orders from You. Some transactions and services may not be performed online; in which case, You must call to place that order or engage in the service. You must place all orders electronically at **vanguard.com**, through VBS' mobile app, or verbally with one of VBS' brokerage associates. VBS will not accept written orders for the purchase or sale of Securities or Other Property. Online trading is the most efficient way to place trades.

Important information about written orders.

If VBS receives a written instruction accompanied by funding (for example, a check), VBS may treat the instruction as a request to Fund Your Account, with the proceeds (after satisfying outstanding debits) to Sweep.

VBS may accept wires to Fund Your Account, with the proceeds to Sweep. Generally, however, VBS will not accept wires for the purchase of a Security or Other Property. If You send VBS a wire with instructions to purchase a Security or Other Property, including a Vanguard Fund, VBS may, at its discretion, treat the wire as a request to Fund Your Account with the proceeds (after satisfying outstanding debits) to Sweep.

Orders for the purchase or sale of Vanguard Funds may require notification to and/or approval from the Vanguard Fund's manager. Please contact VBS before attempting to place a large transaction.

If a check, EBT, or other item You send to pay for a purchase or to Fund Your Account is returned to Us unpaid, recalled, or otherwise not honored, We may debit Your Account and/or charge Your Account a fee.

VBS may restrict Your ability to withdraw funds recently credited to Your Account.

Additional information for clients who transferred their Vanguard Funds into their Accounts:

Instructions to purchase received by The Vanguard Group, Inc., after You have transferred Vanguard Fund holdings to Your Account. You agree that VBS, at its

sole discretion, may treat any instruction received by The Vanguard Group, Inc., for the purchase of a Vanguard Fund as an instruction to VBS to Fund Your Account, with the proceeds to Sweep. VBS' decision to treat any such instruction received by The Vanguard Group, Inc., as an instruction to VBS does not mean that VBS has an ongoing obligation to treat all such instructions received by The Vanguard Group, Inc., as instructions to VBS.

You agree to instruct Your bank to pay on checks made payable to "Vanguard," or any reasonable variation of "Vanguard," when presented to Your bank for payment by either The Vanguard Group, Inc.'s, or VBS' bank. If Your bank dishonors the check for any reason, You agree to pay for the transaction in accordance with the **Payment, Equity Deposit, and Settlement** information noted below.

You agree that The Vanguard Group, Inc., or VBS, or any affiliate of any of them, will not be responsible for any costs or fees incurred by You as a result of Vanguard's or VBS' bank's presentment of a check as described above.

Checkwriting drafts received by The Vanguard Group, Inc., and presented for payment within the 45 days following the transfer of the position to Your Account. If You have transferred Your Vanguard Fund(s) into Your Account, *immediately upon the transfer of Your Vanguard Funds into Your Account You agree to cease using the checkwriting drafts (CWRs) issued on Your Vanguard Funds and to destroy any unused checks.* If You have written CWRs on a Vanguard Fund You transferred to Your Account, and those CWRs remain outstanding and unpaid at the time of the transfer, You agree that VBS may, in its sole discretion, elect to treat any CWR received by The Vanguard Group, Inc., or its bank in the 45 calendar days following the date You transferred Your Vanguard Funds into Your Account as an instruction to sell sufficient shares of the same Vanguard Fund that the CWR was originally written against to cover the CWR. If at the time the CWR is presented for payment, the balance in the Vanguard Fund that the CWR was originally written against is not sufficient to cover the entire CWR, You agree that VBS may, at its sole discretion, elect to treat the CWR as an instruction to redeem shares from the Vanguard Fund that the CWR was written against and cover the remainder of the CWR by free credit, if any, and/or a redemption of shares from Your Sweep Program or any money market position in Your Account. You agree that You are responsible for any tax consequences associated with the sale of shares to cover the CWR.

If You do not have sufficient shares in the Vanguard Fund the CWR was originally written against, Your Sweep Program and/or any money market position in Your Account to cover the check in its entirety, VBS will not accept the instruction and will direct The Vanguard Group, Inc., to instruct its bank to dishonor (that is, "bounce") the check. You agree that The Vanguard Group, Inc., VBS, or any affiliate of either, will not be responsible for any costs or fees You may incur as a result of a dishonored CWR.

VBS may accept instructions from The Vanguard Group, Inc., to reverse and/or reprocess the transfer of Your Vanguard Funds to Your Account to correct errors made prior to the transfer.

d. Changing or Canceling Orders

It may not be possible for You to cancel an order once You have placed it. If You attempt to cancel or change an order, Your attempt will be treated as a request to cancel or change. VBS will process Your request on a best-efforts basis, but if VBS cannot cancel or change Your order, You will be bound to the terms of the original order You placed.

For VBS' protection, it may, without prior notice, decline, cancel, modify, or reverse Your orders or instructions.

e. Payment, Equity Deposit, and Settlement

You agree to pay for all authorized transactions made in Your Account, including any commissions or fees that VBS might charge. When You purchase Securities and Other Property on a cash basis, You agree to pay for the Securities and Other Property by settlement date.

VBS may use available funds in Your Account or in any account with its affiliates to settle a transaction. Note: For Retirement Accounts, VBS will only use available funds in Your Retirement Account, or in an account with its affiliates that is part of the same Retirement Account or plan, to settle a transaction.

VBS may require an equity deposit or a full payment before it accepts Your order. When You place an order to sell a Security long, You must own the Security at the time You place the order. You agree to make Good Delivery of the Security You are selling by settlement date. Proceeds of any sale will not be paid until VBS has received Good Delivery of the Security and settlement is complete.

You agree to promptly satisfy any Debit Balance or other indebtedness or outstanding obligation related to Your Account, including, but not limited to, any such indebtedness that results from instructions provided to VBS by You or an Investment Advisor authorized to make transactions in Your Account. We may elect at any time, with or without notice, to make any Debit Balance or other indebtedness or outstanding obligation related to Your Account immediately due and payable.

VBS generally will not accept a tender of Control or Restricted Securities as collateral for an obligation You owe VBS. VBS may, however, in its sole discretion, approve such a tender. VBS' approval of such tenders must be in writing and be given in advance.

If You incur a debt to VBS or any of Our affiliates as a result of activity in Your non-Retirement Account, and You do not promptly pay VBS the amount due, You agree that VBS may, at its discretion and without prior demand or notice, sell Securities in the non-Retirement Account that incurred the debt to satisfy the debt, or if there are not enough assets in that non-Retirement Account to satisfy the full amount of the debt, to sell or otherwise liquidate any or all Securities or Other Property in any other non-Retirement Account with VBS or with any of its affiliates in which You have an

interest, regardless of whether there are other Account Owners on the other Account(s), and use the sale proceeds to cover the debt. You understand that the other Account(s) may be charged a commission as a result of any such sale.

VBS may also transfer Securities and Other Property from any non-Retirement Account or in any non-Retirement Accounts with its affiliates in which You have an interest to any other non-Retirement Account in which You have an interest, regardless of whether there are other Account Owners on the other Account(s), in order to satisfy deficiencies in any such Account or if VBS thinks Your obligations in any such non-Retirement Account are not adequately secured.

Important Note for Retirement Accounts: If You incur a debt to VBS or any of Our affiliates as a result of activity in Your Retirement Account, You must promptly pay VBS the amount due. If You do not, You agree that VBS may sell Securities in the Retirement Account that incurred the debt to satisfy the debt. If You do not have sufficient assets in the Retirement Account held with VBS to satisfy the amount You owe, You agree that VBS may require You to sell Securities that You hold in other Accounts (and You agree to execute all documents and/or give oral instructions as necessary to effect the sale), whether held with VBS or with its affiliates, that are part of the same retirement plan and use the proceeds to cover the debt. You understand that Your Account may be charged a commission as a result of any such sale.

To satisfy a Debit Balance in a Retirement Account with a sale of Securities or Other Property, VBS may only sell Securities that are in the Retirement Account that incurred the debit or in an Account held with an affiliate that is part of the same retirement plan. If VBS sells Securities or Other Property to satisfy a Debit Balance, Your Account and any other account held with an affiliate that are part of the same retirement plan are responsible for all costs and commissions related to such a sale.

You understand that VBS may apply any dividends, capital gains payments, interest payments, or other funds coming into an Account, such as incoming funds by check, EBT, wire, or Direct Deposit, or funds that would otherwise be invested through the Vanguard Brokerage Dividend Reinvestment Program, to cover fees or other indebtedness to VBS. Debit Balances in Your Account will be charged interest in accordance with VBS' established rules and policies.

f. Security for Indebtedness for Non-Retirement Accounts

As security for the repayment of all present or future indebtedness owed to VBS by each Account Owner, each Account Owner grants to VBS a first, perfected, and prior lien on, a continuing security interest in, and right of set-off with respect to, all Securities and Other Property that is, now or in the future, held, carried, or maintained for any purpose in or through VBS, and, to the extent of such Account Owner's interest in or through, any present or future account with VBS or its affiliates in which the Account Owner has an interest. You agree to indemnify VBS for any loss or expense that VBS may incur in enforcing its lien or any other remedies available to it, including the reasonable cost of collection. This paragraph does not apply to assets in a Retirement Account.

g. Checks, EBTs, or Other Items Returned Unpaid, Recalled, or Dishonored

If a check, EBT, or other item You remit to VBS is returned unpaid, recalled, or otherwise not honored, VBS may charge a fee to Your Account. Also, for VBS' protection, VBS may restrict Your ability to withdraw funds recently added to Your Account by check or EBT.

VBS may restrict Your ability to use recently added funds to settle a Securities transaction, and/or decline to accept orders until recently added funds are collected or otherwise validated.

h. Foreign Securities Trading, Settlement, and Custody

VBS may act as agent for foreign Security orders traded in local markets. VBS' policy is to settle foreign Security trades in U.S. dollars. Foreign account activity, including, but not limited to, trades, transfers, dividends, and other corporate actions, may be automatically converted to or from U.S. dollars. VBS has an established partnership with JPMorgan (JPM) to facilitate the clearance and settlement of foreign Security transactions. Your foreign Securities may be custodied through JPM or any subcustodian located in a particular local market and will be held at JPM in an account established for the exclusive benefit of customers of VBS.

i. Securities Held in Street Name

Securities and Other Property that are held for Your Account and that are in "street name," or are being held by a Securities depository in VBS' name, may be commingled with the same Securities being held for other clients of financial organizations and for VBS' own clients. VBS is responsible for ensuring that its records reflect Your ownership of these Securities and Other Property. For Accounts other than a Retirement Account, You have the right at any time to request delivery to You of any such Securities or Other Property that are fully paid for or are in excess of margin requirements.

j. Corporate Actions

Securities in Your Account may be subject to mandatory or voluntary corporate actions.

- Mandatory corporate actions, including but not limited to, stock splits, name changes, and liquidations, affect all Security holders and require no action from You. Due to the unpredictability of mandatory corporate actions, VBS may not be able to provide advance notice of such events.
- Voluntary corporate actions are those that require an election from You. You are responsible for knowing the features, rights, and terms of the Securities You own, and are responsible for making election decisions for corporate actions.

VBS may, but is not obligated to, send notification of corporate actions and their expiration dates. In addition, VBS reserves the right to set response deadlines that differ from an offer's expiration date. In

the event of a cashless right or warrant, VBS may, but is not obligated to, auto exercise Your shares.

Mutual Fund Mergers, Share Conversions, Mutual Fund to ETF Conversions, or Other Fund Actions

A mutual fund, or its transfer agent, may merge fund shares held in Your Account to another fund pursuant to a fund merger, convert shares from one class to another, or take other actions as described in each fund's prospectus. VBS will accept the direction of a fund or its transfer agent to merge, convert Your shares, or take other action.

Interest, Dividend, and Other Payments

If You are entitled to receive dividend, interest, or other payments on the Securities held in Your Account, VBS will allocate distributions to Your Account according to Your elections, where feasible. If VBS is unable to reinvest a dividend, it will be credited to Your Account as cash.

Optional Dividends

When a company offers its shareholders a choice to receive a dividend either in cash or stock, VBS will use Your position level dividend election as Your preference for receiving the payment of the optional dividend. However, optional dividends are subject to pro-ration by the issuer and You may receive both cash and shares upon payment, regardless of your election. VBS receives the reinvestment price, cash and shares directly from the issuer to distribute to You upon allocation. If VBS is unable to reinvest an optional dividend, it will be credited to Your Account as cash.

Callable Securities

Many bonds allow the issuer to redeem or call an issue prior to maturity date. Certain preferred Securities can also be called by the issuer. Some calls are "partial" calls, meaning that only some shares are called. If a Security that You and other clients of VBS hold is partially called, VBS will determine, through a random selection procedure, which Securities will be submitted for redemption without regard to unsettled sales. If the Securities held in Your Account are selected and redeemed, Your Account will be credited with the redemption proceeds.

Should You wish to opt Your non-Retirement Account(s) out of this random selection process, You must instruct VBS to deliver Your Securities to You. We will deliver the Securities provided that they are unencumbered and they have not already been called by the issuer. The probability of one of Your Securities being called is the same whether they are held by You or by VBS for You. You cannot opt out of the random selection process for assets held in a Retirement Account because We cannot deliver assets held in a Retirement Account to You.

To review VBS' procedures for the client allocation of Securities selected for redemption in the event of a partial redemption, visit **vanguard.com**. To request a free, printed copy of VBS' allocation procedures, contact VBS.

Nontransferable Securities

If You hold nontransferable Securities in Your Account, VBS may remove the Securities from Your Account without prior notice. Securities become nontransferable when there are no transfer agent services available for the Securities. If nontransferable Securities are removed from Your Account, You should consult with Your tax advisor about any potential tax implications. The position removal will be documented on Your subsequent Account statement.

Credits to Your Account

VBS will credit Your Account with funds belonging to You, such as dividends, interest, redemption proceeds, or the proceeds of corporate reorganizations, in a timely matter and if such funds are received by VBS. An issuer or an intermediary may earn interest on funds held in its possession. If an issuer or intermediary passes such interest along to VBS, VBS will keep the payment.

VBS can provide further information on when Your Account will be credited with funds due to You, when those funds are available to You, and/or when You begin earning interest on those funds.

k. Dividend Reinvestment Program

Our Dividend Reinvestment Program ("the Program") provides a no-fee, no-commission reinvestment program that allows You to reinvest dividend, capital gain, and other eligible distributions from any or all eligible stocks, closed-end mutual funds, exchange-traded funds ("ETFs"), FundAccess® funds, or Vanguard mutual funds in Your Vanguard Brokerage Account or Vanguard Cash Plus Account, as applicable, in additional shares of the same Securities. The Program is available for all Vanguard Brokerage Accounts and Vanguard Cash Plus Accounts except those that are subject to either backup or nonresident alien income tax withholding.

When You choose to reinvest dividends for an eligible Security, VBS combines the cash distributions from all participating clients and purchases additional shares of that same security in the open market. These new shares are then allocated proportionately to participating Accounts.

You may update or cancel Your dividend reinvestment instructions at any time by going online, calling VBS, sending a secure email, or sending a written request. To ensure changes apply to an upcoming distribution, instructions must be received at least two business days before the payable date. VBS provides an applicable tax information statement after year-end with the regular tax filing schedule.

Eligibility Requirements

Not all Securities qualify for dividend reinvestment. Eligibility for the Program is based on the following:

- The Security must be held in street name by VBS, pay a dividend, and have an adequate trading volume.
- The Security must not be otherwise restricted by policy, or involved in certain corporate actions (e.g., mergers or reorganizations).
- Some Security types—such as Unit Investment Trusts ("UITs"), certain foreign or domestic equities, and select American Depository Receipts ("ADRs")—may be excluded.

- Eligibility requirements for mutual funds are determined by each fund; review the fund profile on Our website.

VBS reserves the right to modify or discontinue the Program, remove Securities from the Program, change eligibility standards, or restrict participation at any time, consistent with the Vanguard Brokerage Account Agreement, without notification to You.

Program Limitations and Restrictions

- In rare cases, technological, operational, other limitations, or market conditions may prevent reinvestment.
- VBS may place restrictions on Your account or specific Securities as permitted under the Vanguard Brokerage Account Agreement. You must comply with all Program-related policies and guidelines.
- Fractional Shares created through the Program will appear on Your Account statement and are rounded to 3 (three) decimal places. In certain cases, fractional entitlements may be paid in cash.

  Note: If You are an "affiliate" or "insider," You should consider consulting with Your personal legal advisor before enrolling in the Program.

l. Disclosure of Account Information

We may disclose Your name and/or information about Your Account or Your transactions consistent with Our Privacy Policy and pursuant to Rule 14b-1 under the Securities Exchange Act of 1934 to companies whose Securities You hold unless We receive Your written objection. Different rules apply for European Union domiciled Securities, and You may not be able to opt out of Our disclosure of shareholder identifying information to issuers of such Securities. For additional information about Our Privacy Policy, visit Our Privacy Center at https://investor.vanguard.com/privacy-center. In addition, if You request a large transaction in a Vanguard Fund, including in Your Settlement Fund (if it uses the money market fund Sweep Program option), VBS may disclose information about You and/or Your Vanguard Fund holdings to The Vanguard Group, Inc.

m. Delivery of Required Information, Other Documents, and Notices About Your Account

You may consent to receive Confirmations, Statements, and other Account information electronically. E-delivery is an efficient way to facilitate Account communications. If You do not consent to e-delivery, VBS or its mailing vendor will use the USPS to deliver the information to You. However, to receive certain VBS products, services, or offers, including, for example, online trading, VBS may require that You consent to electronic delivery of certain documents.

VBS will direct communications to the mailing address, email address, or phone number We have for You in Our files (that is, the contact information You provided on Your Account Application or later updated). Any communication (required or otherwise) that VBS directs to such a mailing address, email address, or phone number will be considered delivered to You, whether or not the communication is actually received.

n. Provision of Market Data

We may provide You with Market Data, and may charge You a fee for doing so. We obtain Market Data from Securities exchanges, markets, and from parties that transmit Market Data ("Market Data Provider" or collectively, "Market Data Providers"). All Market Data are protected by copyright laws. We provide Market Data for Your personal noncommercial use; You may not sell, market, or distribute Market Data unless You have entered into a written agreement with the appropriate Market Data Provider that allows You to sell, market, or distribute the data.

We receive the Market Data from industry sources that are believed to be reliable. However, neither VBS nor the Market Data Providers can guarantee the accuracy, completeness, timeliness, suitability for use, or correct sequencing of the Market Data. Neither VBS nor the Market Data Providers will be liable to You for interruptions in the availability of Market Data or Your access to Market Data.

The Market Data are provided "as is" and on an "as available" basis. There is no warranty of any kind, express or implied, regarding the Market Data. We are not responsible for, and You agree not to hold VBS liable for, lost profits, trading losses, or any other damages resulting from Market Data or Your use of the data. In any case, Our liability arising from any legal claim (whether in contract, tort, or otherwise) relating to the Market Data will not exceed the amount You have paid for use of the services or Market Data. You agree that We may correct any execution reported to You that was based on inaccurate Market Data provided to VBS by an exchange, market center, or other provider.

o. Extended-Hours Trading

VBS offers online access to participate in extended-hours trading sessions. If You participate in extended-hours trading, You agree to terms and conditions that are disclosed on the website and that You understand the risks of trading in an extended-hours session.

p. Best Execution and Payment for Order-Flow Practices

Order Routing

VBS routes equity and options orders to various market participants to seek favorable pricing in light of prevailing market conditions in accordance with its best execution obligation. VBS uses a top-down approach in selecting the market participants with which We will establish a relationship. This approach includes a review of system availability, quality of service, and financial and regulatory standing. The designated market participants to which orders are routed are selected based on the consistent high quality of their executions in one or more market segments. In analyzing quality of executions, VBS considers factors such as liquidity enhancement, price improvement, execution speed, and overall effective price compared with the national best bid and offer ("NBBO"). VBS regularly conducts analysis and reviews reports for quality of execution purposes. VBS does not receive compensation for directing order flow in equity Securities. Similarly, VBS does not receive compensation for directing options Securities to Our execution partners, but reserves the right to do so in the future.

Reporting

VBS is required to make publicly available reports that disclose information related to order execution such as the venues to which orders were routed, the nature of Our routing relationships, any payment for order-flow arrangements, and trade execution quality. You can access this information by visiting **vanguard.com**. In addition to the publicly available reports, VBS, upon written request, will provide information related to Your orders that were routed for execution in the past six months. This information will include the venue to which Your order(s) was (were) routed, whether the order(s) was (were) directed or nondirected, and the time of the transactions.

Access to Tradeweb Markets LLC

VBS provides availability to the Alternative Trading Systems operated by Tradeweb Markets LLC ("Tradeweb") and to other content provided by Tradeweb. Tradeweb provides access to certain municipal bond information from DPC DATA. Tradeweb and DPC DATA are third parties and are not affiliated with VBS. While VBS provides access to Tradeweb's Alternative Trading Systems, VBS has no control over actions taken by Tradeweb. All content, with the exception of new issue municipal content, is provided by Tradeweb and DPC DATA; VBS is not responsible for the accuracy of content provided by Tradeweb or DPC DATA. New issue municipal content on the Tradeweb pages is provided by VBS. All offerings are subject to market conditions, prior sale, and/or Tradeweb's clearly erroneous policy.

q. Brokered Certificates of Deposit (Brokered CDs)

VBS offers brokered CDs. For additional information about CDs, including eligibility for FDIC insurance and risks, visit **vanguard.com** and read the **Certificate of Deposit Disclosure Statement**. **You are solely responsible for monitoring the aggregate amount that You have on deposit with each bank to determine Your FDIC insurance coverage, including through other accounts with VBS and other financial institutions.** For additional information regarding FDIC insurance, visit **fdic.gov**. See also **Eligibility for FDIC Insurance**.

r. Fractional Shares

As a VBS client, You may purchase or sell eligible Securities by entering an order either in dollars (where available) or in whole shares (only). In addition, You may acquire a fractional share interest in a single Security (a "Fractional Share") in several ways. For example, You may receive a Fractional Share: (a) as a result of the reinvestment of a dividend if You participate in VBS' Dividend Reinvestment Program; (b) as a divided interest in whole shares resulting from a life event, such as divorce or inheritance; (c) as a result of placing a Dollar-Based Trade (as defined in the Dollar-Based Trading and Recurring Transactions Terms of Use ("Dollar-Based Terms")); or (d) as a result of the conversion of conventional mutual fund shares issued by a Vanguard Fund to ETF Shares of equivalent value of the same Vanguard Fund (refer to Vanguard ETF prospectuses for further information regarding the conversion privilege). Fractional Shares present unique risks and have certain limitations and may have different rights from a whole share of the same Security. For information on Dollar-Based Trading, refer to the Dollar-Based Terms available at **vanguard.com** and provided to You in connection with Your Dollar-Based Trading activity. In the event of a conflict between this Agreement and the Dollar-Based Terms, the Dollar-Based Terms shall prevail.

IMPORTANT: The Fractional Share rights and limitations described below apply to certain Securities, but do not apply to mutual funds. For further information on rights and restrictions related to mutual funds, please consult the mutual fund prospectus and statement of additional information.

i. *Eligibility for Dividends and Dividend Reinvestment.* You will receive dividends on Fractional Shares unless Your prorated amount of the dividend is less than one half of one cent ($0.005). Fractional Shares are eligible for automatic reinvestment of cash dividends received, at Your election, subject to the terms of VBS' Dividend Reinvestment Program. If You participate in dividend reinvestment and Your dividend for a particular Security would equate to less than .0005 shares, then You will receive Your dividend in cash.

ii. *Corporate Actions.* The eligibility of Fractional Shares to participate in corporate actions may be different than whole shares. For mandatory reorganizations, such as mergers and acquisitions, stock splits, or spin-offs, typically VBS will distribute Your interest in proportion to Your ownership interest, inclusive of any Fractional Shares. However, only whole shares, and not Fractional Shares, are eligible to participate in voluntary corporate actions, including tender offers and certain rights offerings. If You are to receive Fractional Shares as the result of a stock split or other corporate action for a Security held in Your Account, VBS, in its sole discretion, may either sell the shares on the open market or to the issuer or transfer agent. You are entitled to receive Your pro rata portion of the proceeds of any such sale. If sold on the open market, Fractional Shares may receive a price that differs from that offered to certain registered owners by the issuer or transfer agent. In some cases, such Fractional Shares cannot be sold and will be deposited into Your Account.

iii. *Voting Rights.* VBS works with a vendor that aggregates any proxy votes for Fractional Shares of VBS' clients with all votes reported to the issuer or issuer's designated vote tabulator. VBS' vendor will report such proxy votes on Fractional Shares, but the issuer or tabulator may not fully count such Fractional Share votes.

iv. *Stock Certificates.* Stock certificates cannot be issued for Fractional Shares.

v. *Transferability.* Fractional Shares are not currently transferrable to another brokerage firm. If, for example, You own 10.5 shares of an individual stock and You transfer the Securities in Your VBS Account to another firm, 10 shares will be transferred. The Fractional Share will be liquidated by VBS and You will receive the closing price from the trading day

immediately preceding the transfer of Your whole share(s). VBS will distribute these proceeds in Your Account and will provide confirmation of the Fractional Share liquidation on Your periodic statement. Since Your Fractional Shares cannot be transferred, Your overall SIPC coverage may be affected.

vi. *Auto-Liquidation of Certain Fractional Shares.* Unless You have established a position in an Eligible Security (as defined in the Dollar-Based Terms) via Dollar-Based Trading, You generally may not hold only a Fractional Share without at least one corresponding whole share. VBS will automatically sell ("Auto-Liquidate") Your Fractional Share under the following two scenarios:

1. "Sell-All" Trades. If You own whole shares and a Fractional Share in a Security, VBS will Auto-Liquidate Your Fractional Share holding when You place a share-based order to sell Your entire whole share position. For example, if You own 10.2 shares in a Security and sold all 10 shares, VBS would Auto-Liquidate the remaining .2 Fractional Share. That Fractional Share will receive the same price as the corresponding whole shares, and will be prorated to reflect Your Fractional Share ownership in that Security. VBS will place the proceeds of that sale into Your Account.
2. "Orphaned" Fractional Shares. As a result of certain actions, Your Account may own a Fractional Share without a corresponding whole share position in a Security. For example, as described above, Fractional Shares cannot be transferred to another firm. Therefore, if You own whole shares and a corresponding Fractional Share in a Security and transfer those Securities to another firm, VBS will transfer the whole shares, and then Auto-Liquidate any remaining Fractional Share. Under these circumstances, You will receive the previous day's closing price for any "Orphaned" Fractional Share Auto-Liquidated by VBS, prorated to reflect Your Fractional Share interest.

vii. *Auto-Liquidation and Dollar-Based Trading.* If You establish a new position in an Eligible Security through a Dollar-Based Trade, VBS will turn off the Auto-Liquidate feature for that Eligible Security in the Account in which You placed that trade. Further information on Auto-Liquidation and its impacts on Fractional Shares acquired by Dollar-Based Trades is available in the Dollar-Based Terms.

viii. *Capacity.* VBS may trade the Fractional Share component of any Security trade You place (or placed on Your Behalf) on a principal or agency basis. Refer to Your trade confirmation or periodic statement, as applicable, for specific capacity.

s. Class Action Service (applicable only if You have not opted out)

You understand and agree that eligible Vanguard Brokerage Accounts are automatically enrolled in VBS' Class Action Service (the "Class Action Service"). Closed Account(s) are not included in the Class Action Service. If Your Account(s) is excluded from the Class Action Service, You may still participate in a class action lawsuit by submitting a claim directly to the claims administrator.

In order to be eligible for a claim, all information necessary for the filing must be available in Vanguard systems prior to any deadline. In order for VBS to file a claim and collect proceeds on Your behalf, VBS provides third-party claims administrators with the information collected as necessary and in accordance with Our Privacy Policy.

You understand and agree that the enrollment of Your Account(s) in the Class Action Service authorizes VBS to automatically file a claim on Your behalf if We receive notice of a potential class action lawsuit that applies to Securities in Your Account(s). You may cancel Your enrollment in the Class Action Service at any time by notifying Us. If You cancel Your enrollment in the Class Action Service and wish to reenroll, You will be required to agree to the terms and conditions for the Class Action Service in effect at the time of reenrollment. Canceling Your enrollment in the Class Action Service or closing Your Account(s) after a claim has been filed on Your behalf will not remove You from participation in that particular class action lawsuit. VBS reserves the right to terminate the Class Action Service at any time with notice to You.

VBS uses a vendor to facilitate the Class Action Service. Generally, the gross proceeds awarded from class actions paid out through the Class Action Service will be deposited into Your impacted Account(s). Awards resulting from the Class Action Service are subject to fees, as permitted, which are assessed subsequent to the deposit of the gross proceeds.

You agree that You cannot separately bring a lawsuit on Your own behalf, either contemporaneously or in the future against any named defendant in connection with such particular class action lawsuit. You further understand and agree that once the Class Action Service files a class action claim on Your behalf, You will not assert: (a) any dissenters' or similar rights under any applicable law, rule, or regulation; (b) any right to require partition or appraisal of any company that is the subject of a potential class action lawsuit or of any of said company's assets or to cause the sale of such company's property; or (c) any right to maintain any action for partition or to compel any sale with respect to shares held by other shareholders or with respect to any of said company's property.

Enrollment of Your Account(s) in the Class Action Service does not guarantee that You will receive a settlement payment for any given class action lawsuit. Neither VBS nor our affiliates take any responsibility for the outcome of any given class action lawsuit, and You agree to indemnify and hold VBS and Our affiliates harmless for all actions taken in connection with the Class Action Service. You further understand that the greater the number of clients enrolled in the Class Action Service, the smaller any potential settlement payment received by each client will be in the event of any class action settlement recovery.

t. Foreign Dividend Tax Withholding

We have contracted with a third-party vendor to provide foreign tax relief and reclamation services which may allow You to receive a favorable foreign tax withholding rate at the time dividend or interest is paid

to You in connection with Securities of issuers based in certain foreign jurisdictions. The vendor charges certain fees for this service, including a percentage of the amount of the tax reclaim. *Fractional shares are not eligible for foreign dividend tax withholding relief.*

u. Vanguard ETF Conversions

VBS Account Owners with eligible conventional Vanguard mutual fund shares (i.e., not exchange-traded shares) may convert those shares to ETF Shares of the same fund (if available). You may request a conversion through **vanguard.com** or by calling VBS.

When a conversion occurs, You receive shares of one class in place of shares of another class of the same fund. At the time of conversion, the dollar value of the "new" shares You receive equals the dollar value of the "old" shares that were converted. In other words, the conversion has no effect on the value of Your investment in the fund at the time of the conversion. However, the number of shares You own after the conversion may be greater than or less than the number of shares You owned before the conversion, depending on the NAVs of the two share classes.

Please note that investors who own conventional shares through a 401(k) plan or other employer-sponsored retirement or benefit plan generally may not convert those shares to ETF Shares and should check with their plan sponsor or recordkeeper. ETF Shares, whether acquired through a conversion or purchased on the secondary market, cannot be converted to conventional shares by a shareholder. Also, ETF Shares of one fund cannot be exchanged for ETF Shares of another fund.

Requests received prior to the New York Stock Exchange close (usually 4:00 PM EST) are typically processed that day. Requests received after this time are typically processed on the next trading day. Mutual fund dividend payments at or near the time of the conversion may delay the process.

An ETF Conversion is irreversible: Vanguard ETF shares cannot be converted back to Vanguard mutual fund shares. If You decide in the future to sell Your Vanguard ETF Shares in order to repurchase Vanguard mutual fund shares, that transaction could be taxable in a non-retirement account and fund minimums must be met.

Certain mutual fund Account options will not apply or will need to be reestablished following the conversion (including dividend reinvest election, distribution and money transfer options).

7. Your Role

As an Account Owner, or his, her, or its properly authorized Representative, You have certain responsibilities. For example, You are responsible for providing accurate and complete information on the Application and for submitting any additional information documents that VBS requires at the time Your Account is opened or at any time afterward. Once Your Account is opened, You are responsible for fulfilling the obligations arising under this Agreement or relating to the Account. If there is more than one Account Owner, each Account Owner is jointly and severally liable for obligations arising under the Agreement or relating to the Account. You should also refer to the confirmations and statements We send to You, the current VBS commission and fee schedules, as well as any brochures and forms that describe Account services and features for additional information about Your responsibilities and obligations.

a. Account Information

If any of the information You provided on Your Application changes, You should notify VBS immediately. You should notify Us, for example, if You, or a household member, becomes a control person or an affiliate of a public company (such as, but not limited to, a member of a board of directors, a 10% shareholder, or a policymaking executive), or if You or Your spouse becomes an employee of FINRA or becomes associated with a member of a stock exchange, a FINRA member firm, or a municipal securities dealer.

You should also notify Us immediately if Account authority changes. For example, if You revoke a third party's authority or a Representative's authority to act or inquire on Your Account, or for jointly owned Accounts, a joint Account Owner dies, You should notify Us.

If You provide Us with a mobile phone number, You agree and consent that VBS and its affiliates may contact You at that mobile number with telephone calls that may use an autodialer and by text messages for the purposes of servicing of Your Account(s) or investigating and preventing fraud. We will not use texts or autodialed calls to contact You for marketing purposes unless we receive Your prior express written consent. You do not have to agree to receive autodialed calls or texts to Your mobile phone number in order to use the products and services offered by Vanguard. You may unsubscribe from receiving text messages by texting STOP, CANCEL, UNSUBSCRIBE, QUIT or END in response to a Vanguard text. Additionally, You can contact VBS and an associate can unsubscribe You. Standard telephone minute and text charges may apply. Vanguard's Activity and Appointment Alerts Terms and Conditions of Use are available at **vanguard.com**.

Please pay close attention to the mailing address, email, phone number, and banking information on Your Account. You are responsible for keeping this information up to date.

Delivery of Distributions/Withdrawals: If You request a distribution or withdrawal from Your Account, but We are unable to deliver the funds as instructed or We have questions on the instructions, VBS will attempt to contact You for corrected or revised delivery instructions or for additional information. If You do not provide VBS with corrected or revised delivery instructions or the requested information promptly, VBS may return the funds to Your Account with the proceeds to Sweep. For Retirement Accounts: If funds are returned to Your Account for one of the reasons noted above, Your Account records (including the tax records) will reflect that the distribution did not occur.

Dormant Accounts, Uncashed Checks: You should periodically log on to Your Account at **vanguard.com** or through Our mobile app or contact VBS about Your Account, even if You do not wish to make a transaction.

If Your Account has had no activity in it for a period of time or You have not contacted Us about the Account, VBS may be required to transfer it to a state under the state's unclaimed property laws. Also, VBS may be required to turn over uncashed checks as unclaimed property; You should promptly cash or deposit any check VBS sends to You and instruct third parties to whom You direct VBS to send checks to do the same.

b. Paying for Transactions and Account Fees, and Return of Overpayments or Uncollected Fees

You are responsible for paying for all authorized transactions made in Your Account, including paying the commissions, processing, or other fees associated with the transactions. You are also responsible for paying any service or other fees imposed on Your Account. Our commissions and fees are detailed in the VBS commission and fee schedules, which are incorporated by reference into this Agreement. Certain clients, individually or as members of a category of clients, may pay different commissions and fees, or none at all, based on assets or activities in their Accounts. VBS reserves the right to change or waive commissions and fees at its discretion. If VBS is required to do so, it will notify You of the changes made to its commissions and fees. Your continued use of Your Account or the applicable service following such a change means You have consented to the change.

If You do not pay for Your transactions or the fees associated with Your Account, then VBS may, on its own behalf and on behalf of its affiliates, cancel Your transactions, liquidate assets, restrict Your Account, or take other appropriate action in seeking payment for the transactions and/or fees. You will be responsible for satisfying all debits and for paying all costs, commissions, and/or losses arising from any actions VBS takes to liquidate or close transactions in Your Account, or from Your failure to make timely Good Delivery of Securities and Other Property. See also **Payment, Equity Deposit, and Settlement** for additional information.

And, if You know or suspect that You have received an overpayment of funds or Securities or Other Property, or if You know or suspect that VBS failed to collect a fee You incurred, You are responsible for notifying VBS as soon as You learn of the overpayment or uncollected fee. You agree not to remove the overpayment or the uncollected fee from the Account. If You do remove the overpayment or uncollected fee, You are responsible for promptly returning the full overpayment or fee to VBS. If You fail to do so, You will be liable to VBS not only for the amount of the overpayment or uncollected fee, but also for all interest, costs, and expenses associated with its recovery.

c. Authority to Transact on an Account

Generally, an adult Account Owner, and/or his, her, or its properly authorized Representative, has authority to transact on his, her, or its individually owned Account. If an Account has more than one Owner or authorized person, each Account Owner or authorized person has authority, acting individually and without notice to any other Account Owner or authorized person, to deal with VBS as fully and completely as if they were the sole Account Owner or authorized person. Notwithstanding the preceding, VBS may, from time to time or for certain transactions, require the consent of all Account Owners.

If Your Representative's authority is revoked, You are responsible for providing VBS with notice of the revocation in a form and manner acceptable to VBS. You remain responsible for transactions effected in Your Account after a Representative's authority has been revoked but before VBS was provided notice and had an opportunity to react to the notice of revocation. See also the **Authorized Access to an Account** and **Liability** sections of this Agreement for additional information.

d. Compliance With Policies, Rules, and Laws

You, and the transactions You make or attempt in Your Account, must comply with the policies that can be found on **vanguard.com** or in other documents We provide to You, such as VBS' Privacy Policy, Frequent-Trading Policy, Market-Timing Policy, After-Hours Trading Policy, Dividend Reinvestment Program Policy, etc., as well as Our house trading rules and policies, as these rules and policies may be amended from time to time. In addition, You and the transactions You make or attempt in Your Account must comply with all applicable rules, regulations, customs, and usage of any exchange, market, or clearinghouse where the transactions are executed, as well as with any applicable laws, rules, regulations, or guidance of any international, federal, state, or self-regulatory agency or instrumentality. If VBS places a restriction on Your Account, You agree to abide by the restriction and to not initiate trades or transactions that would violate the restriction.

e. Control or Restricted Securities

You agree to tell Us whether You hold any Control or Restricted Securities in Your Account, and their statuses, and whether there are any other contractual lockup, blackout, or additional restrictions on Your ability to sell such Securities. You also must furnish promptly whatever information and documents VBS might need to comply with its regulatory obligations relating to the Control or Restricted Security. Furnishing the information that VBS might need on Control or Restricted Securities is not an order to sell those Control or Restricted Securities; orders to sell must be placed by available order entry means, preferably online, meaning through **vanguard.com** or Our mobile app. You agree that You are responsible for all costs, including the cost to repurchase Securities if You sell Securities that are later found to be restricted or nontransferable. You further acknowledge that proceeds from the sale of Control or Restricted Securities will not Sweep or be made available to You for withdrawal or trading purposes until VBS receives what it, in its sole opinion, considers to be adequate verification that Your Control or Restricted Securities have been transferred or cleared for transfer.

Because Control or Restricted Securities transactions require special handling by both VBS and third parties, processing transactions in those Securities could take several weeks or longer, during which time the price of the Control or Restricted Securities may fluctuate. You understand that You, and not VBS, bear the risk of such market fluctuations, as well as for delays in the sale (or

settlement of such sale) of Your Control or Restricted Securities that result from the failure of issuer's counsel to issue or approve any necessary legal opinion, the failure of the transfer agent to process Your shares, or any other third-party's action or failure to act.

You cannot tender Control or Restricted Securities as collateral for an obligation You owe VBS unless You have obtained, in writing, VBS' preapproval of such a tender.

f. Identifying as a Large Trader

If Your transactions in NMS Securities equal or exceed two million shares or $20 million a day, or 20 million shares or $200 million a month, You are a Large Trader and You must identify Yourself as such with the Securities and Exchange Commission ("SEC"). These thresholds apply not only to Accounts that You own but also to Accounts over which You have investment discretion, such as agent authority. If You meet the definition of a Large Trader, You must file Form 13H with the SEC (available at **sec.gov**) to obtain from the SEC a unique ID number. You must provide this unique ID to all broker-dealers with which You trade, including VBS, so that the broker-dealers can provide accurate trading records to an appropriate regulator upon the regulator's request.

g. Understanding Terms of Securities and Other Property, and for Trading Decisions

You are responsible for knowing the rights and terms of Your Securities and Other Property, and for taking action to realize the value of Your Securities and Other Property. You agree that VBS is not liable for any decrease in the value of Your Securities and Other Property or other losses resulting from Your failure to give VBS instructions on how to respond to a tender offer, exchange offer, or other offer or transaction.

However, VBS may, at its discretion, and absent instructions from You, take action for You if Your Security or Other Property is about to expire worthless or be redeemed, exchanged, converted, tendered, or exercised in such a way that You would be left with significantly less value than if You had given VBS instructions to take action. VBS may charge You a brokerage commission and/or fee for any such action it takes on Your behalf, and You are responsible for paying the commission and/or fee.

You understand that, notwithstanding the above, You are solely responsible for investment decisions made in Your Account, including whether to buy or sell a particular Security, unless You receive a recommendation from VBS. VBS provides point-in-time recommendations for certain brokerage products, accounts and services, and relating to account transfers and rollovers. However, unless VBS affirmatively states that it is making a recommendation, it is not providing a recommendation. You are responsible for using multiple sources of information in making investment decisions for Your Account and, absent an explicit affirmative recommendation from VBS, You should not rely solely on any information VBS may provide for any decisions You make or that are made on Your behalf. You agree and understand that any research, analysis, news, or other information VBS may make available to You in any format does not constitute a recommendation to You to buy, sell, or hold a particular Security. Further, You are responsible for monitoring Your Account on an ongoing basis and for modifying Your trading decisions accordingly.

You are responsible for obtaining Your own legal and tax advice as needed, and understanding that VBS does not, and will not, provide personalized legal or tax advice.

h. Account Security, Fraud, Identity Theft, and Reporting Inaccuracies and Discrepancies in Your Account

VBS may ask any caller, any online user, or any person who sends in written requests to provide certain information or to answer security questions to verify that he or she is authorized to act on an Account, and You agree (if You are the person calling, writing, or using Our online systems) to provide the information We request, including, for example, signature guaranteed, notarized, or otherwise authenticated instructions, or other supporting documentation. You understand that VBS is not obligated to approve or open any Account, take any trade, or effect any transaction, and that it may place trading, disbursement, or other restrictions on Your Account and close Your Account. See also **Trading or Disbursement Restrictions** and **Satisfying Legal Obligations by Releasing Account Information or Liquidating Securities and Other Property; Account Restrictions and Closing an Account**.

You are responsible for taking precautions to protect against fraud and identity theft, such as creating strong online credentials, protecting Your device(s), and taking other steps to ensure the confidentiality and security of Your Account number and other Account information. If You use a number or password to access Your Account, You agree to keep the access number or password private and secure.

You are responsible for reviewing any statements, confirmations, or other Account information (collectively, "Account Documentation") that VBS provides to You, and for promptly notifying Us if You become aware of any inaccuracies or discrepancies, including, but not limited to, unauthorized use, theft, or loss of any of Your Account information (including, without limitation, Your Account number); orders or transactions that You did not place; or any other type of error. If You do not promptly notify Us of such inaccuracies or discrepancies, neither VBS nor any of its affiliates will be liable for any consequences in accordance with applicable laws. In addition, if You do not promptly, and in writing, report inaccuracies or discrepancies (including unauthorized trading) to VBS, You may waive any objection to the trade, position, funds transfer, check, disbursement, fee, and/or other information set forth on the Account Documentation. **Important note:** While You may call VBS to discuss inaccuracies and discrepancies, You must confirm Your verbal communication in writing to fully protect Your rights, including any right You might have under the Securities Investor Protection Act ("SIPA").You are responsible for any Account losses resulting from activity that VBS reasonably believes was authorized, as well as any losses associated with restrictions VBS may have placed on Your Account to protect against unauthorized activity.

i. Lost Securities

If You do not receive Securities that VBS forwarded to You, You should notify VBS immediately. If VBS receives Your notice within one hundred and twenty (120) days after the date the Securities were mailed to You (as reflected on Your statement), We will replace the Securities free of charge. If VBS does not receive notice within 120 days, it may charge You a fee to replace the Securities.

j. Liens and Levies

If a lien or levy on Your Account has been stayed or quashed, or the levying authority has otherwise released the levy or lien, it is Your responsibility to promptly provide VBS with appropriate documentation of the release. See **How to Contact VBS** for mailing instructions.

k. Complaints

If You have a complaint about the services VBS provides You, You should direct it to:

Complaints
Vanguard Brokerage Services
P.O. Box 982901
El Paso, TX 79998-2901

l. Closing an Account, Terminating Optional Features or Services

You may close Your Account, or terminate any optional features or services of the Account, by notifying VBS. The closure of Your Account or termination of a feature or service will be effective after VBS has received Your notice and had a reasonable period of time to act upon it. Even after You have notified Us that You wish to close Your Account or terminate an optional feature or service, You remain responsible for satisfying all obligations associated with the Account (for example, a Debit Balance) or with the feature or service. See also **Satisfying Legal Obligations by Releasing Account Information or Liquidating Securities and Other Property; Account Restrictions and Closing an Account**.

m. Trusted Contact

If You provide Us with a trusted contact, You authorize VBS to contact that person and disclose information about Your Account for emergency purposes such as addressing suspected financial exploitation or to confirm Your contact details, health status, legal Representative, or as otherwise permitted. Your trusted contact designation will apply to all Your Accounts unless You specify otherwise. You are responsible for keeping Your trusted contact information up to date. For Accounts held by non-natural persons (e.g., organization), Your Representative will serve as the trusted contact unless You specify otherwise. Without separate Account authorization, a trusted contact cannot view Your Account information nor can they transact on Your behalf.

8. VBS' Role

VBS will provide its clients with services relating to their Accounts, including, among others, order entry, online Account access, and tax and cost basis reporting. VBS is under no obligation to approve any Application, open any Account, take any trade, or effect any transaction.

VBS is required by federal law to obtain from each person who opens an Account certain personal information that will be used to verify that person's identity. If You do not provide this information, VBS will not be able to approve Your Application.

a. Identity and Credit Verification and Reporting

By submitting an Application to open a Vanguard Brokerage Account, You authorize VBS to verify the identity, creditworthiness, and other information for You, Your authorized Representative(s), and Your spouse, if You live in a Community Property state. To verify creditworthiness, We may, but are not obligated to, contact employers, obtain consumer and credit reports, and/or make other inquiries. VBS may refuse to open an Account or may decline to offer You certain features and services otherwise available for any lawful reason.

After Your Account has been opened, We may also obtain consumer and credit reports at any time for reasons including, but not limited to, the following:

- To collect a Debit Balance in Your Account;
- To investigate, detect, and help prevent fraud involving You or Your Account;
- To help VBS evaluate whether to grant, extend, or modify the terms and conditions of any credit You have applied for or received;
- If a deposit of Securities or Other Property to Your Account is returned.

We may report information about Your Account to credit bureaus, regulators, or law enforcement. If We report information to a credit bureau, items such as late payments, missed payments, or other defaults on Your Account may be reflected in Your credit report. On Your written request, VBS will inform You whether it has obtained credit reports and, if so, will provide You with the name and address of the reporting agency that furnished the reports.

If VBS approves Your Application and opens Your Account but later is not able to verify Your identity, it reserves the right to close Your Account or take other steps it deems reasonable.

VBS may require a minimum balance to open an Account. If VBS does require a minimum balance, You can submit the minimum amount needed by methods acceptable to VBS, including by sending a check, an EBT, a Federal funds wire, and/or a deposit of Marginable Securities.

b. Confirmations, Courtesy Trade Status Reports, Statements, and Other Account-Related Communications

VBS is required to send You written confirmation of the trades that are executed in Your Account. VBS may elect to send a monthly statement, in lieu of an immediate confirmation, for transactions executed pursuant to a periodic plan or an investment company plan, for transactions executed in shares of any open-end registered money market mutual fund, or for certain Fractional Share liquidations. VBS will send confirmations to You by USPS or e-delivery, depending on Your delivery elections. E-delivery is an efficient way to facilitate Account notifications.

VBS is not legally obligated to provide any trade status report other than a written confirmation, but it may choose to do so as a courtesy. VBS is not liable for any loss that You claim resulted from a delay or failure by VBS to issue a courtesy trade status report.

If, in any given month, there is Brokerage Activity in Your Account, VBS will send You a statement reflecting the activity at month-end. VBS will send You a statement after each calendar-year quarter-end, regardless of whether there was Brokerage Activity in Your Account. Information about all systematic or scheduled transactions, such as mutual fund dollar-cost-averaging purchases or reinvestment of dividends, will be included on Your statement for the periods during which the transactions occurred.

VBS may combine communications sent by USPS—such as statements, confirmations, and other written communications related to Account activity for Your Accounts—in a manner that reduces the number of envelopes mailed to You. Combined communications for Accounts that have more than one Account Owner may be, and generally will be, addressed to the name listed first in Your Account registration. VBS also may choose to address combined communications to any one Account Owner, or to all Account Owners.

c. Satisfying Legal Obligations by Releasing Account Information or Liquidating Securities and Other Property; Account Restrictions and Closing an Account

VBS is responsible for complying with its legal obligations, including those obligations that involve or relate to a client or his, her, or its Account. This means that VBS may release information about You and Your Account and the activity in it pursuant to a properly authorized request, or liquidate Securities and Other Property in Your Account to satisfy a court order, garnishment, tax levy, or other legal obligation imposed by a court, government agency, or office or self-regulatory body that has jurisdiction over VBS or You. VBS may also impose trading or disbursement restrictions on Your Account if it is required to do so, or if VBS knows or suspects that a violation of any applicable laws, regulations, orders, or similar restrictions may or could occur without the restriction. VBS will not be liable for trading losses, lost profits, tax obligations, or other damages You claim may or did result from any trading or disbursement restriction placed on Your Account. Nor will VBS be liable for any trading losses, lost profits, tax obligation, or other damages resulting from liquidations made in connection with a court order, tax levy, garnishment, or other legal proceeding. You agree We may debit Your Account a reasonable processing fee for each garnishment, subpoena, court order, tax levy, or other legal process on Your Account. See also **Trading or Disbursement Restrictions**.

VBS reserves the right to close Your Account, terminate any Account feature or service, or remove an agent from Your Account, at any time, for any reason, and without prior notice as permitted by law. Reasons for Account closure may include, but are not limited to, a zero or low balance Account with no Account activity; use of inappropriate or abusive statements or actions; not meeting Our Digital Interaction Expectations as stated in this Agreement; and use of algorithmic or software programs to facilitate Account opening or trading. Such closure or termination may include order rejections, cancellation of open orders, and liquidation of Your Account. Any action that VBS takes may affect any or all Account Owners even if not all Account Owners are aware of, or involved with, the activity that prompts VBS to take action. If VBS closes Your Account or terminates any Account feature or service, You agree that VBS is authorized to liquidate Securities and Other Property in Your Account and provide You with the proceeds of such transactions, net of any outstanding debits or obligations. VBS is not responsible for any losses or lost profits resulting from such liquidation. You remain responsible for any outstanding charges, debits, or other transactions pending or settling in Your Account. We may delay a final disbursement, if any, until all obligations You owe Us are satisfied.

d. Investment Advice

VBS will not act as Your investment advisor, and VBS does not have any Discretionary Trading Authority over Your Account or any obligation to review or make recommendations for the investment of Securities and Other Property in Your Account. VBS provides point-in-time recommendations for certain brokerage products, accounts and services, and relating to account transfers and rollovers. However, unless VBS affirmatively states that it is making a recommendation, it is not providing a recommendation. Further, VBS has no affirmative duty to monitor Your Account on an ongoing basis. Any research, analysis, news, or other information that may be made available to You is not a recommendation to You to buy, sell, or hold a particular Security or Other Property.

e. Tax Information, Including Withholding and Mandatory Cost Basis Reporting

VBS may, but is not required to, provide its clients with general tax information. The information VBS provides may not apply to Your specific circumstances and will not take into account Your comprehensive tax or estate planning situation. For information or advice specific to Your situation, You must consult with Your own tax or legal advisor.

VBS may be required to withhold taxes from certain distributions and may require that the amount of tax be available at the time the distribution is processed. It may also restrict the types of distributions available to certain clients; for example, clients who are nonresident aliens may not be permitted to take in-kind distributions.

VBS is required to report to You and the Internal Revenue Service ("IRS") cost basis and certain other information (jointly, "Cost Basis Information") for stock and investment company shares held in Your non-Retirement Accounts (the "Cost Basis Rules"). The IRS requires VBS to report Cost Basis Information for: (i) stock acquired on or after January 1, 2011; (ii) shares of a Regulated Investment Company ("RIC") or stock acquired through a Dividend Reinvestment Plan ("DRP") if the acquisition was on or after January 1, 2012; (iii) options and certain less complex debt instruments (such as an obligation with a fixed yield and maturity) acquired on or after January 1, 2014; and (iv) most other debt obligations acquired on or after January 1, 2016.

You may choose a preferred cost basis method for Your Securities. The options may vary based on the Account and Security. It is Your responsibility to choose the appropriate cost basis method for Your tax situation. For more information, visit vanguard.com.

Cost basis information reported to You and the IRS may differ in some cases from the information You are required to report on Your income tax return. You are responsible for correct tax reporting of Your Securities transactions.

VBS Cost Basis Accounting Service

While VBS may also track the cost basis of some or all of Securities or Other Property held in Your Account that are not subject to the Cost Basis Rules (for example, stock acquired before December 31, 2010, which You purchased through VBS or transferred in to VBS from another broker), it does so for information purposes only and does not report the information to the IRS.

The VBS Cost Basis Accounting Service may also be provided for Retirement Accounts for information purposes only.

Access to Cost Basis Information

You must be a registered user of vanguard.com to view cost basis information and information provided under the VBS Cost Basis Accounting Service.

f. Phone Access to VBS Associates; Recording Phone Calls and Monitoring Conversations

Although many account servicing needs can be accomplished online, VBS offers clients toll-free phone access to its associates that is generally available during regular business hours. However, at times when call volumes are high (such as during times of Market Volatility), You may have to wait to speak to an associate. At such times, VBS may route Your call to an automated answering system and/ or play recorded informational messages. During nonbusiness hours, or when VBS is upgrading or performing maintenance on Our systems, You may not be able to reach an associate. Note that most tasks, including but not limited to Account management (i.e., checking Account balances), transaction execution, and research can be accomplished using vanguard.com, Our mobile app, or Our automated answering system. On vanguard.com VBS provides FAQs, tutorials, and more to assist You. VBS frequently receives client inquiries by phone that are not likely (in VBS' judgment) to require immediate market access ("Non-Trade Inquiries"). VBS reserves the right to limit the communication channels in which it resolves Non-Trade Inquiries and may choose to resolve such inquiries primarily or exclusively on its website or other available channels, as opposed to providing such information by phone. VBS may also implement call routing strategies to support resolution of clients' Non-Trade Inquiries based on external factors (such as during times of Market Volatility), or client-specific factors, such as client assets under management, their trading history, and/or the nature and volume of their calls to VBS.

VBS may, but is not required to, record any or all incoming or outgoing phone calls, or monitor conversations Our associates may have relating to Your Account. We may record or monitor calls to gauge the quality of the service We provide, and/or to verify Securities transaction information. Not all phone calls are recorded, and not all conversations are monitored, and We do not guarantee that We will make or retain recordings of any particular phone call. Any recording that VBS does make is VBS' property, and VBS will decide, at its sole discretion, whether to, at Your request, play back any recording.

g. Authorized Access to an Account

VBS generally will take trades and other instructions only from the Account Owner, and his, her, or its Representative. VBS may require You to provide certain information, answer security questions, or use a number or password to access Your Account. Additional information on authority is noted below.

Agents

VBS may accept trading instructions from a limited or a full agent but will accept instructions to change the beneficial Owner of the assets in an Account only from an agent who has been granted that authority (generally, only full agents have such authority). VBS will recognize an agent's authority if it has appropriate documentation in its files. You may go online at vanguard.com or contact Us for additional information on agents and for a description of appropriate documentation (for example, appropriate documentation could include a letter of Account approval).

We reserve the right, but are not obligated, to place trading, disbursement, and other restrictions on an Account in the event We receive notice of a dispute among, or in Our opinion, conflicting instructions from, any of the multiple agents You may have appointed, or if VBS reasonably believes that the agent's authority has been revoked and that notice of the revocation is forthcoming.

Any notice VBS sends to an agent will be deemed to be notice to the Account Owner.

Fiduciary(ies)

If You, or a court, has appointed a fiduciary with authority to transact on Your Account, VBS may accept the fiduciary's instructions without responsibility for determining whether the fiduciary's action or inaction satisfies the standard of care applicable to such fiduciary's handling of the Account. If two or more fiduciaries have been authorized to make decisions on Your Account, VBS will accept and honor instructions received from one of the fiduciaries. Notwithstanding the foregoing, VBS may, at its discretion, require that instructions be signed or otherwise authorized by some or all of the fiduciaries named on Your Account before it takes any action requested by one or some of the fiduciaries. We reserve the right, but are not obligated, to place trading, disbursement, and other restrictions on an Account in the event We receive notice of a dispute among, or in Our opinion conflicting instructions from, jointly serving fiduciaries.

Any notice We send to one fiduciary will be deemed to be notice to all fiduciaries.

Joint Account Owners

VBS may accept trading instructions from any one Account Owner of a jointly owned Account or his, her, or its Representative. We may, however, require all Account Owners to authorize, in writing or otherwise, certain transactions.

We reserve the right, but are not obligated, to place trading, disbursement, and other restrictions on an Account in the event We receive notice of a dispute among, or in Our opinion conflicting instructions from, joint Account Owners.

Any notice VBS sends to one Account Owner will be deemed to be notice to all Account Owners.

Plan Sponsors and Vanguard Fiduciary Trust Company

If Your Account is held in a Vanguard Brokerage Option of an employer-sponsored Plan, VBS may accept transaction and other instructions from the Plan Sponsor, named fiduciary or investment manager of the Plan, or from the Plan trustee.

h. Market Volatility

During periods of Market Volatility, there may be disruptions, suspensions, or halts in trading in individual Securities, or at some or all exchanges or other venues that trade Securities. As a result, there may be delays in executing Your order, and there may be a significant difference in the quote You receive at the time You place Your trade and the execution price You receive. You may also receive partial executions of Your order at different prices. During periods of Market Volatility and at all other times, VBS may, in its sole discretion and without prior notice, limit the types of orders it accepts, or prevent Securities from being traded through VBS' services. If You attempt to place a trade during periods of Market Volatility, then You accept the risks that may result, and waive any claim to losses, including, without limitation, lost profits or damages resulting from the inability to place orders for impacted Securities.

i. Use of Online Trading Services and System Outages

"Online Trading Services" means VBS' wireless, internet computer services and systems, including, but not limited to, services accessible through proprietary software, websites, externally accessible computers, networks and any other wireless, internet, computer services for placing trades provided to You by VBS, whether established by VBS directly or through other service providers. VBS provides access to Online Trading Services "as is": it does not guarantee that the Online Trading Services will be available to You at a particular time, and reserves the right to suspend or terminate the Online Trading Services at any time without prior notice. Access to the Online Trading Services may be limited or unavailable during periods of peak demand, Market Volatility, systems upgrades or maintenance, or for other reasons, including reasons beyond VBS' control. If Online Trading Services are unavailable or delayed at any time, You agree to use alternative channels to place Your orders, such as calling Our phone associates. See also **Market Volatility**.

j. Correction of Errors

VBS reserves the right to correct any error relating to Your Account or any transaction, including by reversing, canceling, modifying, or adjusting transactions, balances, holdings, fees, or other Account records. Such corrections may be made with or without notice and within a reasonable period after We become aware of the error. You are not entitled to retain any benefit resulting from an error. VBS' right to correct an error is not limited by any delay in discovering the error or by any statement, confirmation, or other Account communication.

9. Contractual Terms

This section outlines the contractual terms of this Agreement. If You do not understand them, please contact Us or seek advice from Your attorney. This Agreement and its terms and conditions are binding, without limitation, on You and Your heirs, executors, administrators, successors, and assignees.

a. Governing Law

Disputes arising out of or in connection with Your Account or this Agreement will be decided and enforced in accordance with the laws of the State of New York (regardless of the choice of law rules thereof).

b. Assignment

VBS may assign its rights and obligations under this Agreement to its parent or to any of its subsidiaries, affiliates, or successors without notice to You. VBS may assign its rights and obligations under this Agreement to any other entity (one that is not a parent, subsidiary, affiliate, or successor) after giving You thirty (30) days' written notice. You may not assign Your rights and obligations under the Agreement without first obtaining VBS' written consent.

c. Liability

VBS is not responsible for determining whether a person is eligible to serve, or capable of serving, as Your Representative, or for transactions effected in an Account after a Representative's authority has been revoked but before VBS was provided written notice of the revocation and an opportunity to react to the notice. You agree to hold VBS, and any of its parents, subsidiaries, or affiliates, and any of their officers, directors, employees, and agents harmless from any liability, claim, or expense (including attorneys' fees) for the actions or inactions of Your Representative. Neither VBS nor any of its parents, affiliates, or subsidiaries, or any of their respective employees, officers, or directors (collectively, "VBS entities"), will be liable to You or to any third parties for any expenses, losses, damages, liabilities, demands, charges, and claims of any kind or nature whatsoever (including without limitation any legal expenses and costs or expenses relating to investigating or defending any demands, charges, and claims) (collectively, "Losses") by or with respect to the Account arising from Your or any of Your Representative(s)'s violation of this Agreement, state or federal securities laws or regulations, or any third-party's rights, including, but not limited to, infringement of any copyright, violation of any proprietary right and invasion of any privacy rights. The VBS entities will not be liable for any Losses or damages caused directly or indirectly by: government restrictions; exchange or market

rulings; Market Volatility or disruptions in order trading on any exchange or market; suspension of trading; war; strikes; severe weather; natural disasters; other acts of God; failure of computer or other electronic or mechanical equipment or communication lines; acts of terrorism; failure of third parties to follow instructions; or other conditions and occurrences beyond the VBS entities' control. Nothing in this provision, or any other provision within this Agreement, restricts or limits in any way Your ability to bring any claim in any forum, including in arbitration, against the above entities that You would otherwise be entitled to bring.

d. Severability

If any provision or condition of this Agreement shall be held to be invalid or unenforceable by any court or regulatory or self-regulatory agency or body, such invalidity or unenforceability shall attach only to such provision or condition. The validity of the remaining provisions and conditions shall not be affected thereby, and this Agreement shall be valid and enforceable as if any such invalid or unenforceable provision or condition were not contained herein.

e. Entire Agreement, Amendment, and Waiver

This Agreement incorporates into it any terms and conditions contained on the Brokerage Account Application as well as any terms and conditions contained on the Account statements and confirmations We send to You. This Agreement, with incorporations and as amended from time to time, contains the entire understanding between You and VBS concerning Your Account and Our provision of services relating to that Account.

Upon notice to You, VBS may modify or rescind existing provisions or add new provisions to the Agreement, and You agree to abide by the Agreement, as amended from time to time. VBS may waive its rights under this Agreement. Any waivers by VBS under this Agreement must be expressed in writing and signed by VBS. A waiver will apply only to the particular circumstance giving rise to the waiver and will not be considered a continuing waiver in other similar circumstances unless VBS states, in writing, that its intention is to make the waiver a continuing waiver. VBS' failure to insist on strict compliance with the Agreement is not a waiver of any of its rights under the Agreement.

f. Arbitration

This Agreement contains a predispute arbitration clause. Under this clause, which You agree to when signing Your Account Application, the parties to this Agreement agree as follows:

- **All parties to this Agreement are giving up the right to sue each other in court, including the right to a trial by jury, except as provided by the rules of the arbitration forum in which a claim is filed.**
- **Arbitration awards are generally final and binding; a party's ability to have a court reverse or modify an arbitration award is very limited.**
- **The ability of parties to obtain documents, witness statements, and other discovery is generally more limited in arbitration than in court proceedings.**
- **The arbitrators do not have to explain the reason(s) for their award unless, in an eligible case, a joint request for an explained decision has been submitted by all parties to the panel at least 20 days prior to the first scheduled hearing date.**
- **The panel of arbitrators may include a minority of arbitrators who were or are affiliated with the securities industry.**
- **The rules of some arbitration forums may impose time limits for bringing a claim in arbitration. In some cases, a claim that is ineligible for arbitration may be brought in court.**
- **The rules of the arbitration forum in which the claim is filed and any amendments thereto shall be incorporated into this Agreement.**

You agree to settle by arbitration any controversy between or among You and VBS arising out of or relating to VBS' business or this Agreement. Such arbitration will be conducted by, and in accordance with, the securities arbitration rules and regulations then in effect of FINRA or any national securities exchange that provides a forum for the arbitration of disputes, provided that VBS is a member of such national securities exchange at the time the arbitration is initiated. Your demand for arbitration shall be made within the time prescribed by those rules and will be subject to the applicable state or federal statutes of limitations as though filed in court. The award of the arbitrator will be final and binding, and judgment on the award may be entered in any court having jurisdiction. No person shall bring a putative or certified class action to arbitration, nor seek to enforce any predispute arbitration agreement against any person who has initiated in court a putative class action; or who is a member of a putative class who has not opted out of the class with respect to any claims encompassed by the putative class action until: (i) the class certification is denied; (ii) the class is decertified; or (iii) the customer is excluded from the class by the court. Such forbearance to enforce an agreement to arbitrate shall not constitute a waiver of any rights under this Agreement except to the extent stated herein.

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Vanguard

Vanguard Brokerage Services
P.O. Box 982901
El Paso, TX 79998-2901

Connect with Vanguard® > vanguard.com

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VBSAACP 102026

Vanguard

Business continuity planning and disaster recovery at Vanguard

Continuity planning

To help give our clients the best chance at investment success, we must plan carefully to ensure a quick, smooth recovery in the event of a disaster or emergency. To that end, Vanguard has invested significant resources and is committed company-wide—from Chief Executive Officer Salim Ramji to each employee—to ensuring that your assets are safe and that our service to you continues without interruption.

Our approach is simple. We think of our business as a chain. Each link—people, facilities, processes, infrastructure, and information—is an important component of our operation and must be accounted for and tested as part of an overall business continuity program.

To accomplish this, we established the Business Continuity Management (BCM) group within our Global Risk & Security Division. BCM is dedicated to leading and coordinating the planning, development, testing, and execution of a global program designed to ensure the recovery and restoration of critical operations following a disruption.

Our approach includes three major components: (1) detailed, written business continuity plans; (2) data security and technology recovery; and (3) business continuity tests.

Business continuity plans

- We have developed plans to respond to a range of incidents, from worst-case scenarios—such as loss of a data center, buildings, or staff—to occurrences such as a power outage or unprecedented phone volumes.
- We work with our business teams to refine plans for short-, medium-, and long-term pandemic, cybersecurity, and other crisis scenarios.
- We assess the design of mission-critical suppliers' and business partners' business continuity programs.
- We assist Vanguard's human resources, global security, environmental and occupational safety, and compliance teams in coordinating with appropriate law enforcement authorities, health and safety experts, and regulators.
- We work with Vanguard's communications teams and others to ensure employees receive training and necessary information about alternate business processes, external developments, and health and safety precautions.

Data security and recovery

- The security of our clients' accounts is extremely important to us. Accordingly, we have implemented a variety of sophisticated measures designed to protect client assets and account information.
- We ensure, through our information technology team, that systems are equipped with the appropriate hardware and software to secure client data. To mitigate computer virus attacks and other acts of cyberterrorism, we have implemented best-in-class controls monitored by a dedicated team of information security specialists.
- We maintain redundant systems and off-site data storage to enable recovery of data in the event of a disaster.

Business continuity tests

- We conduct continuity and recovery plan tests for all Vanguard business units throughout the year. These include communication (phone tree and emergency notification system) tests and alternate workspace exercises.
- We conduct workshops, tabletop exercises, and crisis response drills. These have included Vanguard-only exercises as well as collaborations with law enforcement to validate our ability to work together during an emergency.
- We assess response and recovery capabilities with key business partners, critical vendors, and other external organizations through periodic tabletop exercises.
- We conduct disaster recovery simulations with our information technology division to gauge readiness in the event of an emergency.

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Continuing to serve our clients

In addition to regular planning and testing, Vanguard uses several routine business strategies to ensure that our operations are both flexible and resilient.

To reduce the potential impact of global events (such as a pandemic) and regional incidents (such as a major power outage or winter storm), we maintain geographic diversification by running key business functions from multiple locations. Moreover, significant buildings have redundant or emergency power-supply capabilities. In addition, we are able to shift our operating model to manage business-as-usual operations remotely in most cases. This ensures that our most critical resources and employees are available to all clients during an emergency.

Complementing our multiple campus locations in the U.S., Vanguard offices outside the country offer additional recovery options. Our global locations can transfer trading between regions as necessary (which also facilitates follow-the-sun trading) and provide critical support functions such as IT incident management and cybersecurity operations. Should events warrant, our executive enterprise incident response and communications teams are prepared to lead our response.

Business disruption information

In the event of a significant business disruption, please visit our website at **vanguard.com** for additional information, including instructions to ensure prompt access to your assets.

Ascensus notice of business continuity plans

For clients of Vanguard Brokerage Services® and Ascensus College Savings

The following is a general description of plans in place at Vanguard, Vanguard Brokerage Services ("Vanguard Brokerage"), and Ascensus College Savings to maintain business continuity in the event that operations are disrupted for any reason.

At Ascensus*, we know how much our Plan participants rely on our systems and services. We also recognize that business disruptions of varying scope can and do occur. Ascensus wants you to know that we have plans in place designed to safeguard your assets and protect vital account information in the event of a business disruption. Ascensus and its affiliated companies have developed plans to recover from situations including, but not limited to, facilities failures that may cause business interruption. These plans are designed to account for interruptions of varying length and scope and require that Ascensus is able to recover mission-critical functions according to their time criticality. In order to maintain secure and effective plans, Ascensus does not provide the specific details in this notice, but you should be aware that Ascensus' corporate disaster recovery planning includes the following:

1. Identification of all mission critical systems and backup facilities and arrangements
2. Setup of alternate communication channels between Ascensus and its customer and business partners
3. Alternate physical site location and preparedness
4. An ongoing review of financial and operational risks
5. Employee safety strategies and communications

Ascensus mitigates risks to reduce the potential impact on customer service. In addition to following the guidelines stated above, Ascensus' recovery plans also include the following:

- **Designated Contingency Sites**–Ascensus maintains designated contingency sites to provide for the backup and recovery of data, maintenance of mission critical applications, and business resumption activities for key business personnel and critical business functions.
- **Notification to Clients**–Procedures for notifying Plan Participants have been established for Ascensus associates to follow in the event of an outage. Notification will include information regarding length of outage, instructions for contacting Ascensus, and support information (e.g., where to send faxes, and issues pertaining to data transmissions and communications).
- **Access to Your Funds**–A site outage should not impact your ability to access your investments, as Ascensus' business continuity plans are designed to facilitate sustained service. However, your ability to trade securities may be impacted by market events outside of Ascensus' control. Our business continuity plans are designed to ensure necessary personnel are available to assist with transactions.

Please note that Ascensus' business continuity plans are reviewed as necessary, and at least annually, to ensure they account for technology, business, and regulatory changes, operations, structure, or location. The plans are subject to change, and material changes will be reflected in an updated "Notice of Business Continuity Plans" that will be posted on our website, **https:// www2.ascensus.com/college-savings-plans-ascensus-inc/**. You may obtain a current written copy of this notice by contacting an Ascensus representative or by visiting our website at **https://www2.ascensus.com/college-savings-plans-ascensus-inc/**.

*Ascensus consists of Ascensus College Savings Recordkeeping Services, LLC, Ascensus Investment Advisors, LLC, and Ascensus Broker Dealer Services, LLC.

Important update for owners of IRAs held through Vanguard's legacy mutual fund investment platform

Effective April 17, 2026, your Disclosure Statement and Custodial Account Agreement are amended as follows:

Vanguard Traditional and Roth IRA Disclosure Statement - Introduction:

Important change

For IRAs held through Vanguard's legacy mutual fund investment platform, all contributions, exchanges, and transfers into these IRAs are no longer permitted as of April 17, 2026.

Vanguard Traditional and Roth IRA Custodial Account Agreement - Article VIII:

8.8 Account restrictions. Notwithstanding anything in the Agreement to the contrary, for owners of IRAs held through Vanguard's legacy mutual fund investment platform, until further notice, all contributions, exchanges, and incoming transfers will be prohibited in the Account.

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Vanguard®

The Vanguard Traditional IRA, SEP-IRA, and Roth IRA

Disclosure Statement and Custodial Account Agreement

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Vanguard Traditional and Roth IRA Disclosure Statement

Introduction

This Disclosure Statement describes the general requirements and features of both a traditional and a Roth IRA, as well as the specific features of the Vanguard Traditional and Roth IRA Custodial Account Agreement. This Disclosure Statement is provided in accordance with Internal Revenue Service (IRS) regulations. (Where the requirements for a traditional and a Roth IRA are the same, this Disclosure Statement refers to both types of accounts as an "IRA.")

Section I

Revocation

You may revoke your Vanguard IRA® at any time within seven days after it is established by mailing or delivering a written notice of revocation to Vanguard, P.O. Box 982901, El Paso, TX 79998-2901. Any notice of revocation will be deemed mailed on the date of postmark (or if sent by certified or registered mail, the date of certification or registration) if it is deposited in the U.S. Postal Service in an envelope or other appropriate wrapper, first-class postage prepaid, properly addressed. Upon revocation, you will be entitled to a full refund of your entire IRA contribution without adjustment for administrative expenses, sales commissions (if any), or fluctuations in market value. If you have any questions concerning your right of revocation, please call 800-662-2739 during normal business hours.

Section II

Establishment of your account

A. Statutory requirements

An IRA is a trust or custodial account established for the exclusive benefit of you and your beneficiaries. The Internal Revenue Code of 1986, as amended, provides for several types of IRAs, including a "traditional" IRA and a "Roth" IRA. You must clearly designate on the forms establishing your IRA that your account is either a traditional IRA or a Roth IRA. An IRA must be created by a written document that meets all of the following requirements:

1. **Bank trustee or custodian.** An IRA must be established with a qualified trustee or custodian, such as Vanguard Fiduciary Trust Company, which is a bank or other person approved by the IRS. You cannot be your own trustee or custodian.
2. **Cash contributions up to annual contribution limit.** All contributions to your IRA, excluding rollover or conversion contributions as described in Sections V and VI, must be made in cash. The total amount of contributions, other than rollover or conversion contributions, for any taxable year to your traditional and Roth IRAs may not exceed the contribution limit in effect for such taxable year as described in Section III[A].
3. **Nonforfeitability.** The balance of your IRA account must be fully vested and nonforfeitable at all times.
4. **Prohibitions against life insurance and commingling.** No part of your IRA assets may be invested in life insurance contracts, nor may your IRA assets be commingled with other property except in a common trust fund or common investment fund.
5. **Distribution rules.** Your IRA must comply with certain minimum distribution requirements, which are described in Section VIII. (No age 70½ distribution requirements apply for Roth IRAs.)

B. Tax consequences of traditional IRA

In general, the federal income tax consequences of establishing a traditional IRA are the following:

1. **Tax-deferred earnings.** Earnings and gains on your traditional IRA contributions will not be subject to federal income taxes until they are actually distributed.
2. **Deductible contributions.** You may be permitted to make contributions to your traditional IRA that are deductible for federal income tax purposes in an amount up to the lesser of the contribution limit in effect for such year or 100% of your current-year compensation. You are permitted to make deductible traditional IRA contributions if neither you nor your spouse is an active participant in an employer-maintained retirement plan, or if your adjusted gross income for the taxable year does not exceed certain dollar limits. To the extent that your traditional IRA contributions are not deductible, they may be treated as "nondeductible contributions" that must be reported on your federal income tax return. See Section III[D] for more information.
3. **Taxable distributions.** Distributions from your traditional IRA will generally be taxable as ordinary income in the year of receipt, with the exception that if you have made any nondeductible contributions or after-tax rollover contributions to your traditional IRA, part of your traditional IRA distributions may be treated as a nontaxable return of your nondeductible traditional IRA contributions or after-tax rollover contributions. Any distributions you receive from your traditional IRA prior to age 59½ may be subject to an additional 10% tax (although exceptions may apply—see Section VII[C]). You must start receiving certain minimum distributions from your traditional IRA beginning by April 1 of the year following the year in which you attain age 70½ (see Section VIII[B]).
4. **Tax-free rollovers.** You may be eligible to make a rollover contribution to your traditional IRA of cash or other assets you receive from another individual retirement plan or employer-maintained retirement plan. In addition, you may be eligible to roll over the taxable amount you withdraw from your traditional IRA to another individual retirement plan or an employer-maintained retirement plan. See Sections V and VI for more information.

5. **State taxes.** The state tax consequences of your traditional IRA will vary from state to state. You are strongly encouraged to consult a tax advisor to determine the state tax consequences of establishing a traditional IRA.

C. Tax consequences of a Roth IRA

In general, the federal income tax consequences of establishing a Roth IRA are the following:

1. **Tax-deferred earnings.** Earnings on contributions to a Roth IRA will accumulate on a tax-deferred basis and may ultimately be tax-free if the earnings are part of a "qualified distribution." (A "qualified distribution" is generally a distribution made to you after age 59½ and after you have held your Roth IRA account at least five years [see paragraph 3, below].)
2. **Nondeductible contributions.** Contributions to a Roth IRA are not deductible for federal income tax purposes.
3. **"Qualified" distributions are completely tax-free.** A distribution from a Roth IRA will be tax-free for federal income tax purposes as long as it is a "qualified distribution." A qualified distribution is a distribution from a Roth IRA: (1) made after a five-year holding period, and (2) made after age 59½, due to death or disability, or for the first $10,000 of "qualified first-time home purchase expenses." See Section VII[B] for more details.
4. **"Nonqualified" distributions are tax- and penalty-free return of contributions first; taxable earnings last.** Any distribution that is not a qualified distribution (for example, a distribution taken before you hold your Roth IRA for five years) is first considered a tax- and penalty-free distribution of your contributions to your Roth IRA. Once an amount equaling the cumulative contributions to your Roth IRA has been recovered tax-free, all further distributions that are not qualified distributions will be subject to both ordinary income tax and possibly an additional 10% penalty tax (if you are under age 59½). See Section VII[B] for more details.
5. **Rollovers.** You may be eligible to make a rollover contribution to your Roth IRA of cash or other assets you receive from another Roth IRA or from a designated Roth account in a 401(k) or 403(b) plan. In addition, you may be eligible to roll over the amount you withdraw from your Roth IRA to another Roth IRA. See Section V for more details.
6. **Conversions.** If you are not a married individual filing a separate tax return, you may roll over or "convert" a traditional IRA into a Roth IRA. Distributions from an eligible retirement plan can be rolled over (i.e., "converted") directly to a Roth IRA (subject to the same income limits as traditional to Roth IRA conversions).
7. **State taxes.** The state tax consequences of your Roth IRA will vary from state to state. You are strongly encouraged to consult a tax advisor to determine the state tax consequences of establishing a Roth IRA.

Section III

Contributions

A. Amount and timing of traditional and Roth IRA contributions

Maximum annual contributions to all IRAs. The total amount of contributions to all of your IRAs (both traditional and Roth IRAs) for any taxable year (excluding any rollover or conversion contributions as described in Sections V and VI) may not exceed the lesser of the contribution limit in effect for such taxable year as described below or 100% of your compensation for the taxable year. If you reach age 50 before the close of the tax year for which you are making a contribution, your annual contribution limit is increased by $1,000 as described below. In addition, the maximum contribution permitted under a Roth IRA is phased out to $0 for individuals earning above a certain level of adjusted gross income (see Section III[C]).

The annual IRA contribution limits for 2018 are shown below.

Maximum annual contribution Individuals under age 50	Maximum annual contribution Individuals age 50 or older
$5,500	$6,500

The contribution limit will be periodically adjusted for cost-of-living increases in $500 increments.

Definition of compensation. For purposes of the IRA contribution limits, your compensation includes all wages, salaries, tips, professional fees, bonuses, and other amounts you receive for providing personal services, and any earned income from self-employment. It does not include earnings and profits from property such as dividends, interest, or capital gains, or amounts received as a pension or annuity, or as deferred compensation. Your compensation includes any taxable alimony or separate maintenance payments you may receive under a decree of divorce or separate maintenance. Compensation includes nontaxable combat pay received by a member of the U.S. Armed Services.

Repayment of qualified reservist distribution. If you receive a qualified reservist distribution from an IRA, 401(k), or 403(b) plan, you may repay the amount of the distribution at any time during the two-year period beginning on the day after the end of your active duty period or by August 17, 2008, if later. Such repayment contributions are not subject to the annual IRA contribution limits described above. No deduction is permitted for repayment contributions.

IRA for your spouse. If both you and your spouse earn compensation for a taxable year, you may each make contributions to a traditional or Roth IRA up to the lesser of the contribution limit in effect for such taxable year or 100% of your compensation for the taxable year, although your Roth IRA contribution limit may be phased out based on your adjusted gross income for the year (see Section III[C]). In addition, under a special rule for spousal IRAs (explained in Section III[E]), contributions of up to the contribution limit in effect for such taxable year may be made to either a traditional IRA or, subject to the adjusted gross income phase-out discussed in Section III[C], a Roth IRA of a spouse, regardless of the income level of the spouse,

provided the married couple files a joint tax return and has total compensation at least equal to their combined IRA contributions.

Contributions in cash. All contributions to your traditional or Roth IRA (other than rollover contributions as described in Section V) must be made in cash, check, or by electronic transfer. If you wish to use shares of a previously established Vanguard fund account for your annual IRA contribution, you must first redeem the amount of shares you wish to invest, and then use the cash proceeds as your IRA contribution.

Contributions up to the date your return is due (April 15). You may make contributions to your traditional or Roth IRA for a taxable year at any time during the year, either periodically or in a lump sum, or in the next year, up to the due date for filing your federal income tax return for the taxable year, not including extensions. For taxpayers who file on a calendar-year basis, the latest date for any year is April 15 of the following year. If you do not inform the Custodian of the year for which an IRA contribution is made, the Custodian will assume that the contribution is made for the year in which it is received.

Only Roth IRA contributions permitted after age 70½. If you are otherwise eligible, you are permitted to make Roth IRA contributions even after you have attained age 70½. You are not permitted to make traditional IRA contributions (either deductible or nondeductible) for the year you attain age 70½ and any subsequent years (other than rollover contributions as described in Section V).

Maximum contributions not required. You do not have to contribute to your traditional or Roth IRA every year, nor are you required to make the maximum contribution for any year. However, if you decide in any year not to make the maximum IRA contribution, you may not make up the missed contribution amount in later years. For both the Vanguard traditional and Roth IRAs, there is a minimum initial contribution required when you establish your account, as described in Section XII[D].

B. Traditional IRA: Deductible contributions

Contributions to your traditional IRA may be deductible in whole or in part for federal income tax purposes, as determined by the rules summarized below. Contributions to a Roth IRA are not deductible for federal income tax purposes. You should contact your tax advisor to determine the deductibility of IRA contributions for state income tax purposes.

Fully deductible contributions if you do not participate in an employer plan. If neither you nor your spouse is an active participant in an employer-maintained retirement plan, your annual traditional IRA contributions up to the lesser of the contribution limit in effect for the taxable year (as described in Section III[A]) or 100% of current-year compensation are generally fully deductible (regardless of your level of adjusted gross income).

Phase-out of deduction if you participate in an employer plan. If you are an active participant in an employer-maintained retirement plan for a taxable year, your traditional IRA deduction is phased out as your adjusted gross income approaches the upper limits of the applicable "phase-out range." The phase-out ranges for joint and single filers are as follows:

Year	Joint filers Phase-out range	Single filers Phase-out range
2018	$101,000–$121,000	$63,000–$73,000

The dollar limits above will be periodically adjusted for cost-of-living increases in $1,000 increments.

Formula for deduction phase-out for active participants. As stated on the previous page, if you are an active participant in an employer-maintained retirement plan for a taxable year, your traditional IRA deduction limit for the year is phased out as your adjusted gross income exceeds the applicable adjusted gross income threshold for the tax year. This phase-out is accomplished as follows: Your maximum traditional IRA deduction is reduced by an amount, rounded down to the nearest $10, that bears the same ratio to the maximum deductible amount as your "excess" adjusted gross income for the taxable year bears to $10,000 for single filers or $20,000 for joint filers.

Deduction limit for spouse. If you are married and file a joint return, and neither you nor your spouse is an active participant in an employer-maintained retirement plan, the traditional IRA contributions for each spouse for the taxable year will be fully deductible regardless of the level of your combined adjusted gross income.

If you are married and file a joint return and both you and your spouse are active participants in employer-maintained retirement plans, the traditional IRA deduction limit for each spouse for the taxable year is phased out as your combined adjusted gross income exceeds the applicable threshold (see the joint filer phase-out range above).

If you are married and file a joint return and you are an active participant in an employer-maintained retirement plan but your spouse is not, a higher adjusted gross income phase-out will apply for your spouse's (but not your) deduction limit. The traditional IRA deduction limit for your spouse, who is not an active participant, is phased out if your combined adjusted gross income for the taxable year falls between $150,000 and $160,000, as adjusted for cost-of-living increases (i.e., $189,000–$199,000 for 2018). The deduction limit for you—the active participant in an employer-maintained plan—is phased out as your combined adjusted gross income for the tax year exceeds the applicable threshold (see the joint filer phase-out range above).

Special rules for married couples filing separate returns. If you are a married individual filing a separate return and you or your spouse is an active participant in an employer-maintained retirement plan, the deduction for your traditional IRA contributions will be phased out as your adjusted gross income for the taxable year increases from $0 to $10,000. However, if you and your spouse did not live

together at any time during the taxable year, your filing status will be considered to be single for traditional IRA deduction purposes, and your traditional IRA deduction limit will be determined without regard to whether your spouse is an active participant in an employer retirement plan.

Active participant defined. For purposes of the traditional IRA deduction limits, you are considered an active participant in an employer-maintained retirement plan for a taxable year if you participate in a qualified pension, profit-sharing, stock bonus, or annuity plan (including a Keogh or 401(k) plan), a tax-sheltered annuity plan under Section 403(b) of the Internal Revenue Code, a simplified employee pension (SEP) plan, or a government plan (but not an eligible deferred compensation plan under Section 457(b) of the Code) during any part of the plan year ending with or within the taxable year. The determination of whether you are an active participant in an employer-maintained retirement plan is made without regard to whether your rights under the plan are nonforfeitable or vested. Under a defined contribution plan, you are generally considered an active participant if any employer contribution or forfeiture is allocated to your account during the taxable year. Under a defined benefit plan, you are considered an active participant if you are not excluded by the plan's eligibility requirements during any part of the plan year ending with or within your taxable year. However, if you have not satisfied the plan's minimum age or service conditions required for participation, you are not considered an active participant in the plan. The Form W-2 you receive from your employer each year should indicate whether you are an active participant in an employer-maintained retirement plan.

Adjusted gross income. For purposes of the traditional IRA deduction limits, your adjusted gross income is calculated by taking into account any taxable Social Security benefits and taxable IRA distributions you may receive for the year. However, your adjusted gross income is not reduced by any deductible traditional IRA contributions you may make for the taxable year, employer-provided adoption assistance that is otherwise not taxable, or proceeds of U.S. Savings Bonds used for higher-education expenses that are otherwise not taxable.

Minimum deduction limit of $200. There is a special rule providing that if your adjusted gross income for any taxable year is within the "phase-out range," your traditional IRA deduction limit is never less than $200.

C. Roth IRA: Maximum annual contribution

The maximum contribution permitted under a Roth IRA (excluding any rollover or conversion contributions as described in Sections V and VI) is generally the lesser of the contribution limit in effect for the taxable year as described in Section III[A] or 100% of your compensation for the taxable year (less any amount you also contribute to a traditional IRA). Furthermore, your Roth IRA contribution maximum is phased out as your adjusted gross income approaches the upper limits of the applicable "phase-out range." The applicable phase-out ranges are as follows:

- If you are single—your phase-out range is adjusted gross income of between $120,000 and $135,000 for 2018.
- If you are married filing jointly—your phase-out range is adjusted gross income of between $189,000 and $199,000 for 2018.
- If you are married filing separately—your phase-out range is adjusted gross income of between $0 and $10,000. However, if you and your spouse did not live together at any time during the taxable year, your filing status will be considered to be single for Roth IRA contribution purposes.

The dollar limits above for single filers and married filing jointly will be periodically adjusted for cost-of-living increases in $1,000 increments.

Formula for contribution phase-out. As your adjusted gross income approaches the upper limit of the phase-out ranges described above, your Roth IRA contribution limit for the year is phased out to $0. This phase-out is accomplished as follows: Your maximum Roth IRA contribution is reduced by an amount, rounded down to the nearest $10, that bears the same ratio to the maximum IRA contribution limit in effect for the taxable year as your "excess" adjusted gross income for the taxable year bears to $15,000 ($10,000 in the case of a joint return or a married individual filing a separate return).

Adjusted gross income (AGI) for Roth IRA contribution phase-out purposes. For purposes of the Roth IRA contribution phase-out, your adjusted gross income is determined as it is for determining the traditional IRA deduction limit for active participants (see Section III[B]), except that your adjusted gross income is reduced by (1) any deductible traditional IRA contributions you may make for the taxable year, and (2) any amount included in gross income as a result of a traditional IRA to Roth IRA "conversion" discussed in Section VI.

Minimum contribution limit of $200. There is a special rule providing that if your adjusted gross income for Roth IRA contribution phase-out purposes is within the "phase-out range," your Roth IRA contribution limit is never less than $200.

D. Traditional IRA: Nondeductible contributions

You may wish to make "nondeductible contributions" to your traditional IRA to the extent that you are not eligible to make either deductible traditional IRA contributions or Roth IRA contributions (or if you do not wish to make deductible traditional IRA contributions). Remember, however, that the total amount of deductible and nondeductible contributions to all of your traditional and Roth IRAs for any taxable year may not exceed the lesser of the contribution limit in effect for the taxable year or 100% of your compensation for the year.

Tax advantage of nondeductible contributions. The primary tax benefit associated with nondeductible contributions to a traditional IRA is that the earnings and gains on these contributions will not be subject to federal income tax until they are actually distributed to you.

Election to treat deductible contributions as nondeductible contributions. You are permitted to elect to treat your traditional IRA contributions, which would otherwise be deductible for any taxable year, as nondeductible contributions. You may wish to make such an election, for example, if you have no taxable income for the year after taking into account other deductions or tax credits, and you are not eligible for a Roth IRA contribution.

Designation of nondeductible contributions. Your designation of traditional IRA contributions as nondeductible contributions for a taxable year is to be made by filing Form 8606, Nondeductible IRAs, with your federal income tax return for the taxable year. Nondeductible traditional IRA contributions for a taxable year may be made at any time during the taxable year, or in the next year, up to the due date for filing your federal income tax return for the taxable year, not including extensions. If you file an amended return, you may change your designation of your traditional IRA contributions from deductible contributions to nondeductible contributions or vice versa (although such a change may result in an increased or different tax liability).

E. Spousal IRAs

If your spouse has little or no income, your spouse may still be eligible to establish and contribute to an IRA under a special rule for spousal IRAs. To qualify for this rule, you and your spouse must file a joint return for the taxable year.

IRA contributions up to limit in effect. Under the special rule for spousal IRAs, contributions may be made to your spouse's traditional or Roth IRA for any taxable year in an amount up to the lesser of: (1) the contribution limit in effect for the taxable year, or (2) the combined compensation of you and your spouse for the taxable year, less the amount of any contributions you made to your IRA for the taxable year.

Deductibility of contributions for spouse. The deductibility of contributions to a traditional IRA for a spouse is generally determined by the rules discussed in Section III[B]. Thus, for example, if neither you nor your spouse is an active participant in an employer-maintained retirement plan, contributions to a spousal traditional IRA are fully deductible. However, if your spouse is an active participant in an employer-maintained retirement plan, your spouse's IRA deduction limit for the taxable year is phased out as your combined adjusted gross income for the tax year exceeds the applicable threshold provided in the chart in Section III[B].

If you participate in an employer-maintained retirement plan and your spouse does not, the spousal deduction limit is reduced by an amount that bears the same ratio to the contribution limit in effect for the taxable year as your combined adjusted gross income over $150,000, as adjusted for cost-of-living increases (i.e., $189,000 for 2018), bears to $10,000. Thus, for example, the deduction for contributions to a spousal traditional IRA for 2018 would be eliminated if your combined adjusted gross income exceeds $199,000, although nondeductible traditional IRA contributions could still be made.

Contributions after age 70½ to traditional IRAs. You may make contributions to a traditional IRA for your spouse even after you reach age 70½ provided your spouse has not yet attained age 70½. However, you may not make contributions to a traditional IRA for your spouse (other than rollover contributions described in Section V) for the year your spouse reaches age 70½ or any subsequent year. Distributions from the account do not have to begin until April 1 of the calendar year following the calendar year in which the spouse for whom the account is maintained reaches age 70½. With the exception of the contribution limitations, all rules that apply to a traditional IRA generally apply to the traditional IRA for a spouse.

Roth IRA contributions for spouse. The amount of permitted contributions to a spousal Roth IRA is generally determined by the rules discussed in Section III[C]. Thus, if you are married filing a joint tax return, the spousal Roth IRA contribution limit in effect for the taxable year is reduced by an amount that bears the same ratio to the contribution limit in effect as your combined adjusted gross income over $150,000, as adjusted for cost-of-living increases (i.e., $189,000 for 2018), bears to $10,000. For example, the contributions to a spousal Roth IRA for 2018 would be eliminated if your combined adjusted gross income exceeds $199,000, although nondeductible traditional IRA contributions could still be made. Keep in mind that Roth IRA contributions on behalf of a spouse may be made regardless of whether you or your spouse is over age 70½.

F. Return of IRA contributions

Withdrawal of IRA contributions by the date your return is due. If you make a contribution to your IRA for a taxable year, you may withdraw the contribution amount and the earnings thereon at any time prior to the due date for filing your federal income tax return, including extensions, for the taxable year for which the contribution was made, or such later date as prescribed by the IRS. If the excess (and any earnings) is removed prior to the applicable deadline, the return of the contribution will not be includible in your gross income as an IRA distribution. However, the earnings on the contribution will be taxable income in the year in which the contribution was made and may possibly be subject to the 10% tax on early distributions if you are under age 59½ (see Section VII[C]).

G. Recharacterization of IRA contributions

You may elect to recharacterize all or part of a regular contribution (i.e., treat a contribution made to one IRA (the "First IRA") as made to a different type of IRA (the "Second IRA"). You may wish to recharacterize a

contribution because, for example, you realize you were not eligible to make the IRA contribution or for tax-planning reasons. For example, you may wish to recharacterize a contribution to a Roth IRA into a contribution to a traditional IRA because you decide that you want the tax deduction available for a traditional IRA contribution.

A recharacterization is accomplished by requesting, in a form and manner acceptable to the Custodian, a trustee-to-trustee transfer of the contribution made to the First IRA, adjusted for gains and losses, to the Second IRA. The recharacterized amount is treated as having been originally contributed to the Second IRA on the same date and (in the case of a regular contribution) for the same taxable year that the contribution was made to the First IRA. The recharacterization is permissible only if the contribution could have been made originally to the Second IRA. The recharacterization must be accomplished by the due date (including extensions) of your federal income tax return for the year for which the original contribution was made, or such later date as prescribed by the IRS.

Roth IRA conversions made after December 31, 2017, cannot be recharacterized.

H. Excess contributions to traditional or Roth IRA

Generally, an excess contribution is the amount of any contributions to your traditional or Roth IRA (other than a proper rollover or conversion contribution as described in Sections V and VI) for a taxable year that exceeds your IRA contribution limit for the taxable year. An excise tax equal to 6% of the amount of any excess contribution will be assessed for the year for which the excess contribution is made and for each subsequent year until the excess amount is eliminated.

Return of excess contribution by the date your return is due. If you make an excess contribution to your IRA for a taxable year, you may withdraw the contribution and the earnings thereon prior to the due date for filing your federal income tax return, including extensions, or such later date as may be prescribed by the IRS. If this withdrawal is made by the applicable deadline, the return of the contribution will not be subject to the 6% excise tax on excess contributions (assuming the contribution is not deducted on your return).

Return of excess contribution after tax return due date. If you make an excess contribution to your IRA for a taxable year and you withdraw the excess contribution after the due date for filing your federal income tax return (including extensions), the returned excess contribution will not be includible in your gross income as an IRA distribution (subject to possible premature distribution penalties) if (1) your total IRA contributions for the year did not exceed the contribution limit in effect for the taxable year, and (2) you did not deduct the excess contribution on your return (or if the deduction you claimed was disallowed by the IRS). However, you must pay the 6% excise tax on the excess contribution for each taxable year that it was still in your IRA at the end of the year. Under this procedure, you are not required to withdraw any earnings attributable to the excess contribution.

Applying excess contribution to subsequent year. You may also eliminate an excess contribution from your IRA in a subsequent year by not contributing the maximum amount for that year and applying the excess contribution to the subsequent year's contribution. You may be entitled to a deduction for the amount of the excess contribution that is applied in the subsequent year provided you did not previously deduct the excess contribution (or if the deduction you claimed was disallowed by the IRS). However, if you incorrectly deducted an excess contribution in a closed taxable year (i.e., one for which the period to assess a deficiency has expired), the amount of the excess contribution cannot be deducted again in the subsequent year in which it is applied.

Section IV

Transfers

This section discusses options for carrying out a trustee-to-trustee asset transfer of your existing IRA to another IRA of the same type (e.g., a transfer from traditional IRA to traditional IRA). For a discussion of recharacterizations of contributions to a different type of IRA, rollover options, and the option to "convert" your traditional IRA to a Roth IRA, see Sections III[G], V, and VI, respectively.

A. Tax-free transfer from existing IRA to Vanguard IRA

In order to give you greater investment flexibility, you are permitted to transfer IRA assets directly from one trustee or custodian to another on a tax-free basis. If you already have an IRA with another trustee or custodian, you may authorize a direct transfer of your IRA assets to a Vanguard IRA without paying taxes, subject to the rules and restrictions of your existing account. Transfers are only permitted between the same type of IRA plans (i.e., from a traditional IRA to traditional IRA or from a Roth IRA to Roth IRA). You may make such a transfer as often as you wish. Such a transfer of assets is not tax-deductible.

B. Tax-free transfer from Vanguard IRA

If you so direct in a form and manner acceptable to the Custodian, the Custodian will transfer all or any portion of the assets held in your Vanguard IRA directly to the trustee or custodian of another IRA established on your behalf, provided the trustee or custodian certifies that it will accept the direct transfer of assets and will deposit the transferred assets in the same type of IRA established on your behalf. Transfers are only permitted between the same type of IRA plans (i.e., from a traditional IRA to traditional IRA or from a Roth IRA to Roth IRA).

C. Transfer incident to divorce

All or any portion of your IRA assets may be transferred to a separate IRA for the benefit of your former spouse pursuant to a divorce decree or written instrument incident to divorce. The transfer will not result in a taxable increase for you or your former spouse. After the transfer, your former spouse will be considered the Owner of the IRA mentioned for his or her benefit.

Section V

Rollover contribution

A. Rollover from existing IRA to a Vanguard IRA

If you receive a distribution of assets from an existing IRA, you may make a rollover contribution of all or part of the assets you receive tax-free to a Vanguard IRA. Except in the case of a "conversion" from a traditional IRA to a Roth IRA, rollovers are only permitted between the same type of IRA plans (i.e., from a traditional IRA to traditional IRA or from a Roth IRA to Roth IRA). The rollover must be completed within 60 days after you receive the distribution from your existing IRA. You are limited to one such tax-free rollover within a one-year period (beginning on the date you receive the IRA distribution, not on the date you make the rollover contribution). You cannot make a tax-free rollover if you have already made a rollover from any of your IRAs in the preceding one-year period. IRA conversions are not taken into account for purposes of this limit. You may not roll over any minimum distribution amounts you are required to receive from your traditional IRA upon attaining age 70½ (see Section VIII[B]).

Note: A tax-free transfer of funds as described in Section IV[A] is not a rollover (since you do not actually receive any distribution from your IRA). Rather it is a direct transfer of your IRA funds from one trustee or custodian to another that is not affected by the 12-month waiting period applicable to IRA rollovers.

If your distribution consists of property other than cash, you must roll over to your new IRA the same property you received from your old IRA. If you wish to make a rollover contribution of property other than cash to your Vanguard IRA, you must obtain the prior approval of the Custodian.

B. Rollover from employer retirement plan to Vanguard traditional IRA

Eligible rollover distribution from employer's retirement plan to traditional IRA. If you receive an "eligible rollover distribution" from an "eligible employer plan," other than a designated Roth account, you may make a tax-free rollover contribution of the distribution to a Vanguard traditional IRA. An "eligible employer plan" includes a plan qualified under section 401(a) of the Code (including a 401(k), pension, profit-sharing, defined benefit, or stock bonus plan), a governmental 457 plan, a 403(a) annuity plan, or a 403(b) plan. An "eligible rollover distribution" generally includes any distribution or withdrawal from an eligible employer plan other than:

(1) A distribution that is part of a series of periodic payments over a specified period of ten years or more, or over your life (or life expectancy), or over the joint lives (or joint life and last survivor expectancy) of you and your designated beneficiary;

(2) Any portion of a distribution that represents a required minimum distribution to you after age 70½; or

(3) A hardship distribution.

After-tax contributions. If a portion of your eligible rollover distribution from an employer's plan represents a return of after-tax contributions, you may roll over your after-tax contributions to a traditional IRA. If you elect to roll over after-tax contributions, it is your responsibility to keep track of these after-tax rollover amounts and to report such rollover amounts in accordance with IRS guidelines. Once you roll after-tax amounts into a traditional IRA, those amounts cannot later be rolled into an employer plan.

Direct rollover option. If you will be entitled to receive an eligible rollover distribution from an eligible employer plan, you may elect to have the plan roll over all or any part of your distribution directly to your Vanguard traditional IRA as a tax-free rollover contribution on your behalf. If you elect this direct rollover option, no federal income taxes will be withheld from your distribution to the extent it is transferred directly to your Vanguard traditional IRA. The plan administrator of your employer's plan must give you an explanation of your direct rollover option and the other tax rules affecting your eligible rollover distribution.

Payment option; 20% withholding. If you elect to have an eligible rollover distribution from an employer's qualified retirement plan paid directly to you, your distribution will be subject to 20% federal income tax withholding. You will still have the option of making a tax-free rollover contribution of your eligible rollover distribution to a Vanguard traditional IRA within 60 days of receipt of your distribution. You may roll over any amount up to 100% of your eligible rollover distribution (including an amount equal to the 20% that was withheld by coming up with additional money to make up for the withheld amount).

Rollover of property. If you receive an eligible rollover distribution from an employer retirement plan that consists of property other than cash, you may be permitted to roll over the property received to your traditional IRA. As an alternative, you may sell the property received and roll over the cash proceeds to your traditional IRA, in which case no gain or loss will be recognized on the sale if the entire proceeds are rolled over. If you wish to make an in-kind rollover contribution of property other than cash to your Vanguard traditional IRA, you must obtain the prior approval of the Custodian.

Rollover by surviving spouse. A surviving spouse of a deceased employee may be permitted to make a tax-free rollover contribution to a traditional IRA of all or any portion of an eligible rollover distribution from an employer retirement plan upon the employee's death.

Rollover by nonspouse beneficiary. A nonspouse designated beneficiary of a deceased employee may be permitted to make a direct rollover to an inherited IRA of an eligible rollover distribution upon the employee's death (if permitted under the deceased employee's plan).

Rollover pursuant to divorce or similar proceedings. If you are eligible to receive an eligible rollover distribution from an employer's qualified retirement plan pursuant to a "qualified domestic relations order" (within the meaning of Section 414(p) of the Internal Revenue Code) resulting from divorce or similar proceedings, you may have all or part of the distribution rolled over to your traditional IRA on a tax-free basis.

Separate traditional IRA account for certain rollovers from qualified plans. If you were born before January 1, 1936, and you receive an eligible rollover distribution from an employer's qualified retirement plan, you may prefer to roll over the distribution into a separate traditional IRA (frequently called a "rollover" or "conduit" IRA) to which no annual IRA contributions are made. In this manner, if you later roll over the assets to a new employer's qualified retirement plan you preserve any special tax treatment, such as ten-year averaging, that may be available on lump-sum distributions from the qualified plan.

C. Rollover from a designated Roth account under an employer plan to a Vanguard Roth IRA

Eligible rollover distribution from designated Roth account. If you receive an eligible rollover distribution from a designated Roth account under a 401(k) or 403(b) plan, you may roll over all or part of the distribution to a Vanguard Roth IRA. You cannot subsequently roll over from a Roth IRA back to a designated Roth account under an employer plan.

Five-year holding period. The holding period does not carry over from the designated Roth account to the Roth IRA. Assets that are rolled over from a designated Roth account to a Roth are subject to the five-year holding period applicable to the Roth IRA for purposes of determining a qualified distribution. (See Section VII[B] for more information on the five-year holding period.)

Roth IRA ordering rules. For purposes of the Roth IRA ordering rules described in Section VII[B], the entire amount of any qualified distribution rolled over from a designated Roth account is treated as contributions to the Roth IRA. In the case of a nonqualified distribution from a designated Roth account, the portion of the rollover that represents Roth basis is treated as contributions to the Roth IRA and any remaining amount is treated as earnings. If only a portion of a nonqualified distribution from a designated Roth account is rolled over, the amount rolled over is treated as consisting first of the portion of the distribution that represents earnings.

D. Rollovers from a Vanguard IRA

If you withdraw assets from your Vanguard IRA, you may roll over on a tax-free basis all or any portion of the assets you receive within 60 days of receipt to another IRA of the same type. You are limited to one such tax-free rollover within a one-year period regardless of the number of IRAs you own. You cannot make a tax-free rollover from one IRA to another if you have already made a rollover from any of your IRAs in the preceding one-year period. You may also roll over on a tax-free basis all or part of the taxable portion of the assets you receive from your Vanguard traditional IRA within 60 days of receipt to an eligible employer plan that accepts such rollovers. If any portion of a distribution from your traditional IRA represents a return of nondeductible contributions or after-tax rollover contributions, such portion cannot be rolled over into an eligible employer plan. You cannot roll over any minimum distribution amount you are required to receive from a Vanguard traditional IRA (see Section VIII[B]). Distributions from a Roth IRA are not permitted to be rolled over into a designated Roth account under an employer plan. You should seek competent tax advice concerning IRA rollovers to employer retirement plans.

Section VI

Conversions to a Roth IRA

A. Traditional IRA to Roth IRA conversion: General Rules

If you are age 70½ or older, you must satisfy your required minimum distribution for the tax year prior to making a conversion contribution for such year. The conversion will be treated as a taxable distribution from your traditional IRA and a subsequent conversion contribution to a Roth IRA. Distributions from your traditional IRA will be includible in your gross income in the year of the distribution (except for the portion of the distribution that represents a tax-free return of your nondeductible contributions or after-tax rollover contributions) but will not be subject to the 10% additional tax on early distributions, regardless of whether you are under age 59½. If you make a Roth conversion after December 31, 2017, the conversion cannot be recharacterized. (See Section VI[D] for more information.)

B. Conversions of SEPs and SIMPLE accounts

Accounts held in a SEP-IRA or a SIMPLE IRA may be converted to a Roth IRA. However, a conversion of a SIMPLE IRA is permissible only after the expiration of the initial two-year holding period. (See your SIMPLE IRA materials for more details.) If you make a Roth conversion after December 31, 2017, the conversion cannot be recharacterized. (See Section VI[D] for more information.)

C. Employer plan to a Roth IRA

You can roll over (i.e., "convert") a distribution from an eligible employer plan (other than a designated Roth account) directly to a Roth IRA. If applicable, you must satisfy your required minimum distribution for the tax year prior to rolling a distribution from an eligible employer plan directly to a Roth IRA. In general, such rollover will be taxable in the year of the distribution (except for any portion of the distribution that represents a return of after-tax contributions) but will not be subject to the 10% additional tax on early distributions, regardless of whether you are under age 59½. If you make a Roth conversion after December 31, 2017, the conversion cannot be recharacterized. (See Section VI[D] for more information.) Please consult a tax advisor or refer to IRS Publication 590-A for more information.

D. No recharacterization of conversions

A conversion of a traditional, SEP, or SIMPLE IRA to a Roth IRA, and a rollover from any other eligible retirement plan to a Roth IRA, made in tax years beginning after December 31, 2017, cannot be recharacterized as having been made to a traditional IRA.

Section VII

Taxation of distributions

A. Traditional IRA: Tax treatment of distributions

In general, distributions from your traditional IRA are included in your gross income in the year of receipt. Exceptions to this rule include any distribution that is properly rolled over to another individual retirement arrangement or employer retirement plan as described in Section V, or any return or transfer of an IRA contribution as described in Section III[F], [G], and [H] and Section IV. In addition, distributions from a traditional IRA that contains "nondeductible contributions" (see Section III[D]) or after-tax rollover contributions (see Section V[C]) are treated partly as a nontaxable return of the nondeductible traditional IRA contributions or after-tax rollover contributions, and partly as a taxable distribution of traditional IRA earnings and any deductible IRA contributions, as explained below.

Ordinary income taxation. Distributions from your traditional IRA that are included in gross income will be taxed as ordinary income. Traditional IRA distributions are not eligible for the special tax treatment accorded to certain lump-sum distributions from qualified retirement plans, such as forward averaging taxation.

Distribution of nondeductible traditional IRA contributions. To the extent any distribution from your traditional IRA represents a return of your nondeductible contributions (see Section III[D]) or after-tax rollover contributions, the distribution will be treated as a tax-free return of basis. If you withdraw an amount from a traditional IRA during a taxable year and you have previously made nondeductible contributions or after-tax rollover contributions to a traditional IRA, then the amount excluded from income for the taxable year is the portion of the amount withdrawn that bears the same ratio to the amount withdrawn as your aggregate nondeductible traditional IRA contributions and after-tax rollover contributions bear to the aggregate balance of all your traditional IRAs as of the end of the year, plus the amount of the withdrawal. For these purposes, all traditional IRAs maintained on your behalf, including traditional rollover IRAs and SEPs, are required to be aggregated, and all distributions in a given year are treated as one distribution.

Example: An individual withdraws a total amount of $3,000 from several traditional IRAs during a taxable year. At the end of the taxable year, the aggregate balance of all traditional IRAs maintained by the individual is $21,000, and the aggregate amount of nondeductible contributions not previously withdrawn by the individual is $4,000. The amount of the individual's $3,000 traditional IRA withdrawal excludible from income is $500 (or $4,000 ÷ $24,000 x $3,000), and the remaining $2,500 is taxable.

Distributions rolled into an eligible employer plan. The maximum amount of a traditional IRA distribution that you can roll over into an eligible retirement plan (other than an IRA), to the extent the plan permits rollover contributions, cannot exceed the portion of your traditional IRA distribution that would otherwise be taxable (without regard to the rollover). The portion of a distribution from your traditional IRA that represents a return of after-tax rollover contributions or nondeductible contributions is not eligible to be rolled over into an employer plan. If you roll over all or a portion of a distribution from a traditional IRA into an eligible employer plan (such as a 401(k) plan, governmental 457 plan, or 403(b) plan), the formula described above for determining the taxable and nontaxable portion of a distribution will not apply. Instead, the portion of the distribution that is rolled over to an eligible employer plan (other than an IRA) is attributable first to amounts other than after-tax rollovers and nondeductible contributions. Therefore, you may roll over a distribution from your traditional IRA to the extent the amount distributed does not exceed the taxable amount (e.g., deductible contributions and earnings) of all of your IRAs. You may wish to consult your tax advisor before rolling over a distribution into an eligible employer plan.

Note: If you roll over a portion of a traditional IRA distribution to an eligible employer plan, the formula used to determine the taxable portion of any withdrawal you receive during or subsequent to the year of the rollover must be adjusted in accordance with IRS guidelines.

B. Roth IRA: Tax treatment of distributions

Ordering rules. Distributions from your Roth IRA are treated as made in the following order:

(1) From regular contributions;

(2) From conversion contributions on a first-in, first-out basis; and

(3) From earnings.

For this purpose, all distributions for each year are considered together, and all of your Roth IRA accounts are treated as one account.

Tax-free "qualified distributions." Distributions from your Roth IRA that are "qualified distributions" are not included in your gross income and are generally not subject to the additional 10% penalty tax on early distributions. A qualified distribution from your Roth IRA is a distribution that is both (1) made after a five-year holding period (see details, below) and (2) described by any one of the following:

(1) Made on or after the date you reach age 59½;

(2) Made to your designated beneficiary after your death;

(3) Made because of your permanent disability; or

(4) A "qualified first-time homebuyer" distribution (subject to a $10,000 lifetime limit). Qualified first-time

homebuyer distributions are distributions used to pay for certain acquisition costs (including the cost of acquiring, constructing, or reconstructing a residence, and reasonable settlement, financing, or other closing costs) of a principal residence for a first-time homebuyer, including you, your spouse, or any children, grandchildren, or ancestors of you or your spouse. Such distribution must be used within 120 days after it is received. You are considered a first-time homebuyer if you (and, if married, your spouse) have not owned a principal residence during the two-year period ending on the date of acquisition.

Nonqualified distributions first considered tax-free return of contributions. If a Roth IRA distribution is not a "qualified distribution" as just described, the distribution will be treated for tax purposes as first a tax-free return of your Roth IRA contributions (see "ordering rules" on previous page). To the extent a nonqualified distribution, when added to all of your previous Roth IRA distributions (whether qualified or nonqualified), is attributable to earnings (see "ordering rules"), it will be fully taxable.

Five-year holding period for Roth IRA distributions. In order for your Roth IRA distributions to be "qualified distributions," you must have held your Roth IRA for at least five years prior to the distribution. This five-year holding period is generally measured by counting five years beginning with the earlier of the first year for which your first regular Roth IRA contribution relates (not necessarily the year in which your first regular Roth IRA contribution is made) or the first year in which you made a conversion contribution.

Separate five-year holding period for conversions. A separate five-year holding period applies with respect to the portion of a distribution that is properly allocable to a traditional-to-Roth IRA conversion or a rollover from an eligible employer plan (other than a designated Roth account) to a Roth IRA. For this portion of a distribution, the five-year period starts with the year in which the conversion (or rollover) contribution was made (which may be later than the year for which your first Roth IRA contribution was made). Even though the distribution of amounts attributable to the conversion (or rollover) may not be subject to income tax, they are subject to a 10% penalty tax if made during the five-year holding period (see Section VII[C]).

C. Additional 10% tax on early distributions from traditional IRAs and Roth IRAs

Your IRA is intended to provide you with retirement income. For this reason, the law generally imposes a nondeductible additional tax if you receive a distribution from either your traditional IRA or Roth IRA, unless certain exceptions apply. This additional tax is equal to 10% of the taxable amount of the distribution and is in addition to the ordinary income tax on the distribution. (Thus, the 10% penalty tax does not apply to the tax-free portion of any distributions you receive from your IRAs.) The additional tax may also apply if you are deemed to receive a distribution from your IRA before age 59½ as a result of borrowing from your IRA or pledging your IRA as security for a loan, as described in Section XI. The law also imposes a tax equal to 10% of any distribution from a Roth IRA attributable to conversion contributions within the separate five-year holding period for conversion amounts.

The additional 10% tax will not be imposed, however, on the following types of distributions from your traditional or Roth IRA:

- Distributions after you reach age 59½.
- Distributions attributable to your total and permanent disability.
- Distributions made to your designated beneficiary after your death.
- Distributions that are part of a series of substantially equal periodic payments (not less frequently than annually) made for your life or life expectancy, or for the joint lives or life expectancies of you and your beneficiary.
- Distributions you receive to the extent such distributions do not exceed the amount of your deductible medical expenses for the taxable year.
- Distributions you receive following termination of employment to the extent such distributions do not exceed the amount of medical insurance premiums you paid for yourself, spouse, and dependents for the taxable year, provided that you have received at least 12 consecutive weeks of unemployment compensation during the current or prior taxable year.
- Distributions to pay for qualified higher-education expenses. Qualified higher-education expenses include tuition, fees, books, supplies, equipment, and room and board required for enrollment or attendance for you, your spouse, your children, or your grandchildren at an eligible postsecondary educational institution.
- Distributions up to $10,000 used to pay for acquisition costs (including the cost of acquiring, constructing, or reconstructing) of a principal residence for a first-time homebuyer, including you, your spouse, or any children, grandchildren, or ancestors of you or your spouse. Such distribution must be used within 120 days after it is received. You are considered a first-time homebuyer if you (and, if married, your spouse) have not owned a principal residence during the two-year period ending on the date of acquisition. The aggregate amount of distributions you may take under this first-time homebuyer exception for the year of the distribution and all prior years is $10,000.
- Distributions that are qualified reservist distributions. If you were a member of a reserve component and you were called to active duty after September 11, 2001, and before December 31, 2007, for a period of 179 days or more, distributions made after the date you were called to active duty and before the close of your active duty period are considered qualified reservist distributions.
- Distributions made on account of an IRS levy.

Section VIII

Methods of distribution

A. Distributions in general

Under either a Vanguard traditional IRA or Roth IRA, you may elect to have all or a portion of your account distributed in one or a combination of the following ways:

(1) A partial payment.

(2) A lump-sum payment.

(3) Monthly, quarterly, semiannual, or annual installment payments over a period not extending beyond your life expectancy or the joint and last survivor life expectancy of you and your designated beneficiary.

The method of distribution you select for your traditional IRA must comply with the minimum distribution requirements described in Section VIII[B]. You may request a distribution from your Vanguard IRA in a form and manner acceptable to the Custodian.

Distributions in cash or in kind. Distributions from the Vanguard IRA will generally be made in cash, unless you notify the Custodian in writing that you wish to have a designated portion of your IRA assets distributed in kind. However, any assets in your Vanguard IRA that cannot be sold by the Custodian for cash in the ordinary course of business shall automatically be distributed to you in kind.

B. Traditional IRA: Age 70½ minimum distributions

Commencement of distribution at age 70½. You must begin receiving distributions from your traditional IRA no later than April 1 of the calendar year following the calendar year in which you attain age 70½. No age 70½ distribution requirements apply for Roth IRAs.

Minimum amount required to be distributed. The minimum amount required to be distributed from your traditional IRA for each calendar year, beginning with the year in which you attain age 70½, is determined by dividing the entire amount in your account as of December 31 of the preceding calendar year by the applicable distribution period. In most cases, the applicable distribution period is determined using the IRS's Uniform Lifetime Table. However, if your spouse is your sole beneficiary for the entire year, the applicable distribution period is the longer of the distribution period determined using the Uniform Lifetime Table, or the joint life expectancy of you and your spouse determined using the Joint Life and Last Survivor Table. See the Vanguard Traditional and Roth IRA Custodial Account Agreement and IRS Publication 590-B, Distributions from Individual Retirement Arrangements (IRAs), for more details including the life expectancy tables used to calculate required minimum distribution amounts. You must receive the required minimum distribution amount from your traditional IRA for the calendar year in which you attain age 70½ by April 1 of the following year. You must receive the required minimum distribution amount from your traditional IRA for each calendar year thereafter by December 31 of that year.

Penalty tax on insufficient distributions. A 50% excise tax may be imposed if the amount actually distributed from your traditional IRA beginning after you attain age 70½ is less than the minimum amount required to be distributed. The 50% tax is imposed on the difference between the amount actually distributed from your traditional IRA and the amount required to be distributed. This penalty tax may be waived in certain cases provided you can establish to the satisfaction of the IRS that the deficit in the amount distributed was due to reasonable error and you are taking steps to remedy the problem.

C. Distribution upon death for traditional IRAs and Roth IRAs

If you die prior to the complete distribution of your account, the remaining balance in your Vanguard IRA (either traditional or Roth IRA) will be distributed to your designated beneficiary, upon request in a form and manner acceptable to the Custodian, over a period selected by your beneficiary subject to the requirements stated below. In general, distributions your beneficiary receives from your IRA that are included in gross income will be taxed as ordinary income.

General rule for Roth IRAs and where required distributions had not begun from traditional IRAs. For all Roth IRAs and for traditional IRAs where a required distribution of your account had not commenced prior to your death, the general rule is that the balance of your Vanguard IRA must be distributed to your designated beneficiary over his or her life expectancy starting by December 31 of the calendar year following the year of your death, or if your sole beneficiary is your surviving spouse, by the later of December 31 of the calendar year following the year of your death or December 31 of the calendar year in which you would have attained age 70½. Instead of applying the general rule described above, your beneficiary may elect to apply the "five-year rule," under which your beneficiary must distribute the entire balance of your account by December 31 of the calendar year that contains the fifth anniversary of your death. If you do not have a beneficiary, or your beneficiary is not an individual (your estate, or a charity, for example), distributions must generally be made in accordance with the five-year rule. However, if your beneficiary is a trust and the trust meets certain requirements, the beneficiaries of the trust may be treated as designated beneficiaries for the purpose of determining the applicable distribution period. See IRS Publication 590-B or consult your tax advisor for more information on the special trust rules.

Where age 70½ minimum distributions from traditional IRA had already begun. If required minimum distributions from your traditional IRA had already commenced prior to your death, the balance of your Vanguard traditional IRA must be distributed to your designated beneficiary over his or her life expectancy, or if longer, over your remaining life expectancy, starting by December 31 of the calendar year following the year of your death. If you do not have a beneficiary, or your beneficiary is not an individual, distributions must be made over your remaining life expectancy. If your beneficiary is a trust and the trust meets certain requirements, the

beneficiaries of the trust may be treated as designated beneficiaries for the purpose of determining the applicable distribution period. See IRS Publication 590-B or consult a tax advisor for more information about the special trust rules. For these purposes, minimum distributions are considered to have begun prior to your death only if distributions were made on or after April 1 of the calendar year following the calendar year in which you attained age 70½.

Exception where surviving spouse is sole beneficiary. An exception to the preceding two rules is that if the sole beneficiary of your Vanguard IRA is your surviving spouse, your spouse may elect to treat your Vanguard IRA as his or her own IRA. Your surviving spouse may elect to treat your Vanguard IRA as his or her own IRA either by making contributions to the IRA (including a rollover contribution) or by not taking required minimum distributions from the IRA as the IRA beneficiary in accordance with the rules summarized in this Section VIII[C]. In the case of a traditional IRA, if your surviving spouse makes the election to treat your Vanguard IRA as his or her own IRA, your spouse would be required to satisfy any minimum distribution requirements as the traditional IRA owner in accordance with the rules summarized in Section VIII[B].

Designation of beneficiary. Under a Vanguard IRA, you may designate one or more individuals or entities (such as a trust) as your beneficiary. You may also select other beneficiary designations deemed acceptable by the Custodian. You will initially designate your beneficiary by completing the Adoption Agreement for the Vanguard IRA. You may subsequently change or revoke your beneficiary designation at any time by notifying the Custodian in a form and manner acceptable to the Custodian. If you fail to designate a beneficiary or if your designated beneficiary (or each of your designated beneficiaries) predeceases you, your beneficiary will be your surviving spouse or, if you have no surviving spouse, your estate.

D. Federal estate and gift taxation

Gift tax consequences. Your designation of a beneficiary (or beneficiaries) to receive distributions from your IRA upon your death will not be considered a transfer of property for federal gift tax purposes.

Estate tax consequences. Generally, amounts remaining in your IRA after your death will be included in your gross estate for federal estate tax purposes. You are encouraged to consult with a financial planner or tax advisor when considering the estate tax consequences of your IRA assets.

E. Income tax withholding

The Internal Revenue Code requires the withholding of federal income tax on distributions from a traditional IRA unless the recipient affirmatively elects not to have withholding apply. The amount of federal income tax required to be withheld on any payment under the Vanguard traditional IRA will generally equal 10% of the amount of the distributions. Upon a request for a distribution under the Vanguard traditional IRA, the Custodian will notify the recipient of his or her right to elect not to have withholding apply (or to revoke any prior election). If the distribution is delivered outside the United States, you may not elect out of federal income tax withholding. State income tax may be withheld from your IRA distributions, if applicable.

If you are a nonresident alien, the amount of federal income tax required to be withheld is 30% of the amount of the distribution. You may not elect out of withholding. If you are eligible to claim a reduced withholding rate under a tax treaty, you must send a valid Form W-8BEN prior to the distribution.

Section IX

Simplified employee pension

A simplified employee pension, or "SEP," is a special traditional IRA plan that permits employers to make deductible contributions to the separate traditional IRAs established for their employees. If your employer has adopted a SEP, your employer may make deductible SEP contributions directly to your Vanguard traditional IRA each year in an amount up to the lesser of 25% of your current-year compensation, or $55,000 for 2018 ($54,000 for 2017 and subject to cost-of-living adjustments for later years).

Exclusion from gross income. The amount of SEP contributions made by your employer to your traditional IRA will be excludible from your gross income provided it does not exceed the deduction limit described above. In addition, you may make your own annual contributions to either your traditional or Roth IRA each year up to the lesser of the contribution limit in effect for the taxable year or 100% of current-year compensation. For more information on the contribution limits and treatment of excess contributions, see IRS Publication 560.

Determination of compensation. For purposes of the 25% of compensation limit that applies to SEP contributions, you may take into account only the amount of your current-year compensation from the employer making the SEP contribution. In addition, you may not include the amount of the SEP contribution in the determination of your compensation. The maximum amount of compensation that may be taken into account on behalf of any SEP participant is subject to an indexed limit of $275,000 for 2018.

Self-employed individuals. Any sole proprietor or partnership may be eligible to establish a SEP and make deductible SEP contributions to the separate traditional IRAs established by the sole proprietor or partners (as well as to the traditional IRAs of any common-law employees). In the case of a self-employed individual, the term "compensation" includes the individual's earned income from self-employment, reduced by the amount of deductible retirement plan contributions.

Contributions after age 70½. SEP contributions may be made to your IRA by your employer even after you have attained age 70½.

Elective deferrals. Prior to January 1, 1997, certain employers were permitted to establish a Salary Reduction SEP under which employees could make elective deferrals—or "pre-tax" salary reduction contributions—up to an indexed dollar limit each year. However, employers can no longer establish Salary Reduction SEPs after December 31, 1996 (although Salary Reduction SEPs in existence before January 1, 1997, can continue to receive contributions and cover new employees).

Section X

Income tax returns

If you are eligible to make deductible contributions to your traditional IRA, you may claim your deduction for traditional IRA contributions on your federal income tax return (Form 1040 or Form 1040A) even if you do not itemize deductions. Each year, the Custodian will send you IRS Form 5498, showing your preceding-year IRA contributions. If you make nondeductible contributions to your traditional IRA for a taxable year, or if you receive any distributions from your IRA during a taxable year that includes nondeductible contributions, you will be required to provide certain information concerning these transactions on Form 8606, to be included with your federal income tax return for the taxable year. If you make after-tax rollover contributions to your traditional IRA, you should report such contributions in accordance with IRS guidelines.

Penalty taxes. If one or more of the following situations occur, you may be required to file Form 5329, Additional Taxes on Qualified Plans (Including IRAs) and Other Tax-Favored Accounts, with the IRS:

(1) Payment of a 6% excise tax because of an excess contribution.

(2) Payment of a 10% additional tax because of an early distribution before age 59½.

(3) Payment of a 50% excise tax because of an insufficient distribution from your IRA after age 70½.

Form 5329 need not be filed if the only activity in your IRA for the year consisted of proper (i.e., nonpenalized) contributions and distributions. If Form 5329 must be filed, it should be attached to your federal income tax return or should be filed separately if you are not required to file a federal income tax return.

Distributions. When you receive distributions from your Vanguard IRA, the Custodian will send you the required tax form (Form 1099-R). If you are a nonresident alien, the Custodian will send you Form 1042-S.

Unrelated business taxable income ("UBTI"). The Custodian of your IRA is required to file an Exempt Organization Business Income Tax Return ("Form 990-T") on behalf of the IRA for each year in which the IRA has unrelated trade or business income of $1,000 or more. The Custodian will calculate the tax due, if any, with respect to the IRA based on the Schedule K-1 received from the respective partnership(s), file the Form 990-T with the Internal Revenue Service, and pay the tax from funds available in the IRA.

By holding assets, such as partnerships, that generate UBTI, you agree to the following to facilitate the payment of any tax due:

(1) Your IRA will contain funds within the Settlement Fund to pay any tax imposed on UBTI when due as well as the Custodian's charges for preparing tax return, if any.

(2) You will sell assets or contribute sufficient amounts to your IRA, if necessary, to satisfy your tax obligation.

(3) To the extent funds are not available, the Custodian is authorized to sell any investments in your IRA necessary to generate the funds needed to satisfy your tax obligation.

The IRS requires the IRA to make quarterly estimated tax payments when the annual federal tax expected is $500 or more. You understand and acknowledge that the Custodian will not make estimated tax payments. If your IRA has a tax liability, you agree to pay from your IRA any interest and penalties for the underpayment of the tax due.

Section XI

Prohibited transactions

Generally, a prohibited transaction is any improper use of your IRA. Examples of prohibited transactions include borrowing money from your account or selling property to the account.

Effect on IRA. Generally, if you engage in a prohibited transaction, your IRA will lose its tax-exempt status, and you will be required to include the entire value of the account in your gross income. If your account is disqualified before you reach age 59½, you may also be required to pay the 10% additional tax on early distributions referred to in Section VII[C].

Pledging your IRA as security. Pledging your IRA as security for a loan will cause the portion pledged to be treated as a distribution to you, includible in gross income and subject to the 10% additional tax on early distributions if you are under age 59½.

Investment in collectibles. If your IRA is invested in collectibles, the amount invested will be considered a distribution to you in the year of investment. For this reason, the Vanguard IRA specifically precludes investments in collectibles, which include art works, rugs, antiques, metals, gems, stamps, coins (but not certain gold, silver, and platinum coins issued by the United States or gold, silver, platinum, or palladium bullion), alcoholic beverages, and certain other tangible personal property.

Section XII

Other information

A. The Custodian

The Custodian of your Vanguard IRA is Vanguard Fiduciary Trust Company, a trust company incorporated under Pennsylvania banking laws. Vanguard Fiduciary Trust Company is a wholly owned subsidiary of The Vanguard Group, Inc., of Valley Forge, Pennsylvania. The Vanguard Group will perform certain administrative services in connection with the Vanguard IRA, for which The Vanguard Group may be reimbursed at cost by Vanguard Fiduciary Trust Company.

B. Amendments

The Custodian is specifically authorized to make any amendments to the Vanguard IRA necessary to comply with the applicable provisions of the Internal Revenue Code and any other such amendments as the Custodian deems appropriate. The Custodian will inform you of any such amendments. Notice of amendments may be provided electronically, provided that you have consented to electronic delivery thereof.

C. Vanguard IRA investments

Your Vanguard IRA may be invested in shares of the mutual funds offered by The Vanguard Group, Inc. (the "Vanguard funds"), or in other types of investments that the Custodian and/or any Vanguard affiliate may permit to be available investments under the Vanguard IRA from time to time.

Directed investments. Your Vanguard IRA will be invested and reinvested solely in accordance with your directions (or, following your death, the directions of your designated beneficiary). You may change your investment directions at any time by notifying the Custodian in writing or by telephone.

Reinvestment of earnings. All dividends and capital gains received on shares of a Vanguard fund held in your Vanguard IRA that are permitted to be reinvested in additional shares of the Vanguard fund shall, in the absence of investment directions by you to the contrary, be automatically reinvested in additional shares of the Vanguard fund. Otherwise, any distribution of earnings received with respect to assets held in your account shall be reinvested solely in accordance with investment directions furnished by you.

Growth in value. The growth in value of your Vanguard IRA will depend on the investment decisions made by you and is neither guaranteed nor projected.

D. Vanguard IRA minimums

Minimum contributions. For both the Vanguard traditional and Roth IRAs, the minimum initial contribution for each Vanguard fund established for an IRA is the minimum set forth in each fund's prospectus. However, certain Vanguard funds may require higher minimum contributions as set forth in their prospectuses.

E. Custodial fees and other expenses

Vanguard may charge reasonable custodial fees with respect to the establishment and maintenance of your Vanguard IRA at any time during the calendar year. Each Vanguard fund is subject to an account service fee that is described in the fund's prospectus.

F. Vanguard fund information

For complete information about the account service fees, advisory fees, and other expenses as well as the method of calculating the price per share for each Vanguard fund you may select for your Vanguard IRA, you should read the fund's prospectus. You may choose to use Vanguard Brokerage Services® to invest your Vanguard IRA directly in stocks, bonds, or options. A schedule of the brokerage commissions to be charged under Vanguard Brokerage Services will be furnished to you when you establish your account.

G. IRS approval

The form of your Vanguard IRA has been approved by the IRS. IRS approval is a determination only as to the form and does not represent a determination of the merits of the Vanguard IRA. For more information, please refer to IRS Publications 590-A and 590-B or contact the IRS.

Vanguard Traditional and Roth IRA Custodial Account Agreement

Introduction

By executing the Adoption Agreement, the Investor hereby establishes a traditional individual retirement account (Traditional IRA) as described in Section 408(a) of the Code or a Roth individual retirement account (Roth IRA) as described in 408A of the Code, in order to provide for retirement or for the support of his or her Beneficiary after death.

Article I

Definitions

The following terms when used herein with initial capital letters shall be defined as follows:

1.1 **Account** means the custodial account established by the Investor to which contributions may be made in accordance with the terms and conditions of this Agreement. All assets of the Account shall be held by the Custodian for the exclusive benefit of the Investor or, following his or her death, the Beneficiary.

1.2 **Adopted Person** means a person adopted through the legal process of the United States and/or any state, commonwealth, or possession of the United States. An Adopted Person shall be considered to be the descendant or issue of the adopting person.

1.3 **Adoption Agreement** means the agreement executed by the Investor in which the Investor designated the Account as a Traditional IRA or a Roth IRA for purposes of adopting the Agreement. The Adoption Agreement shall be considered an integral part of the Agreement as if set forth fully herein.

1.4 **Agreement** means the Vanguard Traditional and Roth IRA Custodial Account Agreement as set forth herein, including the provisions set forth in the Adoption Agreement and any Beneficiary designation filed with and acceptable to the Custodian, as each such document may be amended from time to time.

1.5 **Authorized Party** means the executor, administrator, or personal representative of the Investor's estate, the trustee of a trust Beneficiary, or any other person deemed appropriate by the Custodian to act on behalf of the Investor's Account after the Investor's death.

1.6 **Beneficiary** means the persons or entities designated in accordance with Article 4.4 to receive any undistributed amount credited to the Account at the time of the Investor's death, as well as persons or entities entitled to roll over a distribution from an eligible retirement plan of a deceased employee into an inherited IRA, as permitted under Section 402(c)(11) of the Code.

1.7 **Children** means descendants in the first generation below the individual, including those born within or out of wedlock, and those legally adopted by the individual. This term excludes stepchildren and foster children.

1.8 **Code** means the Internal Revenue Code of 1986, as amended from time to time, or any successor statute.

1.9 **Custodian** means Vanguard Fiduciary Trust Company, a trust company incorporated under Pennsylvania banking laws, or any successor thereto.

1.10 **Investor** means the individual who has adopted the Agreement by executing the Adoption Agreement. (This term does not include a Beneficiary who opens an account after the Investor's death.)

1.11 **Descendants** means all descendants of all generations of an individual.

1.12 **Grandchildren** means descendants in the second generation below the individual, including those born within or out of wedlock, and those legally adopted by Children of the Investor. This term excludes stepchildren and foster children.

1.13 **Issue** means all descendants of all generations of an individual.

1.14 **Per Stirpes** means a way of dividing the Account as follows: The Account shall be divided into as many equal shares as there are surviving descendants in the generation nearest to the decedent that contains at least one surviving descendant and deceased descendants in the same generation who left surviving descendants, if any. The share of each deceased descendant who leaves surviving descendants is divided in the same manner, with the subdivision repeating until the property is fully allocated among surviving descendants. A descendant who dies before the decedent and who leaves no surviving descendants is disregarded.

1.15 **Roth IRA** means an individual retirement account as described in section 408A of the Code.

1.16 **SEP Contribution** means a contribution on behalf of the Investor by his or her employer under a simplified employee pension as described in Section 408(k) of the Code.

1.17 **Successor Beneficiary** means the persons or entities designated in accordance with Article 4.4 entitled to receive any undistributed amount credited to the Account at the time of the Beneficiary's death.

1.18 **Traditional IRA** means an individual retirement account or annuity as described in section 408(a) of the Code.

1.19 **Vanguard** means The Vanguard Group, Inc., a Pennsylvania corporation, or any successor thereto.

1.20 **Vanguard Funds** means one or more of the regulated investment companies or mutual funds that are members of The Vanguard Group, and that the Custodian permits to be available investments under this Agreement.

Article II

Contributions to Account

2.1 **General limitations.** Contributions may be made by or on behalf of the Investor to the Account for any taxable year subject to the limitations set forth below.

(a) **Regular contributions.** Except in the case of a rollover contribution to a Traditional IRA (as permitted under Code Sections 402(c), 402(e)(6), 403(a)(4), 403(b)(8), 403(b)(10), 408(d)(3), and 457(e)(16), an employer SEP contribution as described in Code section 408(k), a qualified rollover contribution to a Roth IRA, or a recharacterized contribution described in Code section 408A(d)(6), all contributions to the Account by or on behalf of the Investor shall be made in cash, and the total of such contributions to all of the Investor's IRAs shall not exceed the lesser of the Investor's compensation (as defined in Article 2.6) or the applicable amount defined in (b) below. However, notwithstanding the dollar limits on contributions, an Investor may make a repayment of a qualified reservist distribution described in Code section 72(t)(2)(G) during the two-year period beginning on the day after the end of the active duty period or by August 17, 2008, if later.

(b) **Applicable amount.** The applicable amount is determined under (i) or (ii) below.

(i) If the Investor is under age 50, the applicable amount is $5,000 for any taxable year beginning in 2008 and years thereafter. After 2008, the $5,000 amount will be adjusted by the Secretary of the Treasury for cost-of-living increases under Code section 219(b)(5)(D). Such adjustment will be in multiples of $500.

(ii) If the Investor is age 50 or older, the applicable amount under paragraph (i) above is increased by $1,000 for any taxable year beginning in 2006 and years thereafter.

(iii) If the individual was a participant in a 401(k) plan of a certain employer in bankruptcy described in Code section 219(c)(5)(C), then the applicable amount under paragraph (i) above is increased by $3,000 for taxable years beginning after 2006 and before 2010 only. An individual who makes contributions under this paragraph (iii) may not also make contributions under paragraph (ii).

(c) **Income limits for Roth IRA contributions.** If (i) and/or (ii) below apply, the maximum regular contribution that can be made to all of the Investor's Roth IRAs for a taxable year is the smaller of the amount determined under (i) or (ii).

(i) **Regular Roth contributions.** The maximum regular contribution that can be made to all of an Investor's Roth IRAs for the taxable year is gradually reduced to $0 between certain levels of modified adjusted gross income (modified AGI), as defined in Article 2.7 below. For a single Investor (or head of household) the maximum annual contribution is phased out between modified AGI of $95,000 and $110,000; for a married Investor who files jointly (or qualifying widower), between modified AGI of $150,000 and $160,000; and for a married Investor who files separately, between modified AGI of $0 and $10,000. If the Investor's modified AGI is in the phase-out range, the maximum regular Roth IRA contribution for the taxable year is rounded up to the next multiple of $10 and is not reduced below $200. The dollar amounts will be adjusted by the Secretary of the Treasury for cost-of-living increases under Code section 408A(C)(3). Such adjustments will be in multiples of $1,000.

(ii) **Contributions to both Roth and traditional IRAs.** If the Investor makes contributions to both Roth IRAs and Traditional IRAs for a taxable year, the maximum regular contributions that can be made to all of the Investor's Roth IRAs for that taxable year is reduced by the regular contributions made to the Investor's non-Roth IRAs for the taxable year.

(d) **Qualified rollover contributions to Roth IRA.** A "qualified rollover contribution" is a rollover contribution of a distribution from an IRA that meets the requirements of Code section 408(d)(3), except the one-rollover-per-year rule of Code section 408(d)(3)(B) does not apply if the rollover contribution is from an IRA other than a Roth IRA (a "non-Roth IRA"). For taxable years beginning after 2005, a qualified rollover contribution includes a rollover from a designated Roth account described in Code section 402A; and for taxable years beginning after 2007, a qualified rollover contribution also includes a rollover from an eligible retirement plan described in Code section 402(c)(8)(B). A rollover from an eligible retirement plan other than a Roth IRA or a designated Roth account cannot be made to a Roth IRA if, for the year the amount is distributed from the other plan, (i) the Investor is married and files a separate return, (ii) the Investor is not married and has modified AGI in excess of $100,000, or (iii) the Investor is married and together the Investor and the Investor's spouse have modified AGI in excess of $100,000. For purposes of the preceding sentence, a husband and wife are not treated as married for a taxable year if they have lived apart at all times during the taxable year and file separate returns for the taxable year. For taxable years beginning after 2009, the limits in this paragraph do not apply to qualified rollover contributions.

2.2 **Recharacterization.** A regular contribution to one type of IRA can be recharacterized pursuant to the rules in Section 1.408A-5 of the Regulations as a regular contribution to another type of IRA. A regular contribution to a non-Roth IRA that is recharacterized to a Roth IRA is subject to the limits in Article 2.1(c) above.

2.3 **Rollover contributions.** The Custodian may accept rollover contributions within the meaning of Sections 402(c), 403(a)(4), 403(b)(8), 403(b)(10), 408(d)(3), 457(e)(16), or 408A(e) of the Code that consist of cash or of such other assets that are acceptable to the Custodian and that are permissible investments under Section 408(a) of the Code. Before making a rollover contribution, the Investor shall provide any information the Custodian may require, in a form and manner acceptable to the Custodian. The Custodian shall be under no obligation to accept any rollover contribution consisting of assets other than cash. The Investor shall have the sole responsibility for determining whether any contribution to the Account qualifies as a rollover contribution.

2.4 **SEP Contributions.** SEP Contributions may be made to a traditional IRA on behalf of the Investor by his or her employer for any taxable year in an amount not to exceed the amount provided in Section 408(j) of the Code or any successor statutory provision thereto for such taxable year. Before making any SEP Contribution, the Investor shall execute such forms as the Custodian may require to certify that the Investor is covered under a simplified employee pension (as described in Section 408(k) of the Code) established by his or her employer and to provide other information as the Custodian may reasonably request. The Investor shall have the sole responsibility for determining whether any contribution to the Account qualifies as a SEP Contribution.

2.5 **SIMPLE contributions.** No contributions will be accepted under a SIMPLE IRA plan established by any employer pursuant to Code section 408(p). Also, no transfer or rollover of funds attributable to contributions made by a particular employer under its SIMPLE IRA plan will be accepted from a SIMPLE IRA; that is, an IRA used in conjunction with a SIMPLE IRA plan, prior to the end of the two-year period beginning on the date the individual first participated in that employer's SIMPLE IRA plan.

2.6 **Compensation.** For purposes of 2.1 above, compensation is defined as wages, salaries, professional fees, or other amounts derived from or received for personal services actually rendered (including, but not limited to commissions paid to salespersons, compensation for services on the basis of a percentage of profits, and commissions on insurance premiums, tips, and bonuses) and includes earned income, as defined in Section 401(c)(2) (reduced by the deduction the self-employed individual takes for contributions made to a self-employed retirement plan). For purposes of this definition, Section 401(c)(2) shall be applied as if the term trade or business for purposes of Section 1402 of the Code included service described in Subsection (c)(6). Compensation does not include amounts derived from or received as earnings or profits from property (including but not limited to interest and dividends) or amounts not includible in gross income. Compensation also does not include any amount received as a pension or annuity or as deferred compensation. The term "compensation" shall include any amount includible in the individual's gross income under Code Section 71 with respect to a divorce or separation instrument described in Subparagraph (A) of Code Section 71(b)(2). In the case of a married individual filing a joint return, the greater compensation of his or her spouse is treated as his or her own compensation, but only to the extent that such spouse's compensation is not being used for purposes of the spouse making a contribution to a Roth IRA or a deductible contribution to a traditional IRA. Compensation includes nontaxable combat pay received by a member of the U.S. Armed Services.

2.7 **Modified adjusted gross income (modified AGI).** For purposes of Articles 2.1(c) and (d) above, an Investor's modified AGI for a taxable year is defined in Code Section 408A(c)(3)(C)(i) and does not include any amount included in gross income as a result of a rollover from an eligible retirement plan other than a Roth IRA (a "conversion").

2.8 **Timing of contributions.** For purposes of Article 2.1, any contribution to the Account by the Investor shall be deemed to have been made on the last day of the preceding taxable year if the contribution is made on account of such taxable year and is made not later than the date prescribed by law for filing the Investor's federal income tax return for such taxable year (not including any extensions thereof). The Investor shall have the sole responsibility for determining whether any contribution to the Account is deductible for federal income tax purposes.

2.9 **Responsibility of Custodian and Vanguard.** Neither the Custodian nor Vanguard shall have any responsibility for determining whether any contribution by or on behalf of the Investor to the Account qualifies as a Rollover Contribution or SEP Contribution, or whether any contribution to the Account is deductible by the Investor for federal income tax purposes.

Article III

Investment of Account

3.1 **Investment of contributions (including transfers).** The Custodian shall invest and reinvest all contributions to the Account in accordance with the investment directions of the Investor as set forth in the Adoption Agreement or in any subsequent investment directions furnished by the Investor pursuant to Article 3.2. If the Account contains any amount with respect to which the Investor has not furnished investment instructions or if the Investor has furnished investment instructions which, in the opinion of the Custodian, are unclear, incomplete, or otherwise not in good order, the Custodian may request additional investment instructions from the Investor. Pending receipt of such investment instructions, the Custodian may (i) hold all or a portion of such amount uninvested, (ii) invest all or a portion of the contribution amount in a Vanguard Money Market Fund, or a Vanguard Brokerage Services' Settlement Fund option or (iii) return all or a portion of the investment amount to the Investor without liability for loss of income or appreciation pending receipt of proper investment directions.

3.2 **Investment directions; available investments.** The Investor or, following his or her death, the Beneficiary, shall be permitted at all times to direct, or retain an agent, investment advisor, or manager to direct the Custodian as to the investment or reinvestment of the assets of the Account. All such investment directions shall be made in a form or manner acceptable to the Custodian. Assets of the Account may be invested in shares of one or more of the Vanguard Funds or in other investments that are eligible for acquisition under Section 408(a) or 408A of the Code and that the Custodian and/or any Vanguard affiliate permits to be available investments under this Agreement. U.S.-domiciled Vanguard funds generally are not available for sale to Investors that reside outside of the United States. All investments of the Account shall be registered in the name of the Custodian or its nominee, or shall be retained unregistered or in a form permitting transfer by delivery, provided that the books and records of the Custodian shall at all times show such investments to be part of the Account.

3.3 **Prohibitions concerning life insurance, collectibles, and commingling.** Notwithstanding any provision of this Agreement to the contrary, no assets of the Account will be invested in life insurance contracts or in collectibles (within the meaning of Section 408(m) of the Code), nor will assets of the Account be commingled with other property except in a common trust fund or common investment fund (within the meaning of Section 408(a)(5) of the Code).

3.4 **Responsibility of Custodian and Vanguard**

(a) I**nvestments in general.** In making any investment or reinvestment of the assets of the Account, the Custodian shall be fully entitled to rely on the investment directions furnished to it by the Investor or Beneficiary in accordance with the terms and conditions of this Agreement, and shall be under no duty to make any inquiry or investigation with respect thereto. The Investor hereby acknowledges that neither the Custodian nor Vanguard undertakes to render any investment advice in connection with this Agreement, and that the assets of the Account are to be invested, reinvested, and controlled exclusively by the Investor or, following his or her death, the Beneficiary in accordance with the terms and conditions of this Agreement.

(b) **Vanguard Fund shares.** The Custodian shall be responsible for delivering to the Investor or, following his or her death, the Beneficiary, all shareholder notices, reports, and proxies relating to Vanguard Fund shares held in the Account. The Custodian shall vote any such shares at shareholder meetings of the Vanguard Funds in accordance with instructions received from the Investor or Beneficiary. By establishing (or by having established) the Account, the Investor hereby directs the Custodian to vote any Vanguard Fund shares held in the Account for which no timely voting instructions are received in proportionately the same manner as shares timely voted by such fund's other shareholders. By directing that assets of the Account be invested in a Vanguard Fund, the Investor or Beneficiary shall be deemed to have acknowledged receipt of the current prospectus for such Vanguard Fund.

Article IV

Distribution of Account

4.1 **General requirements.**

(a) Notwithstanding any provision of this Agreement to the contrary, the distribution of the Investor's interest in the Account shall be made in accordance with the requirements of Code Sections 408(a)(6) (as modified by Section 408A(c)(5) for Roth IRAs), and the regulations thereunder, the provisions of which are incorporated herein by reference. The Custodian shall distribute the balance of the Account to the Investor or, after his or her death, the Beneficiary, at such times and in such manner as the Investor or Beneficiary shall direct, subject to the requirements set forth below.

(b) **Method of distribution.** The Investor, or after his or her death, the Beneficiary, may elect to have all or any portion of the Account distributed in one or a combination of the following ways:

(1) A partial payment.

(2) A lump-sum payment.

(3) Monthly, quarterly, semiannual or annual installment payments provided that the method of distribution satisfies the minimum distribution requirements of Article 4.2.

The Investor's or Beneficiary's request must be made in a form and manner acceptable to the Custodian. The Investor or Beneficiary may change his or her designated method of distribution upon proper notification to the Custodian.

(c) **Distribution in kind.** All distributions from the Account shall be made in cash, with the exception that the Investor or Beneficiary may elect to have all or any portion of the distribution made in kind, in which case the Custodian shall transfer such specific assets as the Investor or Beneficiary may direct into the name of the Investor or Beneficiary, and with the further exception that any assets held in the Account, which cannot be sold by the Custodian for cash in the ordinary course of business for purposes of making distributions from the Account, shall be distributed to the Investor or Beneficiary in kind.

4.2 **Lifetime minimum distribution requirements**

(a) **Traditional IRA**

(i) **Required beginning date.** The entire value of the account of the Investor for whose benefit the Account is maintained must commence to be distributed no later than the first day of April following the calendar year in which the Investor attains age 70½ (the Investor's "required beginning date") over the life of the Investor or the lives of the Investor and his or her designated Beneficiary.

(ii) **Annual minimum amount.** The amount to be distributed each year, beginning with the calendar year in which the Investor attains age 70½ and continuing through the year of death, may not be less than the amount determined by dividing the value of the IRA (as determined under Article 4.3(c) as of the end of the preceding year by the distribution period in the Uniform Lifetime Table in Q&A-2 of Section 1.401(a)(9)-9 of the Income Tax Regulations, using the Investor's age as of his or her birthday in the year. However, if the Investor's sole designated Beneficiary is his or her surviving spouse and such spouse is more than ten years younger than the Investor, then the distribution period is determined under the Joint and Last Survivor Table in Q&A-3 of Section 1.401(a)(9)-9 of the Income Tax Regulations, using the ages as of the Investor's and spouse's birthdays in the year.

(iii) **Timing of minimum distributions.** The required minimum distribution for the year the Investor attains age 70½ can be made as late as April 1 of the following year. The required minimum distribution for any other year must be distributed no later than December 31 of that calendar year.

(iv) **Aggregation of IRAs.** The Investor may satisfy the distribution requirements under Section 408(a)(6) of the Code by receiving a distribution from one IRA that is equal to the amount required to satisfy the minimum distribution requirements for two or more IRAs in accordance with Q&A-9 of Section 1.408-8 of the Income Tax Regulations.

(b) **Roth IRA.** No amount is required to be distributed from a Roth IRA prior to the death of the Investor for whom the account was originally established.

4.3 **Distributions upon death: Traditional and Roth IRAs.** In the event the Investor dies prior to the complete distribution of the Account, the remaining balance of the Account will be distributed to the Beneficiary at such time and in such manner as the Beneficiary shall direct, in a form and manner acceptable to the Custodian, subject to the following rules:

(a) **Traditional IRAs where investor dies before required beginning date and all Roth IRAs.** For traditional IRAs where the Investor dies before his or her required beginning date and for all Roth IRAs, the Investor's interest must be distributed at least as rapidly as follows:

(i) If the designated Beneficiary is someone other than the Investor's surviving spouse, the entire interest must be distributed, starting by December 31 of the calendar year following the calendar year of the Investor's death, over the remaining life expectancy of the designated Beneficiary, with such life expectancy determined using the age of the Beneficiary as of his or her birthday in the year following the year of the Investor's death, or, if elected, in accordance with paragraph (a)(iii) below.

(ii) If the Investor's sole designated Beneficiary is the Investor's surviving spouse, the entire interest must be distributed, starting by December 31 of the calendar year following the calendar year of the Investor's death (or by the end of the calendar year the Investor would have attained age 70½, if later), over such spouse's life, or, if elected, in accordance with paragraph (a)(iii) below. If the surviving spouse dies before distributions are required to begin, the remaining interest must be distributed, starting by December 31 of the calendar year following the calendar year of the spouse's death, over the spouse's designated Beneficiary's remaining life expectancy determined using such Beneficiary's age as of his or her birthday in the year following the death of the spouse, or, if elected, in accordance with paragraph (a)(iii) below. If the surviving spouse dies after distributions are required to begin, any remaining interest will be distributed over the spouse's remaining life expectancy determined using the spouse's age as of his or her birthday in the year of the spouse's death.

(iii) If there is no designated Beneficiary, or if applicable by operation of paragraph (a)(i) or (a)(ii) above, the remaining interest must be distributed by the end of the calendar year containing the fifth anniversary of the Investor's death (or of the spouse's death in the case of the surviving spouse's death before distributions are required to begin under paragraph (a)(ii) above).

(iv) The amount that must be distributed under paragraphs (a)(i) or (ii) above is the amount determined by dividing the value of the IRA as of the end of the preceding year by the remaining life expectancy specified in such paragraph. Life expectancy is determined using the Single Life Table in Q&A-1 of Section 1.401(a)(9)-9 of the Income Tax Regulations. If distributions are being made to a surviving spouse as the sole designated Beneficiary, such spouse's remaining life expectancy for a year is the number in the Single Life Table corresponding to such spouse's age in the year. In all other cases, remaining life expectancy for a year is the number in the Single Life Table corresponding to the Beneficiary's age in the year specified in paragraph (a)(i) or (ii) and reduced by one for each subsequent year.

(b) **Traditional IRAs where investor dies on or after required beginning date.** If the Investor dies on or after the required beginning date, the remaining portion of his or her interest in the Traditional IRA must be distributed at least as rapidly as follows:

(i) If the designated Beneficiary is someone other than the Investor's surviving spouse, the remaining interest must be distributed over the remaining life expectancy of the designated Beneficiary, with such life expectancy determined using the Beneficiary's age as of his or her birthday in the year following the year of the Investor's death, or over the period described in paragraph (b)(iii) below if longer.

(ii) If the Investor's sole designated Beneficiary is the Investor's surviving spouse, the remaining interest must be distributed over such spouse's life or over the period described in paragraph (b)(iii) below if longer. Any interest remaining after such spouse's death must be distributed over such spouse's remaining life expectancy determined using the spouse's age as of his or her birthday in the year of the spouse's death, or, if the distributions are being made over the period described in paragraph (b)(iii) below, over such period.

(iii) If there is no designated Beneficiary or if applicable by operation of paragraph (b)(i) or (b)(ii) above, the remaining interest must be distributed over the Investor's remaining life expectancy determined in the year of the Investor's death.

(iv) The amount to be distributed each year under paragraph (b)(i), (ii), or (iii), beginning with the calendar year following the calendar year of the Investor's death, is the amount determined by dividing the value of the IRA as of the end of the preceding year by the remaining life expectancy specified in such paragraph. Life expectancy is determined using the Single Life Table in Q&A-1 of Section 1.401(a)(9)-9 of the Income Tax Regulations. If distributions are being made to a surviving spouse as the sole designated Beneficiary, such spouse's remaining life expectancy for a year is the number in the Single Life Table corresponding to such spouse's age in the year. In all other cases, remaining life expectancy for a year is the number in the Single Life Table corresponding to the Beneficiary's or Investor's age in the year specified in paragraph (b)(i), (ii), or (iii) and reduced by one for each subsequent year.

(c) **Value of IRA.** The "value" of the IRA includes the amount of any outstanding rollover, transfer, and recharacterization under Q&A-7 and Q&A-8 of Section 1.408-8 of the Income Tax Regulations.

(d) **Designated Beneficiary for minimum distribution purposes.** The "designated Beneficiary" for purposes of determining the distribution period for required minimum distributions after the Investor's death is determined in accordance with Section 1.401(a)(9)-4 of the Income Tax Regulations. In general, the Investor's designated Beneficiary for required minimum distribution purposes is determined based on the Beneficiaries designated as of the date of the Investor's death who remain Beneficiaries as of September 30 of the calendar year following the Investor's death.

(e) **Spousal election.** If the sole designated Beneficiary is the Investor's surviving spouse, the spouse may elect to treat the IRA as his or her own. This election will be deemed to have been made if such surviving spouse makes a contribution to the IRA or fails to take required distributions as a Beneficiary.

4.3A **Required minimum distribution service**

Vanguard may, but shall not be required to, offer a service for Investors and/or Beneficiaries providing calculation or calculation and distribution of required minimum distributions ("RMD Service"). Any such service shall be provided under the terms and conditions set forth in a Service Agreement, as such may be amended from time to time. By enrolling in the RMD Service, the Investor and/or Beneficiary shall be deemed to have consented to such terms and conditions of such Service Agreement, including any amendments thereto effective after the date of enrollment in the RMD Service.

4.4 **Designation of Beneficiary**

(a) **General rules.** The Investor may designate from time to time any person or persons, entities, such as a trust, or other recipient acceptable to the Custodian as his or her primary and/or contingent Beneficiaries. To be entitled to receive any undistributed amounts credited to the Account at the Investor's death, any person or persons designated as a Beneficiary must be alive and any entity designated as a Beneficiary must be in existence at the time of the Investor's death. The surviving primary Beneficiaries shall be first entitled to receive any undistributed amounts credited to the Account at the Investor's death. If the Investor has designated more than one primary Beneficiary, the Beneficiaries shall be entitled to receive any undistributed amount credited to the Account at the time of the Investor's death in the proportions indicated by the Investor. In the event that the Investor has not indicated the proportions to which multiple Beneficiaries may be entitled or has indicated percentages that do not exactly equal 100%, payment will be made to the surviving Beneficiaries in equal shares. Except as described in the next sentence, if any primary Beneficiary has not survived the Investor, that Beneficiary's share of the Investor's Account will be divided proportionately among the surviving primary Beneficiaries. Notwithstanding anything to the contrary in this paragraph 4.4(a), if the Investor has indicated that any Beneficiary designation is made on a Per Stirpes basis and the deceased primary Beneficiary has surviving Issue, the share of the deceased primary Beneficiary shall be divided into equal shares for each such surviving Issue. In the event that there are no surviving primary Beneficiaries at the time of the Investor's death, the contingent Beneficiaries, in the

order indicated by the Investor (secondary, tertiary, etc.), shall be entitled to receive any undistributed amount credited to the Account at the time of the Investor's death and shall succeed to the rights of a primary Beneficiary in accordance with this Agreement. If multiple contingent Beneficiaries at the same level become entitled to any amounts credited to the Account, distribution shall be made in the same manner as if the Beneficiaries were multiple primary Beneficiaries. If no Beneficiary designation is in effect, or if there are no surviving Beneficiaries, at the time of the Investor's death, the Beneficiary shall be the Investor's surviving spouse, if any. If the Investor has no surviving spouse, the Investor's Beneficiary shall be the Investor's estate.

Any Beneficiary designation by the Investor shall be made in a form and manner prescribed by or acceptable to the Custodian and shall be effective only when received by the Custodian during the Investor's lifetime. The Investor may change or revoke his or her Beneficiary designation at any time prior to his or her death by making a new Beneficiary designation with the Custodian. Any such change will revoke all prior Beneficiary designations submitted to the Custodian in their entirety. Investor agrees that in the event of a dispute as to the Beneficiary of the Account, the Custodian, in its discretion, may rely upon an order of a court of competent jurisdiction determining the beneficiary provided that, all interested parties (1) had notice of and an opportunity to participate in the court proceeding, or (2) executed an agreement resolving the dispute. The Custodian reserves the right to ask a court of competent jurisdiction to resolve any beneficiary dispute and to recover its costs of doing so, including reasonable attorneys' fees, from the Account. Unless the Investor has indicated otherwise on the Beneficiary designation, any designation of a Spouse by name or by relationship shall be deemed revoked by the divorce of the Investor and such Beneficiary; provided that no such revocation shall be deemed final until documentary evidence of such divorce, in form and substance acceptable to Custodian, shall have been provided to Custodian, following the investor's death, and Custodian shall not be liable for any payment or transfer made to a Beneficiary in the absence of such documentation. For purposes of this Agreement, divorce shall mean a final decree of divorce, annulment, or dissolution of the marriage in effect in any jurisdiction.

(b) **Minors.** If upon the death of the Investor, a Beneficiary known to the Custodian to be a minor is entitled to receive any undistributed assets of the Account, the Custodian may, in its absolute discretion, transfer assets to an inherited Account for the benefit of the minor Beneficiary. So long as the Beneficiary is a minor, such inherited Account shall be controlled by such person or persons demonstrated to the Custodian's satisfaction to be authorized to act on behalf of the minor. Any person or entity representing his authority to act on behalf of a minor shall submit such information and documentation to authenticate such authority as the Custodian shall reasonably request. The minor Beneficiary's representative may be the guardian, conservator, or other legal representative of such minor Beneficiary, the natural parent of such Beneficiary (provided that if the minor's parents are divorced, the Custodian may deem only the parent having legal custody of the minor to be authorized to act on behalf of the minor), a custodian appointed for such Beneficiary under a Uniform Gift to Minors Act, Uniform Transfers to Minors Act, or similar act, a person appointed by the Investor to act as an authorized person for such minor Beneficiary with respect to the Account in a writing filed with the Custodian or in the Investor's last will and testament as admitted to probate or trust document as to which the Investor is grantor, or any person having control or custody of such minor Beneficiary.

Any minor Beneficiary shall be deemed to be a minor until the later of such Beneficiary reaching (1) the age of majority under the law of the state of the minor's domicile with respect to the right to own mutual funds and other investments or (2) a later age for termination of minor status, but in no event later than age 25, as designated by the Investor in a Beneficiary designation accepted by the Custodian with respect to the Account.

(c) **Marital trusts.** The Investor or, as permitted by law, the spousal Beneficiary following the death of the Investor, may designate as Beneficiary a trust for the benefit of the surviving spouse intended to satisfy the conditions of Sections 2056(b) (pertaining to qualified terminable interest property trusts or "QTIP" trusts) or 2056A (pertaining to qualified domestic trusts or "QDOT" trusts) of the Code (collectively, referred to as "marital trusts"). To the extent such QTIP or QDOT trust is a Beneficiary of the Account, the following provisions shall apply until the earlier of the death of the surviving spouse or the termination of the Account: (1) all of the income of the Account shall be payable to the marital trust or directly to the surviving spouse, at the direction of the trustee of the marital trust, at least annually or at such more frequent intervals as may be directed by the trustee of the marital trust, and (2) no person, other than the surviving spouse, shall have the right to assign any part of the Account to any person other than the marital trust or the surviving spouse.

(d) **Rights of primary Beneficiaries upon investor's death.** In addition to rights otherwise conferred upon Beneficiaries under this Agreement, all individual Beneficiaries shall be entitled to designate Successor

Beneficiaries of their inherited Account. Any Successor Beneficiary designation by the Beneficiary shall be made in accordance with the provisions of paragraph (a) above. If a Beneficiary dies after the Investor but prior to receipt of the entire interest in the Account and has Successor Beneficiaries, the Successor Beneficiaries shall succeed to the rights of the Beneficiary. If a Beneficiary dies after the Investor but prior to receipt of the entire interest in the Account and no Successor Beneficiary designation is in effect at the time of the Beneficiary's death, the beneficiary shall be the Beneficiary's estate. Upon instruction to the Custodian, each multiple Beneficiary may receive his, her, or its interest as a separate account, within the meaning of Regulation Section 1.401(a)(9)-8, Q&A-3, to the extent permissible by law. The trustee of a trust Beneficiary shall exercise the rights of such trust Beneficiary.

4.5 **Responsibility of Custodian and Vanguard**

(a) **Identification of Beneficiaries.** The Custodian shall not be responsible for determining the identity or interest of any Beneficiary designated by relationship to the Investor. The Custodian is fully entitled to rely on any representations made by the Authorized Party or, if applicable, the Beneficiaries, with respect to the identity of the Beneficiaries of the Account, and shall be under no duty to make any inquiry or investigation thereto. The Custodian and Vanguard have no responsibility to locate or notify any Beneficiary or the personal representative of the Investor or any Beneficiary of the existence of the Account. It is the responsibility of the Beneficiary or personal representative of the Investor or of the Beneficiary to notify the Custodian of the death of the Investor or Beneficiary, and to provide the Custodian with such documentation as the Custodian deems necessary to transfer ownership of the Account. The Investor agrees that the Custodian and Vanguard shall have no liability for, and shall be fully indemnified against, any cost or damage they incur in connection with their good-faith reliance upon such representations.

(b) **Distributions.** In making any distribution from the Account, the Custodian shall be fully entitled to rely upon the directions of the Investor or, following his or her death, the Beneficiary, and shall be under no duty to make any inquiry or investigation with respect thereto. The Custodian shall not have any responsibility to make any distribution, including a required minimum distribution, until it receives such directions in form and manner acceptable to the Custodian. Neither the Custodian nor Vanguard shall have any responsibility for the timing, propriety, or tax consequences of any distribution from the Account, which matters shall be the exclusive responsibility of the Investor or Beneficiary, as the case may be.

(c) **Further obligations.** The Custodian shall not be responsible for (1) the interpretation of any formula clause or trust provision contained in any Beneficiary designation filed with the Custodian, (2) the determination of the legal effect of any disclaimer or renunciation made by any Beneficiary to the Account, or (3) the enforcement of any legal obligation, including tax obligations, of the Investor or any Beneficiary. The mere acceptance of any Beneficiary designation submitted by an Investor shall not limit the Custodian's rights or increase its responsibilities under this Agreement and under law. The Custodian is fully entitled to rely on any instructions or representations made by the Beneficiary or the Authorized Party with respect to any of the responsibilities identified in this Article 4.5(c). The Investor agrees that the Custodian and Vanguard shall have no liability for, and shall be fully indemnified against, any cost or damage they incur in connection with their good-faith reliance upon such representations.

(d) **Additional information.** The Custodian reserves the right to request such additional information and documentation from the Investor, the Beneficiary, or the Authorized Party as the Custodian deems may be needed in respect of establishment, maintenance, and distribution of the Account.

Article V

Transfers

5.1 **Transfers to Account.** Assets held on behalf of the Investor in another individual retirement account may be transferred by the trustee or custodian thereof directly to the Custodian, in a form or manner acceptable to the Custodian, to be held in the Account on behalf of the Investor under this Agreement. Transfers are generally only permitted between the same type of IRA plans (e.g., traditional IRA to traditional IRA) or, if made after the Investor's death, to another inherited IRA of the same type from or in the name of the same decedent. In accepting any such direct transfer of assets, the Custodian assumes no responsibility for the tax consequences of the transfer, the responsibility for which rests solely with the Investor or the Beneficiary, as the case may be.

5.2 **Transfers from Account.** If so directed by the Investor, or in the event of a transfer made after the Investor's death by the Beneficiary, in a form or manner acceptable to the Custodian, the Custodian shall transfer assets held in the Account directly to the trustee or custodian of another individual retirement account established on behalf of the Investor or the Beneficiary, as the case may be. Transfers are generally only permitted between the same types of IRA plans (e.g., traditional IRA to traditional IRA) or, if made after the Investor's death, to another inherited IRA of the same type from or in the name of the same decedent. In making any such direct transfer of assets, the Custodian assumes no responsibility for the tax consequences of the transfer, the responsibility for which rests solely with the Investor or the Beneficiary, as the case may be.

5.1 **Transfers incident to divorce.** All or any portion of the Investor's interest in the Account may be transferred to a former spouse pursuant to a divorce decree or written instrument incident to divorce as provided in Section 408(d)(6) of the Code, in which event the transferred portion of the Account shall be held as a separate individual retirement account for the benefit of such spouse in accordance with the terms and conditions of this Agreement.

Article VI

Reporting, disclosure, and fees

6.1 **Information by Investor.** The Investor shall furnish the Custodian with all information as may be necessary for the Custodian to prepare any reports required pursuant to Section 408(i) of the Code and the regulations thereunder.

6.2 **Annual reports by Custodian.** The Custodian shall render an annual report to the Investor, on or before January 31 of each calendar year, containing all information with respect to the preceding calendar year as is required to be furnished pursuant to Section 408(i) of the Code and the regulations thereunder and such information concerning required minimum distributions as is prescribed by the Commissioner of Internal Revenue. In addition, the Custodian shall submit all other reports to the IRS and the Investor (or, following his or her death, the Beneficiary) as may be prescribed by the IRS including any returns relating to unrelated business taxable income generated by the Account.

6.3 **Fees and expenses.**

(a) **Custodian fees.** The Custodian shall be entitled to such reasonable fees with respect to the establishment and maintenance of the Account as are established by it from time to time, and to reimbursement for all reasonable expenses incurred by it in the management of the Account. The Custodian may change its fees payable under this Agreement at any time upon notice to the Investor. The Custodian shall be entitled to reimbursement for costs, including attorney's fees and other expenses, related to the Account.

(b) **Investment management and advisory fees.** Notwithstanding anything contained herein to the contrary, the Investor may authorize the direct payment of investment management and/or advisory fees from the Account to the Custodian or other third party provided that the Account is solely liable for the payment of such fees.

Article VII

Amendment, termination, and assignment

7.1 **Amendment.** The Custodian is authorized to amend the Agreement in any respect and at any time (including retroactively) to comply with the applicable provisions of the Code and the regulations thereunder. The Custodian is also authorized to amend this Agreement to reflect any other changes to the terms of this Agreement that the Custodian deems appropriate. Any such amendment to the Agreement shall be deemed effective upon the delivery of notice of the amendment to the Investor (or following the Investor's death, the Beneficiary). Notice may be provided electronically, provided that the Investor or the Beneficiary has consented to electronic delivery of the Agreement and any and all amendments thereto. The Investor (or following the Investor's death, the Beneficiary) shall be deemed to consent to any such amendment if he or she fails to object thereto by notifying the Custodian, in a form and manner acceptable to the Custodian, within 30 calendar days from the date the notice is delivered, to terminate the Agreement. The terms of the Agreement in effect at the death of the Investor or the Beneficiary, as the case may be, will control the disposition of the assets.

7.2 **Resignation or removal of Custodian.** The Custodian may resign at any time upon 30 days' written notice to the Investor (or, following the Investor's death, the Beneficiary), which notice may be waived by the Investor (or, following the Investor's death, the Beneficiary), and may be removed by the Investor (or, following the Investor's death, the Beneficiary) at any time upon 30 days' written notice to the Custodian (which notice may be waived by the Custodian). Upon such resignation or removal, the Investor shall appoint a successor custodian under this Agreement. The successor custodian shall be a bank or other person qualified to serve as trustee of an individual retirement account under Section 408(a)(2) of the Code. Upon receipt by the Custodian of written acceptance of such appointment by the successor custodian, the Custodian shall transfer to the successor custodian the assets of the Account and all necessary records pertaining thereto, after reserving such reasonable amount as it deems necessary for payment of its fees and expenses. If within 30 days after the Custodian's resignation or removal, or such longer time as the Custodian may agree to, the Investor (or, following the Investor's death, the Beneficiary) has not appointed a successor custodian, the Custodian may terminate this Agreement in accordance with Article 7.3.

7.3 **Termination.** The Investor may at any time terminate this Agreement by delivering to the Custodian a written notice of termination. The Custodian may terminate this Agreement in the event the Investor (or, following the Investor's death, the Beneficiary) fails to appoint a successor custodian under circumstances described in Article 7.2. Upon termination of the Agreement, the assets in the Account shall be distributed to the Investor in a lump-sum payment of cash or in kind.

7.4 **Assignment of Agreement.** The Custodian reserves the right to assign and/or delegate any and all of its rights and obligations under this Agreement to an affiliate of the Custodian without the prior approval of the Investor or Beneficiary.

Article VIII

Miscellaneous

8.1 **Exclusive benefit and nonforfeitability.** The Account is established for the exclusive benefit of the Investor or, following his or her death, the Beneficiary. The interest of the Investor (or following the Investor's death, the

Beneficiary) in the balance of the Account shall at all times be nonforfeitable.

8.2 **Prohibition against assignment of Account.** Except as may otherwise be provided in Article 5.3 or 8.3, no interest, right, or claim in or to any part of the Account or any payment therefrom shall be assignable, transferable, or subject to sale, mortgage, pledge, hypothecation, commutation, anticipation, garnishment, attachment, execution, or levy of any kind, and the Custodian shall not recognize any attempt to effect any of the preceding, except to the extent required by law.

8.3 **Escrow Account.** The Account may serve as an escrow arrangement to hold restricted distributions from defined benefit plans in accordance with an escrow agreement received by, and acceptable to, the Custodian in its sole discretion. Distributions from such Account will be restricted in accordance with the terms of the escrow agreement.

8.4 **Prohibited transactions.** The Investor (or following the Investor's death, the Beneficiary) shall not engage in any transaction with respect to the Account that is prohibited under Section 4975 of the Code and which, under Section 408(e)(2) of the Code, would cause the Account to no longer qualify as an individual retirement account.

8.5 **Governing law.** This Agreement shall be governed, administered, and enforced according to the laws of the Commonwealth of Pennsylvania, except to the extent preempted by federal law.

8.6 **Agreement controls.** In the event of any discrepancy between this Agreement and any other document or instrument filed with Vanguard in respect of the Account, the terms of this Agreement shall control. Except with respect to section 4.5(c), the terms of any Beneficiary designation accepted by the Custodian shall control over the terms of this printed Agreement to the extent of any inconsistency.

8.7 **Simultaneous death and slayer statutes.** In the event that the order of the deaths of the Investor and any primary Beneficiary or on inherited Accounts, of the Beneficiary and any primary Successor Beneficiary, cannot be determined or are deemed to have occurred simultaneously under the law of the state of Investor's or Beneficiary's domicile, as the case may be, the survivor shall be that person who is determined to survive in accordance with the law of that state at the time of the Investor's or Beneficiary's death, as the case may be. In the event that the death of the Investor or any Beneficiary is the result of criminal act involving any other Beneficiary, Vanguard may look to the law of the state of domicile, including any slayer or similar statute, to determine the rights of the Beneficiaries to the assets in the Account.

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